PHINMA UNIVERSITY OF ILOILO

College of Information Technology Education

A Web-Based Integrated School Management System for Timothy's Cradle Christian International School Inc.

In Partial Fulfillment of the Requirements

For Capstone 2 and Research Subject

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CHAPTER I

INTRODUCTION

1.1 Background of the Study

The rapid and continuing advancement of information and communication technologies has profoundly reshaped how organizations across all sectors manage information, conduct daily operations, and maintain meaningful communication with their stakeholders. Within the education sector, digital transformation has become increasingly central to strengthening administrative efficiency, enhancing academic management, improving institutional communication, and expanding equitable access to information.

However, despite widespread recognition of these benefits and growing adoption of digital systems in larger institutions, many smaller educational organizations — particularly private schools and those located in underserved communities — continue to rely predominantly on manual, paper-based, and fragmented methods for managing student records, attendance monitoring, inventory oversight, report preparation, and stakeholder communication [1].

These traditional approaches persist not out of preference, but due to practical constraints: limited financial resources, insufficient in-house technical expertise, uneven digital infrastructure, and legitimate uncertainty regarding the cost, complexity, and long-term maintenance requirements of implementing and sustaining computerized systems [2].

Manual information management demands that school personnel devote substantial time and effort to repetitive, labor-intensive tasks: writing and updating physical records, maintaining handwritten logbooks, organizing and cross-referencing paper documents, manually computing academic grades, compiling attendance summaries, monitoring stock levels, and distributing printed announcements to parents and teachers.

While these methods may appear manageable during the early stages of a school's development or when the student population and scope of activities remain small, they become progressively cumbersome, inefficient, and error-prone as enrollment grows and institutional responsibilities expand.

As the volume of information increases, the time and physical effort required to locate, update, and preserve records multiply disproportionately. Consequently, manual systems are increasingly likely to produce avoidable delays, duplicated entries, inconsistent or conflicting versions of the same information, human calculation errors, misplaced or damaged documents, and genuine difficulty in monitoring trends and identifying emerging issues before they become critical [3].

Student records represent the institutional memory and core information asset of any educational institution. These comprehensive repositories include personal identifiers, enrollment details, demographic information, parent or guardian contact particulars, attendance histories, academic performance data, and special needs or medical notes — all essential for supporting daily instruction, reporting, and student welfare.

When such vital information is maintained primarily in physical filing systems and disconnected spreadsheets, retrieving specific details becomes inefficient and unreliable. Staff may need to search across multiple cabinets, folders, bound registers, and unconnected computer files before locating the required record — a process that can disrupt scheduled activities, delay decision-making, and strain administrative capacity. Furthermore, physical records are uniquely vulnerable: they can be misfiled, lost, damaged by water or fire, deteriorated through handling, or

accessed without authorization. Similarly, data stored in separate, unlinked spreadsheets creates version-control problems; changes made in one file do not propagate to others, resulting in inconsistent or outdated information being used across different departments.

A centralized digital system directly addresses these challenges by providing a single, structured, and secure database where authorized users can store, locate, and update information rapidly and consistently [1].

Attendance monitoring — a fundamental responsibility linked to student welfare, instructional planning, and regulatory compliance — is similarly affected by the limitations of manual processes. Teachers traditionally record daily attendance using printed sheets, class registers, or handwritten logs.

When reporting periods arrive, these entries must be manually reviewed, tallied, and transcribed into summary documents — a process that consumes valuable instructional and administrative time and introduces opportunities for transcription errors when data is transferred from one format to another.

Over time, assembling longitudinal attendance records for individual students or entire classes becomes increasingly complex and disorganized. A computerized attendance module streamlines this workflow by enabling electronic recording, instant storage, automatic percentage calculation, persistent historical archiving, and on-demand report generation — ensuring that attendance data is accurate, accessible, and permanently preserved for intervention planning and compliance documentation [9].

Inventory management is another critical function directly impacted by manual methods. Schools rely on a continuous supply of classroom materials, learning resources, office equipment, library books, furniture, and consumables to sustain daily operations. In the absence of a centralized tracking system, records of stock levels, issuances, returns, and transfers are often maintained in notebooks, loose forms, or isolated spreadsheets — making it difficult to determine current availability, identify items in use, or anticipate shortages.

This lack of real-time visibility can lead to stockouts that disrupt teaching, delayed procurement, unaccounted-for assets, and inconsistent resource allocation across departments. A digital inventory management component provides a unified, always-accurate registry of resources, complete with transaction logging, low-stock alerts, and accountability trails — empowering staff to monitor, plan, and procure resources systematically and cost-effectively [14].

Efficient school management is essential in delivering quality education institution. In recent years, schools have increasingly recognized the value of digital systems in streamlining administrative processes – from student records management to communication. However, many educational institutions, especially smaller ones, continue to rely on manual or semi-digital methods for managing financial transactions. Among the most time-consuming and error-prone of theses processes is the preparation and distribution of student billing statements.

Traditionally, computing tuition and fees, preparing individual billing statements, and notifying parents of payment obligations are done manually by school personnel. This process is not only labor-intensive but also susceptible to mathematical errors, inconsistent record-keeping, and delays in delivering updated account information to parents. Consequently, parents often face inconvenience in verifying their child's financial status, requiring them to visit the school office just to view current charges and payment updates. This gap in accessibility and transparency can lead to misunderstandings, late payments, and increased administrative workload for school staff.

To address theses challenges, the proposed TCCISI Web-Based School Management System integrates a Billing Statements feature designed to automate the computation and generation of student charges based on the school's official tuition and fee schedules. Through the system, billing statements are generated accurately and made accessible

to parents via a secure online portal, allowing them to view real-time updates on their child's account balances, breakdown of charges, and payment status-anytime and anywhere. By complementing this with a dedicated Tuition and Fees Management module for administrators, the system ensures that fee structures remain consistent, up-to-date, and centrally managed, while changes are automatically reflected in all related billing statements.

Beyond financial management, the broader system also covers essential functions such as student information management, website content administration, user activity tracking, and secure parent access. Together, these features create a unified platform that reduces manual paperwork, minimizes discrepancies, enhances information transparency, and strengthens the overall operational efficiency of the institution. By Digitizing these processes, the school can focus more on it's core mission of providing quality education while delivering better service to students, parents, and stakeholders

Effective communication between the school, parents, and teachers is equally essential to student success and institutional trust. Families require timely, reliable information regarding their children's attendance, academic progress, behavioral updates, school events, and policy changes; teachers and administrators need an efficient, documented channel for sharing announcements and responding to inquiries.

When communication depends primarily on printed circulars, personal messages, or disparate external platforms, important information can be delayed, overlooked, or lost entirely — with no permanent, searchable record for reference. A centralized, role-aware communication portal consolidates all official correspondence within one accessible platform, ensuring that announcements reach intended audiences promptly, preserving conversations for future review, and strengthening the partnership between home and school [8].

At Timothy's Cradle Christian International School Inc. (TCCIS), core administrative and academic workflows — including enrollment, attendance tracking, grading, reporting, inventory monitoring, and stakeholder communication — remain managed largely through manual processes, physical files, handwritten logbooks, and basic, unconnected spreadsheets. While these methods have served the institution faithfully, they create growing challenges as enrollment rises and operational responsibilities expand. Increasing workloads mean personnel must manage more records, process more data, and maintain more documentation than before — creating a clear and pressing need for a system that can organize information efficiently while remaining affordable, intuitive, and aligned with the school's actual resources and technical capacity [4].

Beyond inefficiency, reliance on physical records raises serious concerns regarding information accessibility and long-term preservation. Documents stored in filing cabinets are available only when physically present and can be retrieved by only one person at a time; they are also exposed to risks of accidental damage, loss, deterioration, or unauthorized inspection. Data scattered across unrelated spreadsheets compounds these issues, with version conflicts and inconsistent updates undermining confidence in the information itself.

A centralized, role-governed database resolves these discrepancies by providing a single, authoritative source of information — accessible simultaneously to authorized users from different locations, protected from accidental loss, and consistently updated across all applications.

Report preparation is another area where manual methods place disproportionate demands on staff time. Attendance summaries, progress reports, grade sheets, enrollment rosters, and inventory status updates must all be compiled from disparate sources, manually cross-referenced, and reformatted for distribution — a process that can consume days of work per reporting period. Repeated data entry and manual calculation also increase the likelihood of errors appearing in final documents.

Automated report generation directly addresses this by drawing information from the centralized database, applying predefined formatting and filters, and producing consistent, professional-quality outputs on demand — freeing personnel to focus on educational leadership, student support, and institutional improvement rather than repetitive data processing [5].

The broader context of national educational policy further underscores the timeliness and relevance of this initiative. The Philippine education sector has increasingly recognized technology's vital role in improving service delivery, information management, and equitable access to resources — with frameworks such as the Department of Education's MATATAG Agenda and related digital transformation policies encouraging schools to adopt modern, reliable information systems [4].

However, commercially available school management platforms are not universally suitable; many require substantial recurring subscription fees, include features irrelevant to smaller institutions, demand complex configuration, or depend on infrastructure that the school does not currently possess. There is therefore a genuine and underaddressed need for solutions that are purpose-built, affordable, practical, user-friendly, and closely aligned with the specific operational realities of institutions like TCCIS [3].

A customized, web-based school management system offers a sustainable, cost-effective answer to these challenges. Accessible through standard web browsers on computers, tablets, or smartphones, such a system eliminates the need for specialized software installation and enables authorized users to work from multiple locations.

By consolidating student records, attendance tracking, inventory oversight, parent–teacher communication, and automated reporting into a single integrated platform, the proposed TCCIS School Management System (TCCIS SMS) reduces fragmentation, minimizes redundant data entry, and ensures that all operations draw from the same verified information.

The student records component establishes a unified digital repository where personal details, enrollment status, guardian associations, class assignments, and historical performance data are securely stored and easily retrievable — replacing physical folders and disjointed files with a searchable, consistent information foundation.

The attendance module transforms daily monitoring from a paper-based logbook process into a streamlined digital workflow, enabling real-time entry, automatic calculation, permanent archiving, and instant reporting — making participation patterns visible and actionable without manual compilation.

The inventory component creates a living resource registry, tracking quantities, movements, and usage while alerting staff to approaching shortages — ensuring that supplies and equipment remain available and accountable throughout the institution.

The communication portal centralizes announcements and direct messaging within a secure, school-managed environment, ensuring that important information is shared promptly, preserved for reference, and accessible only to appropriately authorized parties — strengthening transparency and trust between the school and families.

The automated reporting engine converts stored data into standardized documents efficiently and accurately, eliminating labor-intensive manual collation and ensuring consistency across all official outputs. While human review

and oversight remain essential, the system removes the mechanical burden of repeated data processing and computation.

Information security and privacy are embedded throughout the design. Because the system contains sensitive personal, academic, and financial data, role-based access control ensures that administrators, teachers, parents, and support personnel each see only the functions and information relevant to their responsibilities — protecting confidentiality, preventing unauthorized modification, and maintaining the integrity of institutional records.

Realistic operational considerations also guide the development. Recognizing that web-based systems depend on reliable internet connectivity, suitable hardware, and user confidence, the interface is designed to be intuitive and familiar; training and orientation are integral parts of the implementation plan. Graceful handling of connectivity interruptions, regular data backups, and straightforward maintenance procedures are prioritized to ensure sustainability within the school's technical environment.

Ultimately, this project is more than a digital conversion of existing forms: it is a purpose-built, contextually appropriate solution designed to support Timothy's Cradle Christian International School Inc. as it grows. By replacing fragmented manual processes with an integrated, accessible, and efficient digital platform, TCCIS SMS aims to enhance operational productivity, improve data accuracy and security, strengthen home–school communication, and ensure transparent stewardship of school resources.

It also lays a flexible foundation for future digital advancement — demonstrating how a carefully designed, needs-aligned web-based system can empower smaller educational institutions to modernize sustainably, protect their most valuable information assets, and focus on their central mission: nurturing student success [5].

1.2 Statement of the Problem

Timothy's Cradle Christian International School Inc. presently depends on a decentralized mix of manual records, physical filing systems, handwritten logbooks, and basic, unconnected spreadsheets to administer its core academic and operational functions — including student record management, daily attendance monitoring, inventory and resource tracking, stakeholder communication, and the preparation of official administrative and academic reports.

While these established methods have served the institution as a familiar and accessible means of handling information, they demand substantial time, sustained effort, and repeated labor from teaching and administrative personnel. As the school's enrollment grows and the volume of information it manages increases, the inherent limitations of these manual, fragmented, and siloed processes have become increasingly apparent, affecting efficiency, consistency, timeliness, and reliability across daily operations.

First, the reliance on paper-based records and disconnected digital files undermines the organization and consistency of school information. Essential records are scattered across multiple physical locations — filing cabinets, bound logbooks, labeled folders — and separate computer files, with no single, authoritative repository.

When personnel need to locate specific details, they must search through several unrelated sources, delaying access to information and disrupting workflow. Updating records compounds this inefficiency: the same information may need to be re-entered repeatedly into different documents or spreadsheets to serve separate purposes.

This redundant data entry directly increases the likelihood of duplicate entries, incomplete fields, conflicting versions of the same record, and avoidable human error. Because changes made in one place do not automatically appear elsewhere, stakeholders may unknowingly act on outdated or inconsistent information.

Second, manual processes significantly impact the timeliness and availability of critical data. Attendance summaries, student profiles, inventory status, and academic performance records are not instantly accessible; instead, they must be manually compiled, cross-referenced, and organized before they can be reviewed or used for decision-making.

This delays the preparation of reports required for internal oversight, parent consultation, and compliance. It also impedes timely intervention: emerging concerns such as patterns of absenteeism, diminishing stock levels, or delayed academic progress may not be identified promptly because the underlying information is not readily consolidated or visible.

Third, physical and decentralized records are inherently vulnerable to loss, damage, and uncertainty of authenticity. Paper documents are susceptible to misplacement, accidental damage, environmental deterioration, or unauthorized access — risks that accumulate over time and with frequent handling. Digital files stored in isolation present their own challenges: they may be accidentally modified, duplicated, saved under conflicting names, or circulated in outdated versions

. Without a centralized, authenticated information management system, maintaining accurate, up-to-date, and consistent records becomes progressively difficult. The absence of a single, trusted source of information means users cannot reliably determine which version — paper or digital, file or folder — reflects the most current and verified data.

Fourth, institutional and interpersonal communication suffers from fragmentation and inconsistency. Important announcements, student progress updates, event notices, and policy changes are disseminated through multiple channels — printed circulars, personal messages, informal group chats, or individual emails — with no unified, school-managed platform to ensure delivery, preserve context, or confirm receipt. Consequently, messages may be overlooked, delayed, or lost entirely; critical details may be remembered differently by different recipients; and there is no permanent, searchable record for reference. This inconsistency hinders the ability of parents and teachers to remain reliably informed about matters concerning student welfare, academic progress, and school activities.

Finally, report preparation imposes a disproportionate and repetitive administrative burden. Producing attendance summaries, grade reports, enrollment rosters, and inventory status updates requires gathering information from disparate sources, verifying accuracy, performing manual calculations, and reformatting data into standardized documents.

Repeated encoding and computation consume valuable time that personnel could otherwise dedicate to instructional leadership, student support, and institutional improvement — while simultaneously increasing opportunities for transposition and calculation errors. When corrections are necessary, the process becomes even more labor-intensive: staff must revisit multiple original records and update every affected file or document individually.

Collectively, these challenges establish a clear and pressing need for a more centralized, consistent, and efficient approach to managing the school's information and daily operations. A purpose-built, web-based school management system can provide a unified digital environment where authorized users can record, update, retrieve, and manage information according to their assigned roles and responsibilities.

By integrating student records, attendance tracking, inventory oversight, parent–teacher communication, and automated reporting within a single cohesive platform, the proposed system can substantially reduce dependence on scattered manual processes and disconnected files, while strengthening the overall organization, accessibility, and reliability of school information.

Accordingly, this study proposes the development of the TCCIS School Management System (TCCIS SMS), a web-based platform designed to address the specific operational challenges identified in the existing management processes of Timothy's Cradle Christian International School Inc.

The system is engineered to serve as a centralized digital foundation that supports efficient record management, timely and accurate attendance monitoring, systematic inventory tracking, organized and transparent stakeholder communication, and reliable automated report generation. Through this solution, the researchers aim to measurably improve the efficiency, accuracy, accessibility, consistency, and security of the school's information management — delivering a practical, sustainable digital tool aligned with the institution's actual operational needs, resources, and future growth.

The school's reliance on manual and fragmented processes presents five critical challenges:

1.The school's reliance on manual and fragmented processes presents five critical challenges:

Attendance recording is slow, prone to human error, and compiled late, delaying timely intervention for absenteeism or chronic tardiness [1].

2.Student records are scattered across physical files and disconnected spreadsheets, making retrieval difficult, data integrity hard to verify, and information vulnerable to loss or damage [2].

3.Inventory and asset monitoring lack systematic tracking mechanisms, leading to unrecorded movement, unexpected shortages, unnecessary overstocking, and poor accountability during audits [3].

4.Communication between the school and families relies on informal, unstructured channels — resulting in missed announcements, delayed feedback, and limited parental engagement [5].

5.Report generation requires repeated manual data consolidation and computation, consuming significant staff time and producing inconsistent, error-prone outputs [4].

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1.1 Objectives of the Study

General Objective
The general objective of this study is to design, develop, implement, and evaluate a robust, web-based Integrated School Management System tailored exclusively for Timothy's Cradle Christian International School Inc. (TCCIS) — hereafter formally designated as the TCCIS School Management System (TCCIS SMS) — that will comprehensively elevate the efficiency, accuracy, accessibility, systematic organization, and overall governance of the institution's administrative and academic operations.

At present, TCCIS remains heavily reliant on labor-intensive, paper-based workflows, physical logbooks, filing cabinets, and isolated spreadsheets to execute and document its most critical functions — an operational setup inherently vulnerable to recurring data duplication, cross-departmental information inconsistencies, prolonged delays in record retrieval, severely restricted information sharing, and an unnecessarily heavy administrative and instructional workload that diverts time from core institutional priorities.

This study directly and purposefully addresses these well-documented limitations by delivering a unified, cloud-accessible, and role-secured digital platform that consolidates all essential school management functions into a single, cohesive, and user-friendly system. Going far beyond basic digitization or simple conversion of existing paper forms into digital format, the TCCIS SMS is purpose-engineered to: streamline complete end-to-end operational processes; enforce consistent data integrity through a centralized, well-structured, and regularly backed-up database; facilitate transparent, timely, and secure information sharing among school administrators, faculty members, parents, guardians, and duly authorized personnel; automate repetitive, time-consuming clerical and reporting tasks; strengthen structured internal communication and meaningful parental engagement; enhance accountability over school assets and instructional resources; and generate reliable, standardized, data-driven reports that support sound administrative decision-making and ensure full compliance with institutional and regulatory requirements.

Ultimately, this project seeks to systematically modernize TCCIS's entire operational framework, substantially reduce long-standing dependence on physical record-keeping, and establish a sustainable, scalable technological foundation that will actively support the school's continued growth, uphold its high academic standards, and advance its long-term strategic goals for full digital transformation.

Specific Objectives
To achieve the general objective, the study pursues the following specific, measurable, and actionable objectives:

1. Automate Attendance Recording, Monitoring, and Follow-Up Mechanisms

- Replace manual paper-based attendance sheets and delayed encoding with a real-time digital attendance module accessible via web browsers on desktops, laptops, and tablets, allowing teachers to mark daily or period-wise attendance in seconds.

- Automatically compute daily, weekly, monthly, termly, and annual attendance percentages; distinguish between present, absent, late, and excused status; and maintain a continuously updated attendance history for every enrolled student.

- Build in configurable alert and notification systems — including flags for consecutive absences, falling below minimum attendance requirements, and irregular attendance patterns — triggering timely notifications to class advisers, school administrators, and parents through the system portal and integrated messaging features.

- Maintain a complete, timestamped audit trail of all attendance entries, edits, and corrections to ensure accountability and eliminate record discrepancies common in manual updating.

2. Centralize, Standardize, and Secure Student Academic and Administrative Records

- Consolidate all student-related data — personal profiles, enrollment forms, admission records, grade history, attendance logs, disciplinary actions, medical information, and transfer credentials — into a single, relational, searchable database that eliminates scattered files, duplicate entries, and conflicting information across departments.

- Enforce strict role-based access control (RBAC) to ensure data privacy and security: administrators possess full system oversight; teachers access only records relevant to their assigned classes; parents view information exclusively for their own children; and unauthorized persons are blocked from all sensitive data.

- Implement structured data validation, backup protocols, and recovery procedures to prevent data loss, corruption, or unauthorized modification; standardize record formats across all grade levels and academic years; and enable quick, keyword-based or filter-based retrieval of student records for inquiries, counseling, evaluation, or official documentation.

3. Develop a Comprehensive Inventory and Asset Lifecycle Management System

- Create a centralized repository for all school-owned resources — including textbooks, learning materials, classroom furniture, laboratory and IT equipment, office supplies, and facilities — with unique identification, category classification, location tagging, and condition tracking.

- Log every inventory transaction systematically: item issuance, returns, borrowing and lending, restocking, transfers between departments, write-offs, and disposal; maintain a full activity history to establish clear accountability and traceability.
- Provide real-time stock visibility with automated low-stock and reorder-level alerts; generate inventory status reports; monitor item consumption patterns; and support procurement planning to prevent shortages or overstocking.

- Manage the full lifecycle of fixed assets — from acquisition and registration to maintenance scheduling, depreciation tracking, and eventual retirement — ensuring that TCCIS can accurately account for, utilize, and preserve its physical resources.

4. Establish a Dedicated, Secure, and Two-Way School–Parent Communication Portal

- Integrate a centralized communication hub within the system that supports timely, organized, and transparent information exchange between school administrators, teaching staff, and parents, replacing fragmented channels such as personal text messages, social media posts, or occasional paper memoranda.

- Enable the publication of verified school-wide announcements, academic calendar updates, event details, policy reminders, and emergency notifications; support direct, private messaging between individual teachers and parents regarding specific student matters; and facilitate the sharing of real-time academic progress updates, grade releases, and behavioral feedback.

- Store all communications within the system with timestamps and read-status tracking, ensuring a permanent, retrievable record of school–parent correspondence and reducing miscommunication or lost information.

5. Enable Automated, Customizable, and Standard-Compliant Report Generation

- Build a flexible reporting engine that dynamically compiles data from all integrated modules — attendance, student records, grades, inventory, and user activity — to produce accurate, consistent, and professional-quality reports without manual calculation or collation

- Generate essential operational and academic documents including: daily and monthly attendance summaries, individual student report cards, class grade summaries, master lists of enrollees, inventory balance and usage reports, personnel activity logs, and government or regulatory compliance forms.

- Provide users with options to customize reporting periods, filter data by class, grade level, department, or date range, and export outputs in widely compatible formats such as PDF, Excel, and CSV; support printing-ready layouts aligned with TCCIS's existing document standards and official submission requirements.

6. Enhance System Usability, Accessibility, and Role-Based Workflow Efficiency

- Design an intuitive, user-friendly web interface that requires minimal technical proficiency, ensuring that non-technical personnel — including administrative staff, teachers, and parents — can navigate and utilize the system effectively with basic orientation and training.

- Ensure cross-device accessibility, supporting consistent use on desktop computers, laptops, and mobile devices with standard web browsers; optimize page loading speeds and interface responsiveness to accommodate varying internet connectivity conditions.

- Streamline and digitize key administrative workflows — such as student enrollment processing, grade encoding and approval, attendance review and endorsement, and inventory request and approval — replacing manual routing and signature procedures with systematic, permission-controlled digital workflows that reduce processing delays and bottlenecks.

7. Evaluate the System's Performance, Functionality, and User Acceptance

- Conduct a systematic evaluation of the fully implemented TCCIS SMS using established criteria — functionality, usability, efficiency, reliability, information security, and accessibility — based on feedback from the system's intended users: school administrators, faculty members, administrative staff, and parents.

- Gather quantitative and qualitative data through structured surveys, user testing sessions, and direct feedback to assess whether the system meets predefined requirements, resolves identified process inefficiencies, and aligns with TCCIS's operational and academic needs.

- Identify strengths, limitations, and areas for functional improvement or interface refinement; document findings to produce actionable recommendations for system optimization, future feature expansion, and sustainable long-term adoption and maintenance of digital systems at TCCIS.

1.4 Scope and Delimitation

This study centers on the systematic design, development, implementation, and evaluation of a web-based Integrated School Management System tailored specifically for Timothy's Cradle Christian International School Inc. (TCCIS) — formally designated as the TCCIS School Management System (TCCIS SMS). The project is conceived and developed in direct response to the distinct operational challenges currently faced by the institution, including a heavy and resource-intensive reliance on manual, paper-based workflows; physically scattered record-keeping across multiple filing systems and offices; fragmented data storage through disconnected spreadsheets and standalone documents; persistent inconsistencies in shared information across departments; excessively time-consuming processes for data encoding, report preparation, and record retrieval; and the absence of a centralized, structured communication channel linking the school administration, faculty, and families.

To address these gaps comprehensively, the system is engineered to consolidate all essential administrative and academic functions — ranging from student profiling and enrollment management, daily attendance monitoring, academic grade processing, and user account administration to inventory tracking, progress reporting, and parent–teacher correspondence — into a single, unified, and secure digital platform.

Through carefully designed role-based access controls, the system enables authorized users, including administrators, faculty, support staff, parents, and students, to securely record, store, retrieve, update, and monitor relevant school-related information at any time and from any location with reliable internet connectivity. By streamlining these core institutional activities and replacing cumbersome, redundant manual procedures with automated, standardized workflows, the project ultimately aims to significantly reduce administrative and instructional workload, strengthen cross-departmental data consistency and organizational accuracy, improve the accessibility and timeliness of critical information, minimize risks associated with data loss, damage, or unauthorized access, and support faster, evidence-based, and more reliable decision-making across the entire institution.

Scope of the Study

The system encompasses the following integrated modules, user roles, and operational capabilities:

Student Enrollment and Profile Management

The system provides a centralized, searchable digital repository for all student records, replacing physical folders, logbooks, and disconnected spreadsheet files. It supports the creation, updating, viewing, and secure maintenance of comprehensive student profiles, including:

- Personal details — full name, date of birth, address, contact information, and civil status;
- Enrollment specifics — grade level, section, academic year, admission date, and enrollment status;
- Parent or guardian information — names, contact numbers, email addresses, and relationship to the student;
- Supplementary data — emergency contacts, medical notes, and relevant background information.

This module ensures that authorized personnel access consistent, up-to-date information, minimizes duplicate or conflicting entries, and significantly speeds up record retrieval for inquiries, documentation, and decision-making.

Attendance Monitoring and History Tracking

The system digitizes daily attendance through an intuitive web-based interface accessible to class advisers and authorized administrators. Key capabilities include:

- Marking attendance status — present, absent, late, or excused — per student, per date;
- Maintaining a complete, timestamped attendance history organized by student, class, section, and academic term;
- Reviewing and filtering records by date range, grade level, or section;
- Supporting efficient monitoring of absenteeism, tardiness patterns, and compliance with minimum attendance requirements.
- This feature eliminates the need to manually consolidate paper attendance sheets and enables timely visibility into student attendance trends.

Grade Encoding, Viewing, and Academic Record Management

Authorized subject teachers can securely encode, update, and manage student grades directly within the platform. Access to academic information is governed by role-based permissions:

- Teachers view and edit grades only for classes they are assigned to teach;
- Parents access grade information exclusively for their own children;
- Administrators maintain oversight of academic records across all grade levels.

The system centralizes grade data, replacing separate grade sheets and disconnected files, and supports term-based and year-end academic tracking. It is explicitly not designed as a full online teaching or learning delivery environment — grade management is its core academic function.

Inventory and Asset Management

The system serves as a structured record-keeping hub for all school-managed resources, including office and classroom supplies, learning materials, textbooks, furniture, IT equipment, and other institutional assets. Authorized personnel can:

- Register new items with details such as item name, category, quantity, unit of measure, and location;
- Log all inventory transactions — procurement/addition, issuance, return, borrowing, transfer between departments, and write-off;
- View real-time stock levels and transaction history to track resource movement and accountability.

This module strengthens transparency, reduces unrecorded usage or loss, and supports more timely and informed procurement and resource allocation decisions.

Communication Portal — Announcements and Direct Messaging

A centralized communication feature consolidates school–family interactions within the system, replacing fragmented channels such as personal text messages, social media groups, or occasional paper memoranda. The system enables:

- Administrators to publish verified, time-stamped school-wide announcements, academic calendar updates, event details, policy reminders, and general notices;
- Teachers and parents to exchange private, direct messages regarding individual student progress, behavior, attendance, or school-related matters;
- All communications to be retained within the system as a retrievable record, helping ensure clarity, accountability, and continuity of information.

Automated and Exportable Reporting

The system generates standardized, data-driven reports directly from the centralized database, eliminating the need for manual cross-referencing and computation. Available reports include:

- Student master lists and enrollment summaries;
- Daily, weekly, monthly, and term-based attendance reports;
- Individual and class grade summaries and progress records;
- Inventory status reports, transaction logs, and resource usage summaries;
- Additional operational or compliance-related documents as required by TCCIS.

Reports can be exported in widely compatible formats such as PDF, Excel, or CSV for sharing, printing, or official documentation. All generated outputs remain subject to final review and verification by authorized personnel before official submission or distribution.

Role-Based Access Control and Security Framework

To protect confidentiality and ensure appropriate system usage, the system enforces distinct permission levels based on user role:

- Administrators — full system oversight, user management, record maintenance, and report generation across all modules;

- Teachers — attendance marking, grade encoding, class-specific student information access, announcement visibility, and parent messaging for assigned students;
- Parents — view-only access to their children's profiles, attendance records, grades, and school announcements, plus direct messaging with relevant teachers;
- Other authorized staff — limited access to specific modules as assigned by the school administration.

All access requires secure login credentials; the system restricts sensitive data to authorized persons and prevents unauthorized viewing or modification of records.

Platform Accessibility and Deployment Options

The system is web-based and designed to function through standard internet browsers on desktop computers, laptops, tablets, and compatible mobile devices — no separate desktop application installation is required. The platform supports two deployment models:

- Cloud-based deployment — accessible from any location with an active internet connection;
- Local-area network (LAN) deployment — accessible only within the school's internal network infrastructure.

Final deployment configuration will be determined based on TCCIS's existing network capacity, hosting preferences, and technical resources.

Delimitations of the Study

The following boundaries define what is not included in the current scope, ensuring the project remains feasible, focused, and aligned with the school's immediate requirements:

Exclusion of a Full Learning Management System (LMS)

The TCCIS SMS is an administrative and academic records platform, not a comprehensive online education delivery system. It does not include:

- Live virtual classrooms or scheduled online lectures;
- Digital course content hosting, lesson modules, or learning materials distribution;
- Online assignment submission, automated grading, or student portfolio management;
- Student–student collaboration tools or discussion forums.

While grade and attendance functions support academic oversight, they do not constitute an e-learning or distance education environment.

Exclusion of Financial and Payment Processing

The system does not integrate online payment gateways, real-time financial transactions, or full accounting functionality. Users cannot process tuition payments, transfer funds, or conduct card-based transactions through the platform. Any fee-related information that may be recorded is limited to basic record-keeping for internal monitoring and does not include billing invoicing, payroll processing, budget tracking, or general ledger management — these remain the responsibility of the school's separate financial systems and personnel.

Dependence on Internet Connectivity — No Offline Mode

In cloud-based deployment, all system operations — logging in, viewing records, entering or updating data, sending messages, and generating reports — require an active internet connection. The current version does not support offline data entry with later automatic synchronization; interrupted or absent connectivity will limit access until the connection is restored.

External Infrastructure and Hardware Limitations

The study focuses exclusively on software design, development, implementation, and evaluation. The following are outside the scope of the researchers' responsibilities:

- Procurement, maintenance, or repair of physical hardware — computers, servers, tablets, network switches, routers, and cabling;
- Electrical system upkeep, power supply reliability, or backup power provision;
- Internet subscription management, ISP-related troubleshooting, or external network service continuity;
- Installation or configuration of third-party hardware devices not explicitly specified in the system design.

The project assumes the school will provide functional devices, stable electricity, and reliable connectivity as prerequisites for system operation.

No Advanced or External Integrations

The current version does not incorporate emerging or specialized technologies unless explicitly requested by the school, including:

- Biometric, facial recognition, or RFID-based attendance capture;
- Automated SMS or push notification services;
- Artificial intelligence, predictive analytics, or advanced data mining;
- Direct interfaces with external government databases or third-party institutional software;
- Standalone native mobile applications separate from the web-based platform.

Continued Need for Human Oversight and Verification

The system is designed to support, not replace, the professional judgment and responsibilities of school personnel:

- Teachers and administrators remain responsible for verifying the accuracy of attendance entries, grades, and all data entered into the system;
- Inventory records require periodic physical cross-checking to confirm consistency between system data and actual stock levels;
- All generated reports must be reviewed and approved by authorized staff before official use or distribution;

- Inaccurate or incomplete data entered by users will produce unreliable outputs — consistent orientation, responsible encoding, and regular verification remain essential to system reliability.

Limited to TCCIS-Specific Requirements

The system is purpose-built and strictly tailored to the specific, verified operational requirements of Timothy's Cradle Christian International School Inc. (TCCIS), distinguishing it from generic, commercially available, or one-size-fits-all enterprise platforms that often include extensive, unnecessary functionalities irrelevant to the school's context and scale. Every module, workflow, and interface has been designed directly in response to the challenges, processes, and priorities identified through consultations and needs assessment within the institution—ensuring that the solution aligns precisely with how TCCIS functions rather than forcing the school to adapt to a pre-built system.

Consequently, features and capabilities that have not been explicitly identified, discussed, and approved as part of the school's official requirements are intentionally excluded from the current scope. This deliberate limitation is essential to maintain project feasibility, control development and maintenance costs, optimize the use of available time, technical capacity, and financial resources, and prevent scope creep that could compromise the quality, timely delivery, and long-term manageability of the system. It also ensures that the platform remains straightforward for staff and stakeholders to learn and use effectively, avoiding unnecessary complexity. Should new requirements, opportunities, or evolving operational needs emerge in the future, additional features, expanded capabilities, or system enhancements may be evaluated and pursued as separate initiatives, subject to further needs analysis, formal review, and official approval from the school's management and relevant authorities.

1.5 Significance of the Study

The development of the TCCIS School Management System (TCCIS SMS) holds substantial value for Timothy's Cradle Christian International School Inc. (TCCIS) and its stakeholders, as it directly addresses critical limitations of the school's current manual, paper-dependent, and fragmented workflows—including slow record retrieval, inconsistent information across files, heavy administrative workload, and scattered communication channels. By consolidating student profiling, attendance monitoring, grade encoding, inventory tracking, parent–teacher communication, and automated reporting into a single, secure, web-based platform, the system delivers measurable operational improvements and lays a sustainable foundation for the school's digital transition.

Specifically, this study is significant to the school administration as it provides a centralized, reliable data repository that streamlines decision-making, reduces repetitive clerical work, and enhances data integrity and institutional accountability; to the faculty and teaching staff, as it simplifies classroom and instructional management, minimizes paperwork, and facilitates timely collaboration with parents to better support student learning; to parents and guardians, as it offers transparent, real-time access to their children's academic progress, attendance, and school updates, strengthening home–school partnership and trust; to the students, as it ensures consistent, accurate tracking of their academic journey and fosters a more efficient, fair, and responsive educational environment; to administrative and support staff, as it automates routine tasks, lightens manual workload, and minimizes errors and delays in daily operations; and to future researchers and system developers, as it serves as a practical reference documenting the design, challenges, and best practices of digital system implementation for small to mid-sized educational institutions, contributing valuable insights to the broader field of educational information systems.

School Administrators

For school administrators, the system establishes a unified, reliable digital hub for overseeing all core institutional information and daily operations. Instead of cross-referencing physical logbooks, filing cabinets, and disconnected spreadsheets — a process prone to gaps, duplication, and delays — administrators gain instant, role-secured access to consistent, up-to-date records across all departments.

This centralized visibility strengthens oversight of enrollment trends, class sizes, attendance patterns, academic standing, and resource utilization in real time.

The system also enhances administrative efficiency and decision-making. Automated report generation eliminates labor-intensive manual data gathering and consolidation, enabling fast production of standardized attendance summaries, grade summaries, student master lists, inventory status updates, and compliance-ready documents.

These outputs support evidence-based planning, policy review, and timely submission of required internal and external documentation [4]. By reducing clerical bottlenecks, administrators can redirect effort toward strategic planning, faculty support, quality improvement, and student welfare initiatives.

Furthermore, the built-in role-based access control strengthens data governance, confidentiality, and accountability. Administrators can systematically assign and adjust permissions; track system activity through audit trails; and restrict sensitive personal, academic, or financial-related data only to authorized personnel.

This structured approach minimizes the risk of unauthorized access, data leakage, or accidental loss, while ensuring that record-keeping aligns with ethical standards and privacy best practices.

Teachers

Teachers stand to gain significantly through reduced administrative burden and more streamlined academic workflows. The web-based attendance module replaces handwritten rolls and manual consolidation, allowing daily attendance to be recorded in seconds and stored permanently.

Attendance histories are instantly retrievable, and emerging patterns such as chronic absenteeism or repeated tardiness become visible sooner — enabling timely intervention and more accurate counseling or referral.

Likewise, the integrated grade management system centralizes encoding, updating, and review of student academic performance. It eliminates the need to maintain multiple separate grade sheets or repeatedly transfer data between documents, lowering the risk of encoding errors and version mismatches [5].

Teachers can securely input term marks, review class performance trends, and share progress updates without switching platforms — ensuring that academic records remain consistent, current, and ready for verification or reporting.

By lessening the time spent on routine encoding, computation, and paperwork, the system restores valuable hours to core instructional responsibilities: lesson design, differentiated instruction, classroom management, assessment, and one-on-one student support.

The built-in communication portal further supports teachers by providing a structured, professional channel to share updates, clarify concerns, and discuss student progress directly with parents — replacing informal, scattered channels and helping ensure that important messages are acknowledged and preserved.

Parents

For parents, the system transforms how they access information and engage with their children's education. Through secure, role-restricted access, parents gain convenient, anytime visibility into their children's attendance records, grades, class standing, and school announcements — from any internet-connected device, without waiting for periodic paper reports or verbal updates. This transparency fosters stronger home–school partnership by keeping families continuously informed rather than only being notified when problems arise [8].

The direct messaging and announcement features deliver timely, verified information, reducing reliance on unofficial groups or delayed notices. Parents can receive immediate alerts regarding important schedules, policy changes, event details, or attendance concerns; they can also send inquiries or feedback directly to teachers, encouraging responsive, respectful collaboration. This improved flow of information helps parents better monitor academic progress, reinforce learning goals at home, identify areas requiring additional support, and participate more meaningfully in school activities — ultimately strengthening trust and cooperation between the family and the institution.

Students

Although students do not hold independent system accounts in this version, they benefit indirectly but meaningfully from the system's implementation. Centralized, accurate record-keeping ensures that their academic history, attendance status, and enrollment details are reliably preserved and readily available — reducing the risk of missing or misplaced records that may affect promotion, transfer, or certification.

More consistent and accessible data enables teachers and administrators to detect attendance gaps, academic struggles, or behavioral patterns earlier, allowing for timely guidance, intervention, or targeted support. Improved communication between school and home creates a more coordinated support network, where teachers and parents align in encouraging the student's growth and well-being.

Additionally, the inventory management function helps maintain availability of textbooks, learning materials, classroom tools, and equipment — ensuring that resources needed for instruction and activities are properly tracked, replenished, and equitably shared, contributing to a more organized and effective learning environment [3].

School Personnel and Staff

Administrative and support staff also derive clear benefits. The system standardizes data entry workflows, simplifies information retrieval, and automates routine encoding tasks, reducing the physical and mental fatigue associated with manual filing, sorting, and computing. Uniform digital formats across all modules minimize confusion, streamline handover between staff members, and simplify training for new personnel.

The inventory module, in particular, offers clear traceability of supply requests, issuances, and returns — helping staff maintain accurate stock levels, anticipate shortages, and uphold transparency in the management of school resources.

Future Researchers

This study serves as a practical, thoroughly documented, and readily applicable reference resource for academic researchers, system developers, and education practitioners engaged in designing, building, or deploying school management information systems — with particular relevance and direct applicability to small to mid-sized private educational institutions operating with constrained financial budgets, limited existing technical infrastructure, or no dedicated full-time information technology personnel. The comprehensive, transparent account provided within this research — covering clearly defined system requirements, structured database architecture, modular feature design and functional specifications, iterative development methodologies, encountered implementation challenges alongside applied solutions, and verified evaluation outcomes — collectively establishes a replicable, adaptable framework that can guide and accelerate similar digital transformation initiatives across comparable educational settings.

Researchers and development teams may draw directly from this work to recognize and diagnose frequently encountered operational pain points that are characteristic of resource-conscious educational environments — including information fragmentation, manual-record inefficiencies, limited data accessibility, and communication gaps — while distinguishing between essential, high-impact core features and non-essential advanced functions that may be deferred until resources become available. This prioritized perspective supports balanced, needs-aligned system design, allowing subsequent researchers and implementers to adapt, customize, and scale the TCCIS SMS model to fit the unique circumstances, enrollment profiles, and operational priorities of other schools with comparable constraints, community contexts, and improvement goals.

Furthermore, this research establishes a concrete, verified baseline and structured foundation for pursuing targeted future enhancements and extended innovation — identifying specific promising directions such as the development of dedicated native mobile applications, integration of biometric or radio-frequency identification (RFID) attendance-capture mechanisms, implementation of automated short-message-service (SMS) notification and alert systems, incorporation of secure online payment processing capabilities, or expansion into a fully featured learning management system (LMS) component — thereby offering subsequent researchers a clearly defined, practically validated starting point from which to explore, test, and evaluate additional functionalities and emerging technological approaches [1].

this study contributes substantive, evidence-based insights to the broader field of educational technology and digital school transformation, specifically regarding the measurable value and sustainable impact of context-appropriate, cost-effective, purpose-built web-based solutions — demonstrating how carefully scaled, institution-centered platforms can meaningfully advance institutional digital readiness, strengthen cross-departmental operational efficiency, improve data integrity and security, and enhance overall service delivery within educational institutions operating under limited material and technical resources.

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CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 School Management & Student Information Systems

Integrated School Management Systems (SMS) and Student Information Systems (SIS) have emerged as foundational digital tools for modern educational institutions, serving to streamline, secure, and unify the broad spectrum of administrative and academic workflows. At their core, these systems consolidate otherwise dispersed processes into a centralized digital ecosystem, enabling information to be captured, stored, processed, updated, and retrieved from a single, shared platform.

In place of disjointed physical logbooks, filing systems, and disconnected spreadsheets — each maintained independently by different departments — an integrated SMS/SIS establishes a unified database where student profiles, attendance records, academic performance, resource inventories, communications, and reports are interconnected and consistently managed.

This shift from fragmented to centralized information architecture addresses longstanding operational bottlenecks: duplication of effort, data inconsistency, delayed record retrieval, and limited visibility across the institution [6].

Core Functions and Integration Benefits

A comprehensive integrated platform typically unifies student enrollment and profiling, grade management, class scheduling, daily and period-based attendance monitoring, inventory and asset tracking, internal and school–parent communication, compliance documentation, and dynamic reporting within one accessible environment. When these functions operate in isolation — forming what are known as data silos — information entered in one process is not automatically reflected elsewhere, forcing staff to manually transfer, re-enter, and reconcile data across multiple sources.

This fragmentation increases the likelihood of discrepancies: a student's updated contact details may appear in enrollment files but not in attendance or emergency records; a corrected grade may be reflected in one teacher's sheet but not in the official class summary; or inventory adjustments may be recorded in a notebook but not communicated to the procurement office. Over time, these inconsistencies erode trust in institutional records and compound administrative workload [6].

Integration resolves these issues by ensuring that information is entered once and reused across all relevant modules, linked through consistent, unique identifiers — such as a permanent student ID — so that updates propagate systematically throughout the system.

This design substantially reduces repetitive encoding, minimizes transcription errors, and establishes a single, reliable source of truth for all authorized users. Rather than cross-referencing multiple folders or files to determine which version of a record is current, administrators, teachers, and designated parents access the same verified dataset, updated in real time.

This not only improves data reliability but also accelerates information retrieval: records that previously required minutes or hours of manual searching become available in seconds, supporting timely decision-making, counseling, and official reporting [6].

Cloud-Based Deployment and Accessibility

Cloud-based system architectures further extend these advantages, particularly for smaller institutions with limited on-premise infrastructure. By hosting the system on secure, remotely managed servers, schools can reduce capital expenditure on dedicated server hardware, specialized maintenance, and complex networking setups.

Cloud deployment also enables location-flexible access: authorized users — administrators, teachers, and parents — can log in and perform permitted tasks from any internet-connected device, including desktop computers, laptops, tablets, and smartphones, without installing specialized software. This capability is especially valuable for enabling consistent school–parent engagement and allowing administrative oversight outside standard campus hours [3].
For small private institutions such as TCCIS, cost-efficiency and scalability are decisive considerations.

Building and maintaining an on-premise IT infrastructure demands sustained investment in equipment, space, power, and skilled technical personnel — resources that many smaller schools cannot fully support. Cloud-based and web-native solutions distribute these costs over time and scale with the institution: as enrollment grows, academic programs expand, or record-keeping requirements evolve, the system can accommodate increased volume without a complete replacement of core infrastructure.

At the same time, cloud deployment introduces important operational considerations: uninterrupted access depends on reliable internet connectivity, and data privacy and integrity must be safeguarded through robust authentication, encryption, and backup protocols. Service interruptions, whether from internet outages or hosting downtime, can temporarily limit system availability, reinforcing the need for contingency planning and reliable connectivity arrangements [3].

Alignment with Philippine Education Standards and Interoperability

Within the Philippine educational landscape, the push toward integrated data management is reinforced by policy and reform initiatives. The Second Congressional Commission on Education (EDCOM II) has highlighted the critical need for standardized, interoperable education data systems across the country — systems in which information is collected, stored, and shared using consistent formats, definitions, and identifiers, rather than being trapped in department-specific files or incompatible software.

Interoperability ensures that data generated through one process — such as enrollment — can be reliably used in related processes — attendance tracking, grade reporting, and compliance documentation — without manual re-entry. This standardization directly reduces administrative burden, improves data quality, and facilitates timely, accurate reporting to internal and external authorities.

For schools, adopting these principles means moving beyond simple digitization toward systemic integration — designing information workflows so that they reinforce one another rather than operating as separate efforts [1].
The importance of interoperability becomes clearer when considering how schools naturally generate information: enrollment establishes student profiles; daily attendance tracks participation; teachers encode grades to measure progress; and administrators compile these into summaries and reports. When these workflows are disconnected, staff must repeatedly transfer information between formats, creating opportunities for error and delay.

An integrated system eliminates these gaps by enabling data to flow seamlessly between modules — an updated enrollment status automatically reflects in attendance rosters; a finalized grade appears in progress reports; a change

in contact details updates across all relevant parent communications — while maintaining consistent identifiers and shared data structures [6].

Standardized, centralized management also elevates the quality and reliability of institutional reporting. When information is captured and stored consistently, compiling attendance summaries, grade distributions, enrollment trends, or inventory status becomes a matter of selecting parameters rather than manually gathering and cross-checking documents. This capability is particularly significant for institutions that must submit periodic reports to regulatory bodies or accrediting agencies — integrated systems reduce the risk of omission or discrepancy and allow staff to focus on analyzing information rather than merely assembling it [1].

Automation, Efficiency, and Balanced Human Oversight

A central advantage of integrated systems lies in automating repetitive, rule-based tasks without diminishing human responsibility. Once attendance is marked, the system can compute cumulative attendance percentages, flag chronic absenteeism, and populate reports — all without manual calculation. Similarly, inventory transactions can update stock levels instantly, and grade entries can be consolidated automatically into term summaries. These capabilities do not replace professional judgment: teachers and administrators remain responsible for verifying data, interpreting results, and making decisions.

Rather, they shift effort from routine clerical work toward higher-impact activities — instruction, student support, family engagement, and strategic planning. The result is not only greater efficiency but also a more sustainable and supportive work environment for school personnel [6].

Accessibility is equally important: integrated systems make records available on demand through intuitive web interfaces, eliminating the need to physically locate folders or logbooks. This immediate availability is critical during parent meetings, student counseling sessions, or emergency situations.

At the same time, accessibility must be carefully balanced with information security: student personal details, academic performance, and family information are confidential and require protection. Role-based access control addresses this by granting permissions strictly according to user function — administrators manage system-wide settings; teachers view and edit records relevant to their assigned classes; parents access information concerning their own children; and sensitive data is shielded from unauthorized users. This layered approach preserves openness where appropriate and confidentiality where necessary [2].

Empirical and Conceptual Foundations

The practical value of these design principles is clearly demonstrated in the iSercan (2019) web-based student information system, which consolidated grades, class schedules, attendance records, and select financial data into a unified portal. The study confirmed that centralized digital record-keeping substantially reduced dependence on paper archives and cut the time required to retrieve or verify information.

Equally important was the finding that role-based access directly strengthens data security and confidentiality: by tailoring visibility to user responsibilities, the system minimized accidental or unauthorized exposure of sensitive records — a priority given the highly personal nature of student and family data [2].

The study further highlighted usability as a critical success factor. Even functionally rich systems will be underutilized or abandoned if their interfaces are confusing or unintuitive. Clear navigation, consistent layout, and simplified workflows lower the barrier for users with limited digital literacy, ensuring that the system serves the entire school community rather only technically proficient staff.

Automated notifications — for attendance updates, grade releases, or announcements — were also found to improve engagement and reduce communication delays, keeping families informed and involved in their children's education [2].
Security measures — including data encryption during transmission and storage, strong authentication protocols, and activity logging — were identified as essential safeguards, ensuring that the benefits of accessibility do not compromise privacy or data integrity [2]. These findings directly inform the TCCIS SMS design, reinforcing the importance of balancing power with simplicity and openness with protection.

The concept of true integration is further developed by Chen and Huang (2024), who distinguish between loosely connected modules and a genuinely unified system. Mere surface-level sharing of information is insufficient; sustainable integration requires a common data model — consistent definitions, formats, and relational structures — so that a student, teacher, class, or transaction is recognized identically across all functions.

When a student is enrolled and their details entered, that record should populate attendance sheets, grade forms, and class rosters automatically, eliminating the need for teachers to re-identify or re-enter the same information repeatedly. This consistency is what distinguishes a collection of separate tools from a cohesive system: changes made in one area are reflected everywhere relevant, and discrepancies from duplicated entries are systematically eliminated [6].

Chen and Huang emphasize that fragmented systems create cascading inaccuracies: if an address update is entered in enrollment files but not in attendance records, communications may be sent to outdated locations; if a grade correction is made in a personal spreadsheet but not in the official database, permanent records remain incorrect. These inconsistencies compound over time, undermining confidence in institutional information and creating additional work to identify and resolve mismatches.

A unified data architecture prevents these cascades by ensuring that one source serves all purposes, making reliability the natural outcome of design rather than an ongoing manual effort [6].
The value of embedding institutional rules directly into system design is illustrated by a web-based expert system developed at the Technological Institute of the Philippines, which automated academic advising and curriculum evaluation.

By encoding program requirements, grading policies, and progression standards into system logic, the platform reduced multi-day manual reviews to near-instant results — with consistent, fair, and predictable outcomes regardless of staff availability or turnover [7]. This example underscores a key principle for TCCIS: custom-built systems should not merely digitize existing processes but codify and preserve the school's own policies and workflows, ensuring that operations remain consistent even as personnel change and the institution grows.
Relevance to TCCIS School Management System

Collectively, these concepts — centralized data management, seamless functional integration, cloud-enabled accessibility, role-based security, standardized interoperability, and policy-driven automation — form the conceptual foundation of the TCCIS School Management System.

Designed specifically to address the manual, fragmented, and paper-reliant workflows at Timothy's Cradle Christian International School Inc., the proposed system applies these established principles to consolidate student enrollment and profiling, attendance monitoring, grade encoding, inventory tracking, parent–teacher communication, and standardized reporting into a single, secure web-based platform.

By prioritizing consistency, usability, and cost-effectiveness, the TCCIS SMS seeks not only to digitize records but to transform how the school manages information — reducing administrative bottlenecks, strengthening data reliability, improving family engagement, and laying a sustainable foundation for the institution's digital growth.

2.2 Attendance Monitoring Systems

Attendance monitoring stands as one of the most fundamental, routine, and critically essential administrative activities in any educational institution — serving not merely as a standard compliance requirement, but as a vital, ongoing indicator of student engagement, punctuality, regularity of participation, personal well-being, and sustained commitment to academic progress. It offers a clear, measurable reflection of how consistently learners are showing up for their education and whether they are receiving the full instructional experience to which they are entitled.

Consistent, accurate, and reliably documented attendance data empowers school personnel to identify at-risk students early—those showing signs of academic disengagement, declining participation, or emerging personal or household challenges—enabling timely, targeted interventions before absence patterns become entrenched and academic performance suffers.
It also provides an authoritative, verifiable record essential for validating eligibility for scholarships, financial assistance, continued enrollment standing, and mandatory regulatory reporting to education authorities. When managed systematically, attendance records further foster a strong sense of responsibility and accountability among students, uphold full transparency and trust with parents and guardians, and reinforce the operational integrity and professional standards of the institution as a whole.

However, traditional manual attendance methods—including in-class verbal roll calls, handwritten daily sheets, physical logbooks, and manually compiled registers—are burdened by well-documented limitations that grow increasingly pronounced, inefficient, and error-prone as student enrollment expands, class schedules become more complex, and reporting and accountability requirements become more rigorous and demanding [9].

These conventional approaches are susceptible to illegible handwriting, accidental omissions, duplicate or conflicting entries, recording delays, and transcription mistakes when data is transferred from paper to summary forms; they also hinder real-time access to attendance trends, make cross-period or cross-class comparison difficult, and require substantial hours of staff time for filing, tallying, and retrieving records—all while offering little protection against loss, damage, or unauthorized alteration of physical documents.

Limitations of Manual and Paper-Based Practices

Reliance on manual attendance recording imposes measurable inefficiencies and risks across daily school operations:

- Time and administrative cost: Teachers spend valuable instructional or planning time marking, tallying, and consolidating attendance — a task repeated across multiple classes and sections each day. Cumulatively, these minutes amount to substantial lost teaching hours over an academic term.
- Data inaccuracy and inconsistency: Handwritten entries are prone to erasures, unclear markings, arithmetic errors, and incomplete entries. Totals computed manually at the end of a month or quarter frequently require reconciliation, and discrepancies may only surface when reports are due.
- Record fragility: Physical sheets can be misfiled, damaged by moisture or wear, lost during transfer between offices, or discarded inadvertently. Retrieving historical attendance data — whether for counseling, transfer credentials, or parent inquiries — often requires searching through stacked filing cabinets or off-site storage.
- Integrity risks: The manual system is vulnerable to proxy attendance, where peers mark one another present, or to inconsistent interpretation of excused versus unexcused absences across different classes. These issues

undermine the credibility of attendance records and make it difficult to apply policies fairly and uniformly [9].

These challenges are not merely inconveniences; they mean that attendance data — information central to student support — is often delayed, unreliable, or unavailable when most needed.

Digital Alternatives: QR-Code and Web-Based Attendance

Digital attendance systems directly address these gaps by automating capture, storage, calculation, and retrieval. Among available approaches, QR-code-based and web-native platforms offer particularly strong alignment with the resources and infrastructure of small to mid-sized institutions.

In a QR-based workflow, a unique code is generated for a class or session; students scan it using a mobile device; and the system instantly validates, timestamps, and logs the entry in a centralized database.

Even where teachers directly mark attendance through a web interface — rather than student self-scanning — the shift to digital capture delivers immediate benefits:

- Attendance is recorded in seconds rather than minutes;
- Daily, weekly, and term attendance percentages are computed automatically and remain current;
- Records are instantly retrievable by date, student, class, or attendance status;
- Data is stored in a consistent format, eliminating deciphering handwriting or cross-referencing separate sheets [9].

Security and precision can be further strengthened through time-bound or encrypted QR codes that expire after a defined attendance window — preventing reuse, delayed entry, or sharing of codes outside the approved period. Such measures preserve simplicity while substantially reducing opportunities for misuse, making them suitable even in settings with limited technical support [9].

Addressing Connectivity and Reliability

A practical concern in many educational contexts — including parts of the Philippines — is that internet connectivity may be intermittent, inconsistent, or temporarily unavailable. A system that halts entirely during an outage disrupts daily operations and risks data loss. Hybrid offline–online architectures resolve this by allowing attendance entries to be cached locally on a device or within the school's local network when connectivity drops.

Once restored, records synchronize automatically with the central server, preserving both workflow continuity and data integrity. This design ensures that attendance recording — a daily, time-sensitive task — does not depend entirely on stable internet access, making the system far more resilient and suitable for varied infrastructure conditions [10].

Enhanced Verification and Timely Communication

Beyond basic presence–absence logging, modern systems incorporate additional validation layers to improve reliability:

- Geolocation checks can confirm that attendance entries originate from within designated school premises, reducing off-site or proxy marking;
- Role-based workflows ensure that only assigned teachers or authorized staff can submit or edit attendance for specific classes, creating an audit trail of who made an entry and when;

- Automated notifications — triggered by recorded absence, tardiness, or consecutive missed days — alert parents and administrators immediately, rather than weeks later when reports are distributed. This timely awareness enables early, constructive follow-up, transforming attendance monitoring from a record-keeping exercise into an active student support tool [10].

Biometric Options and Resource Considerations

Biometric attendance systems — using fingerprint, facial recognition, or iris scanning — offer arguably the strongest individual identity verification, as biometric traits are unique to each person and cannot be shared or borrowed. However, they also carry significant implementation requirements: specialized hardware devices, dedicated software licensing, consistent power supply, secure on-site server or cloud capacity, maintenance arrangements, and clear privacy policies governing sensitive biometric data.

For smaller institutions operating with limited budgets, technical personnel, and IT infrastructure, these demands may exceed available resources. QR-code and web-based alternatives, by contrast, function using standard devices already in the school or community — smartphones, tablets, laptops — and require no dedicated equipment beyond what is already accessible.
They thus provide a practical, cost-effective path toward digital improvement without overextending the institution [11].
Attendance Analytics and Early Intervention

When attendance data is stored digitally, it becomes more than a static record — it becomes an analytical resource. The system can identify patterns that would remain hidden in manual logs: declining attendance over several weeks, recurring tardiness on specific days, or a sudden shift from regular to irregular participation. These trends can be flagged automatically, bringing at-risk students to the attention of advisers, counselors, and administrators before difficulties worsen.

Instead of reacting only when minimum attendance requirements are not met, the school can respond proactively — contacting parents, assessing underlying causes, and offering targeted support. This shift from reactive to preventive monitoring is one of the most valuable contributions of a well-designed digital system [12].

Integration Within a Broader School Management Platform

The full potential of attendance data is realized only when it exists not in isolation but integrated with other school information systems. When attendance records share a database with student profiles, enrollment status, grade reports, and parent contact information:

- Attendance summaries appear automatically on report cards and progress updates without manual transcription;
- Reports can combine attendance trends with academic performance data to identify correlations between participation and achievement;
- Changes to enrollment — such as transferring sections — are reflected immediately in attendance rosters;
- Parents view their child's attendance alongside grades and announcements in a single, consistent portal, rather than through separate channels.

Integration eliminates duplicate data entry, ensures consistency across records, and creates a more complete picture of each student's school experience — reinforcing the value of developing attendance monitoring as one component of a unified school management system, not as a standalone tool [12].

Application to the TCCIS School Management System

For Timothy's Cradle Christian International School Inc., these principles directly inform the design of the TCCIS School Management System. While the current implementation may not include advanced features such as biometric verification or geolocation tracking — prioritizing feasibility and immediate usability — the system will incorporate proven, practical digital attendance capabilities:

- A straightforward web-based interface allowing teachers to mark daily attendance quickly from any connected device;
- Real-time calculation of attendance percentages and status flags;
- Automated, searchable attendance histories organized by student, class, and academic year;
- Role-restricted access enabling parents to view their children's attendance records and administrators to generate consolidated reports;
- A foundation for future expansion — including time-bound QR codes, offline data caching, and automated absence alerts — as resources and needs evolve.

By moving away from paper sheets and scattered logs toward a centralized, digital attendance module, TCCIS stands to reduce clerical burden, improve record accuracy, strengthen communication with families, and establish a reliable, long-term archive of student participation — all aligned with the school's broader transition to integrated digital management.

2.3 School Inventory & Asset Management

Effective inventory and asset management forms the operational backbone of any educational institution. Schools rely on a wide range of resources — from everyday consumables to long-term capital investments — to deliver instruction, maintain facilities, and support daily administrative functions.

These resources fall into two primary categories: consumable supplies, including stationery, teaching materials, cleaning items, and routine classroom provisions; and durable fixed assets, such as computers, printers, projectors, furniture, library books, laboratory tools, and specialized instructional equipment.

Proper management ensures that these resources are available when needed, accounted for accurately, maintained in good working order, and deployed efficiently across the institution. Without structured systems in place, schools face blind spots in resource availability, uncertain accountability, avoidable expenses, and preventable disruptions to teaching and learning [13].

Limitations of Manual and Paper-Based Tracking

Traditional inventory practices — handwritten logbooks, physical stock cards, and disconnected spreadsheets — may appear adequate when an institution is small and resources are few. However, as enrollment grows, facilities expand, and equipment accumulates, these methods reveal critical limitations:

- Delayed or inconsistent recording: Transactions — issuance, return, transfer, restocking — are often recorded after the fact or in separate notebooks, leading to discrepancies between what is written and what is actually on hand. Information may be incomplete, unclear, or overlooked entirely.

- Difficulty tracking movement: Items circulate constantly — between classrooms, departments, storage rooms, and individual staff members. Without real-time documentation, it becomes increasingly difficult to determine where an item is, who is using it, and when it is due back. Valuable equipment may remain unaccounted for or be reallocated without official record, eroding institutional accountability [13].

- Inefficient procurement and stock imbalances: When stock levels are monitored manually, low quantities may go unnoticed until supplies are exhausted, disrupting classes or administrative work. Conversely, uncertainty about actual stock on hand can lead to duplicate or over-sized orders, tying up limited funds and storage space in unnecessary inventory [14].
- Absence of maintenance and lifecycle oversight: Durable assets require periodic care — servicing, calibration, repair — to remain functional. Manual systems rarely capture maintenance history or schedule reminders, so equipment may operate until failure, shortening usable life and creating costly emergency replacements. Acquisition dates, warranty terms, and depreciation are also difficult to track systematically [15].
- Audit and reporting challenges: Verifying recorded inventory against physical stock becomes labor-intensive and unreliable when records are scattered across documents. Reconciling purchases, usage, and remaining assets for internal or external audits requires cross-referencing multiple sources, increasing the risk of omission or error [15].

Web-Based Inventory Systems: Centralized Digital Management

Web-based inventory and asset management systems address these gaps by consolidating all resource information into a shared, searchable, role-secured digital database. Authorized personnel access the system through standard web browsers, eliminating reliance on physical logbooks and enabling consistent, real-time recording from multiple locations.

Each item is catalogued with comprehensive details: unique identifier, name, category, unit of measure, quantity, location, condition, acquisition date and cost, supplier information, assigned custodian, and status. This unified repository provides a single, reliable reference point for all stock-related inquiries [14].

Transaction Logging and Resource Traceability

At the core of the system is structured transaction management. Every change in inventory is recorded as a distinct, timestamped action:

- Procurement and receiving — increasing stock quantities and updating acquisition records;
- Issuance — releasing supplies or equipment to individuals or departments;
- Returns and transfers — documenting the return of reusable items or movement between locations;
- Adjustments and write-offs — recording damaged items, expired materials, or retired assets.

This continuous audit trail establishes a complete, verifiable history of how resources enter the institution, circulate, and are eventually retired. It answers critical accountability questions: Who signed for this equipment? When was it transferred? Where does it currently belong? Such traceability discourages misuse, simplifies reconciliation, and builds confidence in resource stewardship [14].

Automated Stock Monitoring and Procurement Support

Digital systems transform inventory from a passive record into an active management tool. Administrators can define reorder points or minimum stock levels for each item. When available quantity falls to or below this threshold, the system automatically generates alerts — visible notifications or summary reports — prompting timely replenishment.

This prevents unexpected shortages that disrupt classroom activities while avoiding speculative over-ordering. Historical consumption data further strengthens procurement planning: usage patterns reveal which items are in

highest demand, which seasons or terms require greater stock, and which resources may be underutilized. Procurement decisions shift from guesswork to evidence-based planning, optimizing budget allocation and storage use [14].

Fixed Asset Lifecycle and Maintenance Management

Beyond quantity tracking, durable assets require specialized management throughout their usable life. A comprehensive digital system captures the full asset lifecycle:

- Acquisition — recording purchase details, cost, warranty period, and expected lifespan;
- Deployment — assigning to specific locations, custodians, or departments;
- Maintenance — logging routine servicing, repairs, part replacements, and associated costs; scheduling recurring inspections or preventive maintenance;
- Condition monitoring — tracking deterioration or obsolescence;
- Disposal or retirement — documenting decommissioning, sale, donation, or write-off at end of service.

Maintenance reminders are particularly valuable: equipment that receives scheduled care performs more reliably, lasts longer, and avoids costly emergency failures. Historical maintenance records also inform replacement decisions — if a device requires repeated, increasingly expensive repairs, the data supports a strategic case for replacement rather than continued investment in aging infrastructure [15].

Accountability, Auditing, and Financial Alignment

The centralized database serves as a permanent, retrievable foundation for verification and review. During physical inventory audits, staff can compare system records directly against items on hand, noting discrepancies and updating statuses immediately.

Traceable transactions — showing who handled an item and when — strengthen accountability and reduce loss or misplacement. When integrated with broader institutional financial oversight, inventory data also clarifies spending patterns: procurement costs can be compared against usage rates, budget allocation becomes more transparent, and resource distribution across departments can be reviewed for fairness and efficiency [15].

Security and Role-Based Access

Because inventory records include sensitive details — high-value equipment, assigned personnel, procurement plans — data integrity and confidentiality are essential. Role-based access control ensures that:

- Administrators maintain full oversight: adding items, adjusting stock, reviewing transactions, generating reports, and managing user permissions;
- Department heads or custodians view and manage resources relevant to their areas;
- General staff may submit requests or record issuances within defined limits;
- Unauthorized modification or deletion of records is restricted.

System logs preserve a history of changes, creating an additional layer of accountability and protecting inventory data from accidental or malicious alteration [14].

Integration Within the TCCIS School Management System

For Timothy's Cradle Christian International School Inc., the reviewed principles directly guide the design of an integrated inventory and asset management module within the TCCIS SMS — rather than a standalone or disconnected tool. This integration carries distinct advantages:

- Unified platform experience: Staff access inventory records through the same web interface used for student information, attendance, grades, and communication, reducing the need to learn and maintain separate systems;
- Consistent data architecture: Shared user accounts, role permissions, and interface standards simplify training and ensure familiarity across all modules;
- Foundational readiness: While current implementation focuses on reliable tracking, the centralized structure enables future enhancements — such as automated procurement requests linked to budget monitoring, QR-code tagging for physical verification, or consumption analysis — as needs and resources evolve;
- Comprehensive reporting: Inventory summaries, stock status, and asset assignments can be generated alongside attendance, academic, and enrollment data, providing TCCIS leadership with a holistic operational picture rather than fragmented departmental reports [14].

The system will support the daily practical needs of the school: logging supplies and equipment, tracking issuances and returns, flagging low-stock items, and maintaining clear custodial assignments — all without requiring specialized hardware or advanced technical expertise.

It addresses the specific challenges of TCCIS: scattered stock records, uncertain resource location, delayed replenishment, and limited visibility into equipment condition — converting these into a structured, transparent, and sustainable resource management practice.

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CHAPTER III

METHODOLOGY

3.1 Research Design

This study employs a developmental research design that combines qualitative and quantitative approaches throughout the planning, development, implementation, and evaluation phases of the proposed system [16]. Developmental research is appropriate for the study because its primary purpose is to design, develop, and evaluate a functional web-based school management system that addresses the identified administrative and academic needs of Timothy's Cradle Christian International School Inc. Rather than focusing solely on describing an existing situation, the research involves the systematic development of a technological solution based on actual requirements, observed processes, and feedback from intended users.

The qualitative component of the study is primarily used during the requirements-gathering and planning stages. Interviews, document analysis, and workflow observation are conducted to obtain detailed information about the school's existing procedures for managing student records, attendance, grades, inventory, communication, and reports. These methods allow the researchers to understand how school personnel currently perform their tasks, what information they require, which activities are repetitive or time-consuming, and what difficulties they encounter when using manual records, logbooks, and spreadsheets. The information gathered from these activities serves as a basis for identifying system requirements and determining the functions that should be incorporated into the proposed platform.

Interviews provide an opportunity for the researchers to obtain direct information from stakeholders regarding their experiences, needs, expectations, and concerns about the proposed system. Through guided questions, respondents can describe the current processes they follow and identify specific problems that may not be evident through document review alone. The information gathered from interviews can help the researchers determine which system features should be prioritized and how the proposed system should support actual school operations.

Document analysis is also included in the research design to examine existing forms, attendance records, student files, inventory records, reports, and other relevant documents used by the school. Reviewing these materials helps researchers understand the type of information currently collected, how records are organized, and what data fields or reporting requirements should be considered in the proposed system. Document analysis also provides a basis for designing database structures and system forms that correspond to existing school documentation while reducing unnecessary duplication.

Workflow observation is used to understand how administrative and academic tasks are performed in practice. By observing processes such as attendance recording, student information management, inventory monitoring, and report preparation, researchers can identify unnecessary steps, repeated encoding, delays, and other operational difficulties. Observation also helps ensure that the proposed system reflects actual work practices rather than assumptions about how school personnel perform their responsibilities. This is particularly important when developing a system intended to replace or improve manual procedures.

The development phase follows an iterative approach in which the identified requirements are translated into system features, interface designs, database structures, and functional modules. Instead of developing the entire system based on a single set of assumptions, prototypes can be created and reviewed during different stages of development. Feedback from intended users can then be used to identify unclear interfaces, missing functions, workflow problems, or other issues that require modification. This iterative process allows the system to gradually become more aligned with the operational requirements of the school [19].

Prototyping is particularly useful because it provides stakeholders with a visual and functional representation of the proposed system before the final version is completed. Users can interact with or review selected features and provide feedback regarding their usability, organization, and relevance to their responsibilities. Their comments can guide adjustments to navigation, forms, dashboards, data entry procedures, reports, and other system components. Through repeated cycles of development and feedback, the researchers can improve the system before final implementation and evaluation.

The quantitative component is applied during the evaluation phase to measure the performance and acceptability of the developed system. Evaluation may include structured assessment of characteristics such as functionality, usability, efficiency, reliability, security, accessibility, and overall user satisfaction. Responses gathered through evaluation instruments can be summarized using appropriate descriptive statistical measures to determine how users perceive the system and whether the developed features meet the identified requirements [16].

The combination of qualitative and quantitative methods allows the study to examine both the context in which the system will be used and the measurable performance of the developed solution. Qualitative methods provide detailed information about user experiences, existing workflows, and operational problems, while quantitative evaluation provides measurable evidence regarding the quality and acceptability of the developed system. Using both approaches strengthens the development process by ensuring that technical decisions are informed by actual user needs and that the completed system is evaluated using systematic criteria.

The research design also emphasizes continuous alignment between the identified requirements and the developed system. Information obtained during the initial investigation is translated into functional and non-functional requirements, which guide the design and development of the system. Feedback obtained during prototyping is then used to refine these requirements and improve system features. This process helps ensure that the final system is not based solely on theoretical assumptions but reflects the actual practices, limitations, and needs of the school [19].

Through this developmental research design, the study seeks to produce a practical and useroriented web-based integrated school management system for Timothy's Cradle Christian International School Inc. The design provides a structured process for investigating existing problems, identifying requirements, developing the proposed solution, incorporating stakeholder feedback, and evaluating the resulting system. This approach supports the development of system that is technically functional while also being relevant to the school's actual administrative and academic operations**.**

3.2 System Development Methodology

The Agile Software Development Model is adopted as the system development methodology for the proposed TCCIS School Management System. Agile is an iterative and incremental development approach that emphasizes continuous development, testing, evaluation, and improvement throughout the system development process [17]. This methodology is appropriate for the study because the proposed system contains several interconnected modules and requires continuous consideration of the needs and feedback of its intended users. Instead of developing the entire system at once and evaluating it only after completion, Agile allows the researchers to develop the system in smaller functional increments and make improvements throughout the development process.

The Agile approach generally involves repeated cycles of planning, requirements analysis, system design, implementation, testing, review, and refinement. At the beginning of each iteration, the researchers identify the features and requirements that should be addressed. These requirements are then translated into interface designs, database structures, system functions, and other technical components. The selected features are implemented and tested before

the results are reviewed. Issues identified during testing or feedback sessions can then be addressed in the next iteration. This cycle continues until the required system functions have been developed and refined [17].

The first stage of the development process focuses on analyzing the existing processes and identifying the requirements of the school. Information gathered through interviews, document analysis, and workflow observation is used to determine the problems associated with manual student records, attendance monitoring, inventory management, communication, and report preparation. The researchers then organize these findings into functional and non-functional requirements. Functional requirements describe what the system should be able to perform, while non-functional requirements address characteristics such as usability, security, reliability, accessibility, and performance.

After the requirements have been identified, the researchers proceed with system design. This includes designing the overall system architecture, user interfaces, database structure, navigation, user roles, and relationships among the different modules. The design process considers the responsibilities of different users, including administrators, teachers, and parents. Role-based access is incorporated into the design so that users can access only the information and functions relevant to their assigned responsibilities. The database design also establishes relationships between student records, attendance, grades, inventory transactions, communication records, and other relevant information.

The implementation phase develops the system features according to the approved requirements and design. Development is performed incrementally, beginning with the core components of the system. The student records module can be developed first because student information serves as a foundation for other functions. Once the basic student records functionality has been implemented and tested, subsequent iterations can introduce attendance monitoring, inventory management, parent–teacher communication, and reporting features. This incremental approach allows each major component to be developed and validated before additional functionality is introduced.

The student records module serves as the foundation for the succeeding modules because other system functions require reliable student information. The module can include student profiles, enrollment information, parent or guardian details, and other relevant records. During its iteration, the researchers can test functions such as adding, updating, viewing, searching, and managing student information. Feedback from users can be used to improve the organization of forms, search functions, navigation, and access permissions before the module becomes part of the larger integrated system.

The attendance module is then developed as another functional increment. It provides authorized teachers or staff with an interface for recording daily attendance and allows the system to store attendance information in the centralized database. The module can support attendance statuses such as present, absent, late, or excused, depending on the requirements identified during the analysis phase. Testing can determine whether attendance records are correctly associated with students, dates, classes, and authorized users. Any problems discovered during testing can be addressed through subsequent refinement.

The inventory module is developed in another iteration to support the management of school supplies, equipment, and other assets. The module can include functions for recording inventory items, quantities, item categories, issuance, returns, assignments, and other transactions. By connecting inventory information to the centralized database, authorized personnel can retrieve updated records and generate relevant inventory reports. User feedback can be incorporated to improve item categorization, transaction workflows, search functions, and reporting features.

The communication module is subsequently developed to support interaction between authorized school personnel and parents or guardians. This module can include
announcements and direct messaging functions based on the requirements identified during the planning stage. The iterative approach allows the researchers to evaluate whether users can easily locate messages, compose communications, receive notifications, and access relevant announcements. Feedback from stakeholders can be used to improve the clarity and organization of communication features [19].

The reporting module is developed to consolidate information generated by the different system components. It can provide authorized users with reports related to student records, attendance, grades, inventory, and other relevant school information. Developing the reporting functions after the core modules have been established allows the system to generate reports using centralized and updated information rather than relying on manually consolidated records. The researchers can test whether reports contain accurate information, follow the required format, and can be generated efficiently.

Testing is performed throughout every development iteration rather than being postponed until the end of the project. Functional testing verifies whether individual features perform according to their requirements, while integration testing examines whether connected modules exchange and use information correctly. Usability testing can determine whether intended users can navigate the system and complete common tasks without unnecessary difficulty. Securityrelated testing can also examine whether role-based permissions prevent unauthorized users from accessing restricted functions or information.

Stakeholder feedback plays an important role in the Agile methodology. After a functional increment has been developed and tested, users can review the feature and provide comments regarding its functionality, usability, and alignment with actual school procedures. Feedback may identify missing features, unnecessary steps, confusing interfaces, or differences between the system workflow and existing school practices. The researchers can then prioritize these observations and incorporate appropriate changes into subsequent development iterations [19].

The iterative nature of Agile also allows changes to be accommodated during development. Requirements may become clearer as stakeholders interact with working prototypes and identify additional needs. Rather than treating the initial requirements as permanently fixed, Agile provides opportunities to revise selected features when changes are justified by user feedback or findings from testing. This flexibility is particularly useful for a school management system because actual workflows may contain details that are difficult to identify completely during the initial requirements-gathering stage.

Each iteration therefore follows a continuous cycle of planning, analysis, design, implementation, testing, review, and refinement. The completion of one iteration provides a functional increment that becomes the basis for the next stage of development. Through this process, the researchers can progressively build the integrated system while continuously checking whether each component meets the identified requirements. Problems can be identified and corrected earlier instead of accumulating until the final stage of development [17].

The Agile methodology also supports the integration of the different system modules. Although student records, attendance, inventory, communication, and reporting are developed as individual functional increments, they are designed to work together through a centralized database and consistent data structures. For example, student information entered into the student records module can be used by the attendance and reporting modules without requiring the same information to be repeatedly entered. This reduces duplication and supports consistency across the system.

For the proposed TCCIS School Management System, Agile provides a structured yet flexible methodology for developing the system according to actual stakeholder requirements. The iterative process enables the researchers to begin with essential functions and progressively introduce additional modules while continuously testing and refining each component. Through repeated stakeholder feedback and system evaluation, the methodology supports the development of a web-based platform that reflects the school's actual workflows and operational needs rather than relying solely on assumptions made during the initial planning stage [19**].**

3.3 System Architecture

The proposed TCCIS School Management System adopts a structured, three-layer client–server architecture composed of the Presentation Layer, Application Layer, and Data Layer [16] — a modular design approach that deliberately isolates the system's user interface, core business logic and processing, and data storage and management responsibilities into distinct, self-contained tiers. This clear functional separation fosters superior organization, clarity, and independence across all system components; significantly simplifies maintenance, unit testing, feature updates, and future scalability; and allows developers to refine or expand individual layers without requiring extensive modification or disruption to the others, thereby minimizing unintended side effects and preserving system stability throughout its lifecycle.

At the top tier, the Presentation Layer serves as the direct point of interaction for all authorized users — including administrators, faculty, staff, parents, and students — delivering a responsive, accessible web-based interface through which inputs are submitted and information is displayed. It handles data formatting, role-based view customization, and user navigation while relaying user actions to the middle tier for processing. Occupying the central tier, the Application Layer functions as the system's operational core, executing all business rules, managing workflow logic, processing computations, enforcing access permissions and security protocols, and coordinating all data transactions between the interface and the repository.

It validates inputs, automates processes such as attendance tracking, grade calculation, and report generation, and ensures that all operations adhere to the school's defined procedures and standards. The foundational Data Layer provides a centralized, secure, and persistent repository that stores, organizes, and protects all institutional information — including student profiles, attendance logs, academic records, inventory listings, user credentials, communication archives, and system activity trails — and manages efficient data retrieval, backup, and integrity enforcement.

Beyond structural clarity and maintainability, this architecture establishes a robust, unified framework that seamlessly integrates the system's major functional modules — namely student profiling and record management, real-time attendance monitoring, academic grade encoding and progress tracking, school asset inventory management, secure parent–teacher communication, and automated reporting and analytics — ensuring consistent data sharing, synchronized workflows, and cohesive performance across the entire platform. It also supports secure, web-based accessibility, role-based access control, and reliable performance as the institution grows, laying a flexible and sustainable technical foundation aligned with TCCIS's long-term digital transformation objectives.

Presentation Layer

The Presentation Layer constitutes the visible, user-facing component of the system — the primary interface through which all authorized individuals interact with the platform — and is built as a modern, standards-compliant responsive web interface using HTML5, Tailwind CSS, and JavaScript [24]. HTML5 forms the structural foundation of every page, defining the semantic layout and organizing essential elements including data-entry forms, dynamic data tables, intuitive navigation menus, personalized user dashboards, and all other functional interface components. Complementing this structure, Tailwind CSS governs the visual presentation, styling consistency, and adaptive layout of the system, ensuring a clean, professional, and accessible appearance across all views. Meanwhile, JavaScript powers interactive capabilities, real-time interface updates, and dynamic content handling — delivering smooth, responsive interactions that significantly enhance overall usability and user experience without requiring full page reloads for routine actions.

A core design priority of this layer is its fully responsive architecture, which automatically adapts the interface layout, element sizing, and navigation style to match the screen dimensions and display capabilities of any device. Users may therefore access the system seamlessly through any standard web browser — whether on desktop computers, laptops, tablets, or compatible mobile devices — without loss of functionality or readability.

This cross-device accessibility offers meaningful flexibility for teachers, administrators, and parents who may switch between different devices throughout the day or who prefer using mobile platforms for routine updates and quick checks. The interface is intentionally designed to group frequently used functions into clearly labeled, logically organized menus and dashboard widgets, minimizing unnecessary navigation steps and enabling users to locate, review, and manage relevant information more efficiently and intuitively [24].

Equally important, the Presentation Layer dynamically renders distinct, role-tailored interfaces based on the authenticated user's assigned access level and institutional responsibilities, ensuring that each user sees only the features and information relevant to their role.

For instance, administrators are presented with comprehensive controls for student enrollment and profiling, inventory oversight, system-wide reporting, and user account management; teachers primarily access classroom-focused tools such as student records, daily attendance tracking, grade encoding, and parent communication; and parents are provided with a dedicated, simplified view displaying authorized information specifically concerning their children — including attendance history, academic progress reports, school announcements, and secure messaging features. This role-based presentation not only streamlines each user's experience by eliminating irrelevant options but also reinforces information security by visually restricting exposure of sensitive or non-public data to those properly authorized to view it.

Data-entry forms constitute the primary mechanism through which users submit new or updated information to the system, covering all key workflows: student registration and profile updates, daily attendance logging, grade encoding and review, inventory issuance and restocking, and secure communication submissions. Built directly into these interface forms are client-side input validation routines that verify the completeness, format, and reasonableness of entries before they are transmitted to the Application Layer — helping to prevent incomplete submissions, flag potential typographical errors, and ensure that required information is provided in the expected structure. Every aspect of form design — including clear field labels, logical grouping of related inputs, descriptive instructional text, appropriately positioned action buttons, and consistent navigation patterns — is carefully considered to maximize usability, reduce confusion, and minimize the likelihood of user error during data entry.

Application Layer

The Application Layer constitutes the operational core of the system, housing all server-side logic responsible for receiving, interpreting, and processing requests transmitted from the Presentation Layer. Developed using PHP, this tier functions as a dedicated intermediary between the user-facing web interface and the underlying Data Layer, ensuring that every user-initiated action is handled systematically, validated thoroughly, and executed strictly in accordance with the institution's defined procedures, business rules, and security requirements. By centralizing processing logic in this middle tier, the system prevents the Presentation Layer from interacting directly with raw database operations, establishing a controlled, standardized pathway for all data transactions and creating a single, maintainable location for implementing, updating, and enforcing system-wide policies.

This layer manages a comprehensive set of critical functions, including secure user authentication, role-based access authorization, multi-stage data validation, new record creation, existing record updates, targeted information retrieval, automated report generation, and internal message processing. When a user submits data or initiates an action through the web interface, the request is routed to the corresponding PHP script, which applies the necessary processing logic before retrieving, modifying, or storing information in the database. This structured separation ensures consistency across all modules, simplifies troubleshooting and future enhancements, and isolates the database from unvetted or unauthorized direct access.

All database interactions are implemented using PHP Data Objects (PDO) in conjunction with prepared statements [22] — a robust security practice that structurally separates SQL command syntax from user-supplied values. Rather than embedding input directly into query strings, prepared statements treat user data exclusively as parameterized values, rendering the system highly resistant to SQL injection attacks and unauthorized data manipulation. This protective mechanism is especially critical given the sensitive nature of the information managed by the platform — including confidential student profiles, verified attendance logs, academic grades, parental contact details, and user account credentials — ensuring that all exchanges remain secure and data integrity is preserved.

A central responsibility of this layer is the enforcement of authentication and role-based access control throughout the system. During every login attempt, the application verifies submitted credentials against securely stored records, validates account status, and identifies the specific role assigned to the authenticated user — such as administrator, teacher, parent, or student. Based on that role, the system dynamically enforces permission boundaries, determining which functions, modules, and datasets the user is authorized to view or modify. Crucially, these authorization checks are applied at the processing level for every relevant request, not merely through interface visibility; even if a user attempts to access a restricted path or feature directly, the Application Layer validates permissions and blocks unauthorized attempts, providing a consistent and reliable security safeguard.

The Application Layer also encodes and enforces all institutional business rules, ensuring that data processing adheres to logical, consistent standards across every module. For example: attendance entries are automatically linked to verified student records and specific dates to prevent orphaned or duplicate entries; grade submissions are strictly associated with the correct student, subject, grading period, and academic term; and every inventory transaction triggers corresponding adjustments to stock levels, with built-in checks to prevent negative inventory counts or unrecorded movements. These embedded rules standardize workflows, reduce reliance on manual oversight, and maintain relational consistency across the entire system.

Furthermore, this layer facilitates seamless, reliable interoperability among all functional modules through a unified data architecture. Because components share a centralized database managed through consistent logic, information captured in one module is immediately available to other authorized functions throughout the system. Student details entered during registration are automatically accessible when logging attendance, encoding grades, or generating

progress reports; attendance and academic performance data can be aggregated across periods for official summaries; and inventory updates reflect instantly wherever stock levels are referenced. This integrated flow eliminates redundant data entry, minimizes discrepancies caused by duplicate recording, ensures all users work from a single authoritative source of information, and delivers the unified, cohesive experience central to the system's design.

Data Layer

The Data Layer serves as the secure, persistent foundation of the system, responsible for reliably storing, systematically organizing, and efficiently managing all institutional information processed and referenced across the platform. The proposed TCCIS School Management System employs MySQL as its relational database management system [23], chosen for its proven reliability, robust performance, strong security features, and widespread compatibility with the system's web-based architecture. At its core, the database implements a carefully designed, normalized schema comprising 18 logically interconnected tables, systematically grouped into four distinct functional categories: core records, academic information, operational records, and public content. This modular yet fully integrated structure is purpose-built to eliminate data fragmentation, support seamless information sharing across modules, and ensure long-term maintainability and scalability.

The core records group houses the institution's foundational reference data — the essential identifiers and relational framework upon which every other system function depends. These tables include user accounts with secure authentication credentials, comprehensive student profiles, parent–guardian relationship mappings, class and section assignments, academic term and school year definitions, and other standardized reference entities. By centralizing this foundational information in one authoritative location, the layer establishes consistent, permanent links across the entire system: for instance, a unique student identifier serves as the common key connecting a student's personal profile to their daily attendance logs, grade records, disciplinary updates, and progress reports, ensuring that all related data remains reliably associated and traceable.

The academic group organizes all information generated through instructional and learning activities, including subject and curriculum definitions, detailed daily and period attendance tracking, grading criteria, term and final grade records, and academic progress summaries. Attendance entries, for example, are explicitly linked to specific student profiles, assigned classes, responsible faculty members, exact dates, and standardized attendance status categories — enabling precise filtering, trend analysis, and absence monitoring over time. Similarly, grade records are systematically associated with individual students, specific subjects, designated grading periods, corresponding school years, and the authorized teacher who encoded the assessment data. By structuring these relationships formally rather than through isolated spreadsheets or paper logs, the system can instantly compile comprehensive academic histories, generate consolidated performance reports, and cross-reference attendance patterns with academic outcomes without manual collation or reconciliation.

The operations group manages data supporting the school's administrative, logistical, and resource-related functions, most notably inventory and asset management. Dedicated inventory tables maintain detailed records of school resources — including item descriptions, stock quantities, reorder thresholds, responsible custodians, and a complete chronological log of all inventory transactions such as acquisitions, issuances, returns, and adjustments. Storing every movement as a distinct, timestamped database entry allows the system to maintain a permanent, auditable trail of asset usage, automatically calculate current stock levels, flag low-quantity items, and produce accurate, itemized inventory reports — replacing manual stock ledgers with a dynamic, real-time resource oversight mechanism.

The public content group holds information intended for broader dissemination through school announcements, general notices, event calendars, and public-facing updates. This category is deliberately separated from confidential administrative and student data, creating a clear functional boundary: public content may be viewed by general visitors or authenticated users without special access, whereas sensitive records such as student performance, personal contact details, inventory costs, and user account information remain strictly restricted to authorized personnel through role-

based permissions. This distinction streamlines content management while reinforcing data privacy and security protocols.

Structurally, the entire database is designed following established normalization principles to minimize redundant data storage, reduce update-related inconsistencies, and enhance overall data integrity [23]. Instead of repeating identical details — such as student names, subject titles, or class descriptions — across multiple tables, information is decomposed into logical, distinct entities connected through formally defined primary keys and foreign key relationships. For example, a student's full personal and contact information is recorded exactly once in the dedicated student profile table; all subsequent references — in attendance, grades, or enrollment — link back to that single source using the unique student ID. This architecture ensures that any update to a student's information is made in only one place, immediately reflected across every authorized module, and eliminates the conflicts, mismatches, and inefficiencies inherent in duplicated records.

These formal inter-table relationships are what transform independent system modules into a truly unified platform. When a new student is enrolled and their profile created, that record becomes instantly accessible wherever needed — when taking attendance, assigning classes, recording grades, or generating reports — removing the need for repetitive re-encoding, minimizing human error, and guaranteeing that every stakeholder views the same accurate, up-to-date information. This shared, consistent data structure directly fulfills the system's fundamental objective: replacing scattered, disjointed files with a single reliable source of institutional truth, thereby strengthening information integrity, improving operational efficiency, and laying a solid foundation for TCCIS's ongoing digital transition.

Client–Server Communication

The three distinct layers of the architecture interact seamlessly through a structured client–server communication workflow, ensuring orderly, secure, and reliable exchange of data and instructions across the entire platform. When an authorized user accesses the system via any standard web browser, the browser functions as the client interface, initiating and transmitting all user-generated requests to the designated server. Upon receipt, the server routes each request directly to the Application Layer for validation, processing, and decision-making.

If the requested action requires retrieving, adding, updating, or verifying institutional information, the Application Layer formulates and dispatches a precisely structured query to the Data Layer, which manages the MySQL database and returns the relevant dataset or confirmation status back to the middle tier. The Application Layer then processes, filters, and formats the result according to the user's role and the specific task at hand, before forwarding the finalized response to the Presentation Layer — where it is rendered into a readable, user-friendly view and displayed within the browser.

This systematic request–response cycle can be illustrated through a concrete example: when a class teacher records daily attendance, the necessary details — including student names, attendance status, date, and section — are entered directly into the web-based forms provided by the Presentation Layer. Once submitted, the request is transmitted securely to the PHP-powered Application Layer, where every data field undergoes comprehensive validation: confirming that all required entries are complete, identifying information corresponds to registered records, and formatting adheres to system standards. Only after successful validation does the application generate and execute a parameterized prepared statement to insert the new attendance record securely into the MySQL database within the Data Layer. Upon confirming successful storage, the system returns a clear, timely confirmation message to the teacher, instantly updates the displayed attendance summary, and makes the newly recorded data immediately available to other authorized functions — such as attendance reports, parent portals, and administrative oversight — without requiring duplicate entry or additional processing.

Beyond functional reliability, this clearly defined, layered communication structure delivers significant long-term advantages by allowing modifications or enhancements to be implemented within one layer with minimal or no

disruption to the others. For instance, improvements to the visual design, layout responsiveness, or accessibility features of the user interface can be accomplished entirely within the Presentation Layer through updates to HTML5, Tailwind CSS, or JavaScript — leaving both the underlying application logic and the database structure completely unchanged. Similarly, performance optimizations, indexing adjustments, or schema refinements can be carried out within the Data Layer to improve query speed or storage capacity, while the established data access interfaces remain consistent, ensuring that application operations and user-facing displays continue functioning without interruption. Likewise, business logic updates or security enhancements implemented within the Application Layer can be deployed independently, provided the standardized request and response formats are maintained. This deliberate separation of concerns directly supports ongoing system maintainability, simplifies troubleshooting, streamlines incremental feature development, and provides the architectural flexibility essential for accommodating TCCIS's evolving digital needs in the years ahead [16].

Deployment Architecture

The proposed TCCIS School Management System is engineered to support two complementary deployment configurations, enabling flexible implementation that adapts to the school's operational needs, accessibility requirements, and available infrastructure: a cloud-based deployment hosted via ProFreeHost and a local-area-network (LAN) deployment running on the XAMPP platform [22]. This dual-option design ensures the system can function effectively across different connectivity scenarios, serving both remote and on-campus stakeholders without compromising core functionality or architectural integrity.

Cloud-based deployment makes the system accessible from virtually any location through a standard internet connection and web browser. Hosted on ProFreeHost, the platform is published to the public web, allowing authorized users — including school administrators, faculty members, parents, and guardians — to securely access their respective dashboards, submit or review information, communicate, and generate reports regardless of physical proximity to the school premises. This configuration is particularly valuable for TCCIS because it accommodates distributed access needs: it enables parents to monitor their children's academic progress and attendance from home, allows teachers to encode grades or update records outside regular school hours, and gives administrators oversight capabilities even when off-site. It aligns directly with the system's objective of fostering transparency, timely information sharing, and inclusive parental engagement by removing geographic barriers to access.

In contrast, the local-area-network deployment offers an on-premise alternative, running the complete system within the school's private internal network. Powered by XAMPP — an integrated software stack that provides a local Apache web server, PHP processing environment, and MySQL database management — the system is hosted on a designated on-campus computer or local server, and accessed exclusively by devices connected to the same network. This setup is ideal for situations where internet connectivity is limited, unstable, or unavailable; it ensures full system functionality remains operational during development, testing, and daily on-site use without relying on external connectivity. It also serves as a controlled environment for validating new features, refining workflows, and conducting training sessions before any updates are deployed publicly.

This dual-deployment approach delivers exceptional operational flexibility because the same underlying three-layer architecture supports both environments without requiring extensive modification or separate codebases. During the development and iterative improvement phases, the XAMPP-powered local environment provides a safe, isolated workspace where developers and technical staff can implement, test, and refine system features — from interface adjustments to database optimizations — with minimal risk to live data or public accessibility. Once functions are verified and stabilized, the identical system can be transferred to ProFreeHost for cloud deployment, adapting only to the hosting service's specific server configuration requirements and resource capabilities [22].

The deployment method primarily influences the physical pathway through which users connect to the system, rather than altering how the system itself operates. In cloud deployment, user devices communicate with the hosted web server over the public internet; in LAN deployment, data travels strictly within the school's internal network infrastructure. Crucially, in both configurations, the logical separation of the Presentation, Application, and Data Layers remains fully intact and consistent — preserving the same security controls, processing workflows, data relationships, and user experience regardless of access method. This uniformity ensures that whether accessed remotely or on-site, the system upholds identical standards of reliability, data integrity, and functional performance, while offering the school the freedom to select, transition between, or operate both configurations according to its evolving priorities and circumstances.

3.4 Tools and Technologies

The development of the Timothy's Cradle Christian International School Management System (TCCIS SMS) is grounded in a deliberate, well-considered selection of modern, stable, and widely endorsed web technologies and development tools — each chosen specifically to ensure robust system responsiveness, comprehensive data security, long-term maintainability, consistent cross-device compatibility, and seamless adaptability across both local and cloud-based deployment environments. Every component in the technology stack has been evaluated against established industry standards, reliability records, and alignment with the institution's operational context, ensuring that the resulting platform is not only powerful and capable today but also sustainable, supportable, and scalable as TCCIS continues to grow and evolve.

The system is built upon a proven, standards-compliant web stack comprising HTML5, Tailwind CSS, JavaScript, AJAX, PHP 8.x, and MySQL, complemented by mature, trusted development and hosting tools that support every phase of the full software development lifecycle — from initial requirements gathering, system design, and iterative coding through functional testing, deployment, and ongoing post-launch maintenance and enhancement.

HTML5 provides the semantic structural foundation of all system interfaces, ensuring clean, accessible, and future-proof page architecture that works reliably across all modern browsers. Tailwind CSS delivers a streamlined, utility-first styling framework that enables consistent visual design, fully responsive adaptive layouts, and efficient interface updates while keeping code organized and lightweight.

JavaScript powers dynamic, interactive user experiences within the browser — including real-time form validation, instant dashboard updates, and smooth navigation — significantly reducing unnecessary page reloads and improving overall system speed and usability. AJAX further enhances performance by enabling asynchronous data exchange between the browser and server, allowing specific sections of the interface to refresh with new information without reloading the entire page, thereby creating a fluid, efficient, and professional user experience.

On the server side, PHP 8.x serves as the robust, high-performance processing engine behind the Application Layer, offering modern security features, improved execution efficiency, broad hosting compatibility, and extensive documentation and community support — making it ideally suited for the development and long-term maintenance of mission-critical institutional applications. Complementing this, MySQL provides a powerful, reliable, and widely adopted relational database management system that securely stores, organizes, and retrieves all the school's administrative and academic data, with strong mechanisms for data integrity, access control, and efficient query performance.

these technologies form a cohesive, well-supported ecosystem that is extensively documented, actively maintained, and familiar to a large global community of developers — factors that directly benefit TCCIS by lowering the barrier to future technical support, simplifying system updates, and ensuring that the platform remains technologically relevant and secure for years to come. The selection of this stack is not merely a technical preference but a strategic decision intended to balance power, reliability, accessibility, and sustainability, laying a solid technological foundation aligned with the school's digital transformation goals.

3.4.1 Frontend Technologies

The frontend constitutes the presentation layer, responsible for how users interact with the system, how information is displayed, and how the interface adapts to different screen sizes.

- HTML5 (HyperText Markup Language 5)

HTML5 serves as the structural foundation of all system pages, defining the semantic layout of forms, tables, menus, dashboards, and public-facing content. It provides standardized elements that ensure clarity, accessibility, and consistent rendering across all major web browsers. Using semantic HTML5 also improves search engine visibility for the public website and supports future accessibility enhancements. Every page—from the login portal to attendance records, grade forms, and enrollment pages—is built on clean, well-structured HTML5 markup.

- Tailwind CSS

Tailwind CSS is the utility-first styling framework used throughout the system to create a consistent, responsive, and visually cohesive interface. Unlike traditional CSS frameworks that rely on fixed component styles, Tailwind applies lightweight, reusable utility classes directly within the HTML, allowing precise control over design while maintaining a unified visual language across all modules. This approach ensures that the system automatically adjusts its layout for desktops, tablets, and smartphones without requiring separate page versions. It also simplifies design updates and maintenance—styling changes can be applied consistently across the entire platform rather than edited page-by-page—while keeping file sizes lightweight and page loads fast.

- JavaScript

JavaScript provides client-side interactivity and dynamic behavior, transforming static pages into functional, responsive interfaces. It powers form validation before data is submitted, real-time interface updates, interactive dashboards, dynamic filtering of records, and modal dialogs. By performing tasks directly in the user's browser, JavaScript reduces unnecessary server requests and improves perceived speed. It also enables visual feedback—such as confirming attendance submissions, displaying error prompts, and updating notification counters—so users can interact with the system smoothly without waiting for full page reloads.

- AJAX (Asynchronous JavaScript and XML)

AJAX enables the system to exchange data with the backend server in the background, updating specific sections of a page without reloading the entire interface. This technology is used extensively throughout TCCIS SMS: for example, searching student records, submitting attendance, loading announcement lists, refreshing inventory status, and displaying real-time messages. By fetching and sending data asynchronously, AJAX significantly enhances system responsiveness and creates a smoother, more application-like experience. It also reduces bandwidth usage and server load, as only necessary data is transferred rather than complete page files with every interaction.

3.4.2 Backend and Database Technologies

The backend manages business logic, security, data processing, and communication between the user interface and stored information.

- PHP 8.x

PHP 8.x is the server-side scripting language that drives all core system functionality—processing form submissions, managing user authentication, enforcing role-based access control, computing attendance percentages and grade averages, generating reports, and assembling dynamic page content. Chosen for its maturity, broad hosting compatibility, strong security improvements in version 8.x, and native integration with MySQL, PHP powers the application layer of the system. It validates inputs, executes business rules, prepares data for the database, and

constructs secure responses to frontend requests. All administrative, academic, and communication workflows—from enrollment processing to message delivery—are managed through PHP scripts.

- MySQL

MySQL is the relational database management system used to store, organize, and protect all persistent system data. It holds student profiles, user accounts, attendance logs, grade records, class assignments, inventory ledgers, communication messages, audit trails, and public website content. MySQL was selected for its reliability, efficiency, ability to handle complex table relationships, and seamless compatibility with PHP. Data is organized into structured tables linked by carefully defined relationships, ensuring consistency, minimizing duplication, and enabling fast, accurate information retrieval. The database also enforces data integrity and security at the storage level, supporting the system's reliability and accountability requirements.

3.4.3 Development Tools

These tools support the coding, testing, and version management processes throughout system construction.

- Visual Studio Code

Visual Studio Code (VS Code) is the primary integrated development environment used for writing, editing, and organizing all project code—HTML, CSS, JavaScript, PHP, and related configuration files. It provides syntax highlighting, code completion, error detection, and extensions that streamline development across multiple file types. Its customizable interface and wide range of community extensions helped maintain coding standards, detect issues early, and keep the project structure organized as the system grew from individual modules into a unified platform.

- Git

Git is the distributed version control system employed to track every change made to the source code throughout development. It maintains a complete historical record of additions, modifications, and deletions, allowing developers to review progress, revert to previous versions if problems arise, and safely implement new features without risking existing functionality. Git also supports structured development workflows, enabling incremental improvements and systematic bug tracking, and serves as an additional layer of backup and accountability for the project codebase.

3.4.4 Deployment and Hosting Environment

These platforms enable local testing and live internet accessibility.

- XAMPP

XAMPP is the local development and testing environment used during system building. It bundles the Apache web server, PHP runtime, and MySQL database into a single, self-contained package that runs directly on the developer's computer without requiring an internet connection. This allows full, offline testing of every feature—page display, data submission, user login, report generation—before the system is deployed online. XAMPP ensures that the local environment matches the live hosting environment as closely as possible, reducing compatibility issues when the system is transferred to the public server.

- Pro Free Host

Pro Free Host serves as the live cloud hosting provider, making TCCIS SMS accessible over the internet to authorized users and visitors. It was selected because it fully supports both PHP and MySQL—the core technologies of the system—while providing sufficient storage and bandwidth for the institution's needs. With the system deployed here,

the public website and internal portal are available from any location with an internet connection and a standard web browser, enabling real-world use by administrators, teachers, parents, and prospective families.

LAYER	TECHNOLOGY
Frontend	HTML 5, tailwindCSS, JavaScript AJAX
Backend	PHP 8.x
Database	MySQL

Development Environment	Visual Studio Code, Git
Local Hosting	XAMPP (Apache, PHP/MySQL)
Cloud Hosting	Pro Free Host (PHP/MySQL compatible)

3.5 Data Gathering Techniques

The study employs document analysis and semi-structured interviews as the primary, complementary data-gathering techniques selected to ensure a comprehensive, balanced, and well-grounded understanding of the school's operational landscape and system requirements. These carefully chosen methods are applied systematically to obtain detailed, accurate, and context-rich information about the current administrative and academic workflows, information practices, record-keeping procedures, and communication patterns at Timothy's Cradle Christian International School Inc., while also clearly identifying the functional, performance, security, and usability requirements that guided the design and development of the proposed TCCIS School Management System. The deliberate combination of verifiable documentary evidence and firsthand stakeholder perspectives enables the research team to build a complete picture: examining both the formal, documented standards and existing records that govern school operations, as well as the lived experiences, practical challenges, unmet needs, and valuable insights of the individuals who directly perform, rely upon, and manage these processes every day [16][19].

Document analysis serves as the foundational, evidence-based component of the data-gathering process, providing objective, detailed insight into the school's established information management systems, reporting structures, and operational standards. Researchers systematically reviewed relevant institutional materials including enrollment forms, student registration documents, attendance registers, grade sheets, inventory logs, procurement records, official announcements, policy handbooks, and existing administrative reports. This review revealed the specific types of data currently collected, the format and structure of maintained records, the flow of information across departments, the frequency and format of required reports, and the documented procedures that guide daily tasks. It also exposed tangible limitations inherent in the existing paper-based and fragmented approach — such as inconsistent recording formats, redundant data entry across multiple forms, delays in information consolidation, physical storage constraints, and difficulties in retrieving historical data — establishing a factual baseline of the current state against which the proposed system's capabilities and improvements could be measured.

Complementing this documentary review, semi-structured interviews were conducted with key stakeholder groups — including school administrators, teaching staff, administrative personnel, and parent representatives — to capture the human dimension of school operations that written records alone may not fully convey. Unlike rigid questionnaires that limit responses to predefined options, the semi-structured format used a core set of guiding questions while allowing flexibility to explore participants' experiences, elaborate on specific challenges, and offer suggestions based on their firsthand involvement. Discussions focused on daily workflows, time-consuming tasks, recurring pain points, information reliability concerns, communication gaps, security and confidentiality considerations, and desired features or improvements. This approach ensured that the researchers did not only identify what the school currently does, but also why certain processes are followed, where bottlenecks occur, what information is most critical to each role, and how a new system could better support the school's educational and administrative goals without disrupting established, valued practices.

Together, these two techniques create a robust, cross-validated research framework where findings from one method clarify, confirm, or enrich results from the other. Document analysis provides the structural, procedural, and data-related facts — revealing exactly what information is handled, how it is recorded, and where formal procedures exist — while semi-structured interviews add the practical, experiential context — explaining how procedures are actually carried out in daily practice, where gaps between policy and reality emerge, and what stakeholders identify as priorities for improvement.

This dual perspective ensures that the requirements gathered are not merely technically feasible, but also operationally relevant, user-centered, and aligned with the school's unique culture, capacity, and long-term vision. By grounding the entire development process in both verifiable institutional records and genuine stakeholder input, the study establishes a strong, trustworthy foundation for the TCCIS School Management System — ensuring that the final platform directly addresses real needs, solves actual problems, and delivers meaningful value to Timothy's Cradle Christian International School Inc. [16][19]

Document Analysis

Document analysis involves the systematic review of existing school documents and records to determine the information currently collected, the format in which it is recorded, and the procedures followed in managing school activities [16]. The researchers examine relevant documents such as enrollment forms, student records, attendance logs, inventory ledgers, grade-related records, report templates, communication records, and other official school documents that are available and appropriate for the study.

The review of enrollment forms and student records helps the researchers identify the information required for student registration and profile management. These documents may contain fields such as student identification information, personal details, parent or guardian information, contact information, grade level, and other relevant enrollment data. Examining these fields provides a basis for determining which information should be included in the student records module of the proposed system.

Attendance logs are reviewed to determine how attendance information is currently recorded and organized. The researchers examine the information commonly included in attendance records, such as student name or identifier, date, class, teacher, and attendance status. The review also helps identify the procedures used to record absences, tardiness, or other attendance conditions. These findings can be used to design an attendance module that reflects the school's existing requirements while reducing repetitive manual recording.

Inventory ledgers and related asset records are examined to identify the information required for managing school supplies and property. The researchers may review fields such as item name, category, quantity, condition, acquisition information, assigned personnel, location, issuance, and return records. This information can guide the design of the inventory module and help determine the appropriate data fields and transaction processes required by the school.

Report templates and official school records are also analyzed to determine the information required for generating administrative and academic reports. The researchers examine the structure, required fields, organization, and frequency of reports to identify which system data should be included in automated reporting functions. This can reduce the possibility of developing reports that do not correspond to the school's actual documentation requirements.

Document analysis also helps the researchers identify similarities and differences among existing records. Some information may be recorded repeatedly across different forms or documents, which can indicate opportunities for centralization within the proposed system. Identifying duplicate information can help researchers design relationships between database tables and reduce unnecessary data entry. The review can also reveal inconsistencies in terminology, formats, or recording procedures that should be addressed during system design.

The information gathered through document analysis serves as a reference during database design, interface development, and workflow modeling. Rather than creating forms and system fields based solely on assumptions, the researchers can use existing school documentation as evidence of the information required in actual operations. This helps ensure that the proposed system reflects relevant school processes and documentation requirements [16].

Semi-structured Interviews

Semi-structured interviews are conducted to gather detailed information from individuals who are directly involved in or affected by the school's administrative and academic processes [19]. Unlike highly structured interviews that rely exclusively on predetermined questions, semistructured interviews provide a consistent set of guiding questions while allowing researchers to ask follow-up questions when participants provide information that requires further clarification.

The participants may include school administrators, teachers, administrative staff, and parents, depending on their availability and relevance to the study. Each group can provide different perspectives regarding the existing procedures and the expected functions of the proposed system. Administrators can provide information about record management, reporting, monitoring, and administrative requirements. Teachers can describe their experiences with attendance recording, grade management, student information, and communication. Staff members can provide information about inventory and other operational activities, while parents can provide perspectives regarding access to student information, announcements, and communication with teachers.

During the interviews, participants are asked about their current procedures, difficulties encountered when performing their responsibilities, information they frequently need to access, and features they consider useful in a computerized system. Questions may also address problems related to manual record-keeping, delays in retrieving information, repetitive data entry, report preparation, inventory monitoring, and communication. Follow-up questions are used to clarify responses and obtain specific examples of existing workflows.

The interviews also help identify usability requirements for the proposed system. Participants can describe their expectations regarding navigation, accessibility, ease of use, forms, notifications, reports, and other interface features. Understanding these preferences is important because a technically functional system may still be difficult to adopt if users find its interface confusing or if its workflow does not correspond to their normal responsibilities [19].

Semi-structured interviews are also used to prioritize system features. Not every possible feature can necessarily be implemented within the scope of the project. Participants can therefore identify which functions are most important to their daily activities. The researchers can use these responses, together with findings from document analysis, to determine which features should receive greater attention during development.

Interview responses can also be compared with the findings from document analysis. For example, documents may indicate that certain information is recorded, while interviews may reveal difficulties in maintaining or retrieving that information. Comparing these sources provides a more complete understanding of the existing process and helps the researchers identify differences between documented procedures and actual practices.

The information collected from the interviews is documented, organized, and analyzed according to recurring themes and system requirements. Common concerns, requested features, workflow issues, and usability expectations are identified and considered during the requirements analysis and system design stages. Relevant findings are then translated into functional and non-functional requirements for the proposed TCCIS School Management System [19].

The use of both document analysis and semi-structured interviews provides complementary sources of information for the study. Document analysis establishes what information and formats are currently used by the school, while interviews provide insight into how these processes are experienced and performed by stakeholders. Combining these techniques helps the researchers develop a clearer understanding of existing workflows and provides a stronger basis for designing a system that responds to the actual administrative and academic requirements of Timothy's Cradle Christian International School Inc.

3.6 Testing Procedures

Testing is a fundamental, continuous, and indispensable component throughout the entire development lifecycle of the proposed Timothy's Cradle Christian International School Management System (TCCIS SMS), serving as the critical verification mechanism that confirms whether the built system operates precisely in accordance with the formally defined functional, performance, security, and usability requirements established during the design phase, and whether it maintains consistent, dependable, and accurate performance under the realistic daily operating conditions it will encounter once deployed. Far from being treated as a final, one-time checkpoint conducted only after coding is fully completed, the testing process is systematically integrated alongside development activities — beginning from the earliest implementation stages and proceeding iteratively through every module build, integration milestone, and refinement cycle — so that issues are detected, diagnosed, and resolved promptly, cost-effectively, and before they can compromise system stability or data integrity in later phases.

A structured, multi-level testing approach is applied to comprehensively identify and address a broad spectrum of potential concerns: functional errors that could prevent features from working as intended, integration problems that may arise when separate modules connect and exchange data, usability difficulties that could hinder user understanding and adoption, compatibility issues across different devices and browsers, and security weaknesses that might expose sensitive personal, academic, or institutional information to risk. To achieve thorough, layered evaluation at progressively broader scopes, the study employs three distinct, complementary testing methodologies — unit testing, system testing, and User Acceptance Testing (UAT) — each designed to examine the system from a unique perspective, at different stages of readiness, and with specific objectives that together provide complete assurance of the platform's quality, reliability, and fitness for real-world deployment [18][20].

Unit testing constitutes the first foundational level of evaluation, carried out during the actual construction of individual system components and modules. At this stage, the developers isolate and test discrete, self-contained units of program logic — such as attendance calculation routines, student record validation, inventory threshold checks, or report formatting functions — to verify that each one performs its designated task accurately, returns correct outputs for all anticipated inputs, and handles edge cases or unexpected data appropriately. Because testing occurs while each component is still being built, defects in logic, calculation errors, or inconsistent data handling can be identified immediately, allowing corrections to be made without disrupting other parts of the system; this early validation also simplifies subsequent development by ensuring that the fundamental building blocks of the platform are reliable before they are interconnected.

Once individual modules have been verified, system testing is conducted as the next comprehensive phase, evaluating the fully integrated, complete TCCIS SMS as a unified whole rather than as separate pieces. This stage validates that all modules — including Attendance Tracking, Student Records Management, Inventory Control, Communication Portal, and Reporting Engine — work together cohesively, exchanging data correctly across the shared centralized database, and that actions performed in one area properly propagate to all related functions without creating conflicts, duplication, or data loss. System testing examines end-to-end workflows, overall system performance, response times, cross-browser and cross-device compatibility, and the effectiveness of security and access control mechanisms; it confirms that the platform meets all defined technical specifications and operates reliably under simulated real-world load conditions, ensuring that the integrated environment functions as a consistent, dependable, and secure solution.

User Acceptance Testing (UAT) serves as the conclusive, real-world validation phase, conducted directly with the intended stakeholders — school administrators, teachers, and parents — to confirm that the system is practical, intuitive, and fully aligned with the institution's actual operational needs and daily workflows. Unlike earlier technical evaluations, UAT assesses the system from the end-user perspective: participants perform realistic, everyday tasks using the platform, providing direct feedback on ease of navigation, clarity of instructions, task efficiency, and overall

suitability for their roles. This testing verifies that the system not only works technically but is also understandable, accessible, and capable of being adopted confidently by personnel with varying levels of technical familiarity; it also validates that every requirement identified at the beginning of the study has been meaningfully addressed, ensuring that the final delivered system is genuinely useful, purpose-fit, and ready for sustained operational use at Timothy's Cradle Christian International School [18][20].

Unit Testing

Unit testing involves testing individual functions, components, and modules independently before they are integrated into the complete system [18]. The purpose of unit testing is to determine whether each small component performs its intended function correctly based on its defined requirements. Testing individual components at an early stage makes it easier to identify and correct errors because problems can be isolated to a specific function rather than being investigated within the entire system.

For the student records module, unit testing may include checking functions such as adding a student, updating student information, searching for a student, viewing student details, and validating required fields. Each function is tested using appropriate inputs to determine whether the expected result is produced. Invalid or incomplete inputs can also be tested to determine whether the system responds appropriately and prevents incorrect information from being stored.

Attendance-related functions are also tested individually. These may include recording attendance, changing attendance status when authorized, retrieving attendance records, and generating attendance-related information. The researchers verify whether the correct student, date, class, and attendance status are associated with each record. Boundary and invalid-input cases can also be considered to determine whether the system handles unexpected entries appropriately.

For the inventory module, unit testing can be performed on functions such as adding inventory items, updating quantities, recording item issuance, recording returns, searching for items, and generating inventory information. The researchers verify that inventory quantities are correctly updated when transactions occur. For example, an item issued from available stock should result in the appropriate change to its recorded quantity.

Other individual functions, such as user authentication, role assignment, grade recording, communication, notifications, and report generation, are also tested independently. Errors discovered during unit testing are corrected before the affected functions are integrated with other modules. This helps reduce the possibility of transferring unresolved errors into later stages of development [18].

System Testing

System testing evaluates the complete integrated system after the individual modules have been developed and combined. Unlike unit testing, which focuses on isolated functions, system testing examines whether the entire application operates correctly as a unified system [20]. The researchers perform end-to-end testing by following actual or representative workflows from the beginning of a process to its expected completion.

For example, a student record can be created through the student management module and subsequently used when recording attendance or grades. The researchers verify whether the information is correctly transferred between the relevant modules without requiring unnecessary duplicate encoding. Similarly, inventory information can be entered, updated through transactions, and subsequently reflected in inventory reports. These tests help determine whether the modules communicate correctly through the centralized database.

System testing also examines the major workflows performed by different user roles. Administrator workflows may include managing users, student records, inventory, reports, and system information. Teacher workflows may include viewing authorized student records, recording attendance, managing grades, and communicating with parents. Parent workflows may include viewing authorized student information, attendance, academic information, announcements, and communication features. Testing these workflows helps determine whether each user role can perform its intended activities successfully.

Cross-device and browser compatibility testing is conducted to determine whether the responsive web interface functions properly across different devices and screen sizes. The system can be tested using desktop computers, laptops, tablets, and compatible mobile devices. Different supported web browsers can also be used to identify display or functionality problems. Particular attention is given to navigation, forms, tables, buttons, dashboards, and other interface components that may behave differently depending on the device or browser.

System testing also includes security assessment. Authentication testing verifies whether only users with valid credentials can access protected areas of the system. Authorization testing examines whether users can access only the features and information permitted by their assigned roles. For example, a parent account should not be able to access administrative functions or another student's restricted information. These tests help determine whether the implemented role-based access controls operate as intended [20].

Data protection is also considered during system testing. The researchers verify whether sensitive information is transmitted and stored through appropriate mechanisms and whether unauthorized requests are rejected. Database interactions using prepared statements are also reviewed as part of the security assessment to reduce risks associated with unsafe database queries. The system is tested for common access-control and input-validation problems that could expose or improperly modify school information.

Error handling is another component of system testing. The researchers intentionally provide invalid, incomplete, or unexpected inputs to determine whether the system displays appropriate messages and prevents improper processing. Examples include submitting forms with required fields left blank, entering invalid values, attempting to access unauthorized pages, or performing actions with incomplete information. Proper error handling helps prevent incorrect records and provides users with understandable feedback when an operation cannot be completed.

User Acceptance Testing (UAT)

User Acceptance Testing is conducted to determine whether the developed system meets the practical needs and expectations of its intended users [20]. Unlike technical testing performed primarily by developers or researchers, UAT allows representative users to interact directly with the completed system and evaluate its functionality and usability based on realistic tasks.

Representative users may include administrators, teachers, and parents because these groups have different responsibilities and information requirements within the system. Administrators can evaluate administrative functions such as student management, inventory, user management, and reporting. Teachers can evaluate functions related to attendance, grades, student information, and communication. Parents can evaluate access to authorized student information, attendance, announcements, and communication features.

During UAT, participants are provided with representative tasks that reflect common activities they would perform when using the system. They may be asked to log in, navigate through the system, enter or view appropriate information, perform transactions, generate reports, or communicate through the available features. Their ability to complete these tasks provides information about the system's usability and suitability for actual school operations.

A structured feedback instrument based on the quality characteristics of ISO/IEC 25010 is used to collect user evaluations [20]. The evaluation may consider characteristics such as functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability, as applicable to the developed system. Users can provide ratings and comments based on their experience while performing the assigned tasks.

The feedback obtained during UAT is reviewed to identify features that require further improvement. Issues may include confusing navigation, unclear labels, missing functions, difficulties in completing tasks, slow responses, or other concerns identified by users. The researchers use these findings to determine whether modifications are necessary before the system is considered ready for its intended use.

UAT also provides an opportunity to determine whether the system reflects the workflows identified during the requirements-gathering stage. Users can compare the developed functions with their actual school procedures and identify differences that may not have been discovered during technical testing. This is particularly important for the proposed TCCIS School Management System because the system is intended to support existing administrative and academic activities rather than impose unnecessary changes on established workflows.

The results from unit testing, system testing, and UAT are documented throughout the testing process. Errors discovered during testing are recorded, analyzed, corrected, and retested to verify that the implemented solutions work as intended. Where user feedback identifies usability or workflow concerns, appropriate revisions are incorporated into the system. Repeated testing after corrections helps ensure that changes made to one function do not introduce new problems in other parts of the application [18][20].

The combination of these testing procedures provides multiple levels of evaluation for the proposed system. Unit testing verifies individual components, system testing evaluates the integrated application and its technical characteristics, and UAT determines whether representative users can effectively use the system for their intended tasks. Together, these procedures provide a structured basis for identifying defects, improving usability, validating security controls, and determining whether the TCCIS School Management System satisfies its defined functional and quality requirements.

3.7 Evaluation Criteria

The evaluation framework for the TCCIS School Management System (TCCIS SMS) is systematically designed to provide a rigorous, balanced, and actionable assessment of whether the developed system fully meets the technical, operational, and user-centered requirements established for Timothy's Cradle Christian International School Inc. Rather than relying on general impressions or isolated observations, the assessment is structured around five core dimensions: Performance, Reliability, Usability, Functionality, and Accessibility.

These criteria were selected to capture both the system's technical quality and its practical value to stakeholders — ensuring that the platform not only operates correctly under ideal conditions but also remains responsive, stable, intuitive, and inclusive in real daily use [20].

Each dimension is defined by specific, observable, and measurable indicators, enabling consistent, objective evaluation across different users, tasks, and usage scenarios. The assessment covers both normal single-user workflows and realistic concurrent-access situations — such as multiple teachers recording attendance simultaneously, administrators updating enrollment records, and parents reviewing student progress at the same time — where system responsiveness, stability, and data consistency are most critical. This dual focus on individual experience and system-wide behavior ensures that the evaluation reflects the actual demands the platform will face upon deployment.

Performance

Performance evaluates the system's operational efficiency — how quickly it processes requests, retrieves information, and delivers results — under both typical and peak workload conditions. It addresses whether users can complete routine tasks without unnecessary delays and whether the system maintains acceptable responsiveness as stored records accumulate over time.

Key indicators include:

- Page load and interface response times — measuring the interval between a user action (clicking a link, submitting a form, opening a record) and the system's display of complete, ready-to-use content;
- Data retrieval speed — assessing how quickly the system locates and displays records, particularly for large datasets such as full student lists, term-long attendance histories, or multi-year inventory logs;
- Throughput under load — verifying system behavior when multiple users perform tasks simultaneously, ensuring that increased activity does not cause disproportionate slowdowns or timeouts;
- Scalability — observing how response times change as the volume of student records, inventory entries, and historical data grows, confirming that the platform remains efficient as the school expands.

Performance is assessed quantitatively using standardized timing tools and simulated load testing, ensuring that delays are kept within thresholds that support smooth, uninterrupted workflow and do not discourage consistent use of the system.

Reliability

Reliability examines the system's stability, consistency, and resilience over extended use — determining whether it operates dependably without unexpected interruption, data corruption, or inconsistent behavior, and whether it recovers gracefully when disruptions do occur.

This dimension is especially critical because the system manages permanent, essential institutional records that must remain accurate and available throughout the academic year and beyond.

Key indicators include:

- Operational stability — tracking system uptime and the frequency of unexpected errors, crashes, or service interruptions during continuous daily use;
- Data integrity and consistency — verifying that information entered into the system is stored completely, accurately, and uniformly across all relevant modules; that updates are reflected everywhere they should appear; and that no records are duplicated, truncated, or lost during processing;
- Transaction dependability — confirming that every action — marking attendance, encoding grades, issuing inventory, sending messages — is recorded fully and permanently, with a clear audit trail; that partial or failed transactions do not leave incomplete or conflicting records;
- Recovery and fault tolerance — assessing how the system responds to temporary connectivity interruptions, session timeouts, or planned maintenance; whether it resumes correct operation without manual reconfiguration; and whether data entered before an interruption is preserved or clearly flagged for re-entry.

Reliability is evaluated through continuous monitoring during the testing period, comparison of system outputs against manually verified reference data, and structured recovery testing to simulate realistic disruptions.

Usability

Usability assesses how intuitively, clearly, and effectively the system supports the actual workflows of its intended users — focusing on ease of learning, interface clarity, and alignment with the school's established practices. This dimension recognizes that even technically robust systems provide little value if they are confusing, cumbersome, or difficult for non-specialist users to adopt.

Key indicators include:

- Ease of learning — measuring how quickly first-time users can complete common tasks — logging in, viewing records, submitting standard forms — without step-by-step assistance, and whether system navigation follows consistent, predictable patterns;
- Interface clarity and organization — evaluating whether labels, menus, instructions, and feedback messages are clear, concise, and free of technical jargon; whether information is presented in a logical sequence; and whether priority actions are easily distinguishable from secondary options;
- Task efficiency — assessing the number of steps, time required, and perceived effort needed to complete routine activities such as marking attendance, encoding grades, or checking announcements;
- User satisfaction and confidence — gathering feedback on overall experience, perceived helpfulness, and willingness to use the system regularly; identifying features that users find confusing, difficult, or unnecessarily complex;
- Error prevention and recovery — confirming that the system guides users toward correct input, clearly explains problems when they occur, and provides straightforward ways to correct mistakes without disrupting their work.

Usability is evaluated qualitatively through structured User Acceptance Testing (UAT), direct observation of representative users from each stakeholder group — administrators, teachers, parents — and standardized feedback surveys that capture both perceived strengths and specific areas for improvement.

Functionality

Functionality verifies that every specified feature operates completely, accurately, and in accordance with the requirements defined during system design — confirming that the platform does what it was intended to do, reliably and without omission.

This dimension ensures that all core modules — student enrollment and profiling, attendance monitoring, grade management, inventory tracking, communication, reporting, and role-based access control — meet their stated objectives and integrate correctly with one another.

Key indicators include:

- Completeness of features — confirming that every required function is implemented and accessible: enrollment processing, profile management, attendance logging, grade encoding, inventory transaction recording, messaging, announcement publishing, and report generation in standard export formats;
- Accuracy of outputs — verifying that computed values — attendance percentages, grade averages, stock balances — are mathematically correct and consistent with independently verified results; that generated reports contain complete, properly formatted information; and that data shared between modules — such as student names, class assignments, or academic periods — remains consistent;
- Appropriateness of access control — confirming that role-based permissions function as designed: administrators maintain full oversight, teachers access only relevant classes and records, parents view information exclusively for their children, and unauthorized actions are blocked;
- Integration consistency — ensuring that updates made in one module appear correctly in related areas: for example, a change in enrollment status updates attendance rosters automatically, or a grade release notification becomes visible in the parent portal without separate action.

Functionality is assessed through comprehensive test cases covering every defined workflow, cross-module consistency checks, and permission-level verification to confirm that security restrictions operate as intended.

Accessibility

Accessibility ensures that the system functions reliably and consistently across the full range of devices, screen sizes, and web browsers used by the TCCIS community — enabling equal access to information and features regardless of the hardware, platform, or connection speed available to each user. This dimension is particularly important because families and staff may access the system from different locations and with varying equipment capabilities.
Key indicators include:

- Cross-device compatibility — verifying correct display and operation on desktop computers, laptops, tablets, and mobile phones; that text, buttons, and input fields are appropriately sized and positioned; and that essential functions remain fully usable on smaller screens;
- Cross-browser consistency — confirming reliable performance across widely used standard web browsers, ensuring that no user is excluded or disadvantaged by their software choice;
- Content clarity and navigation — ensuring that text is legible, that contrast levels support readability, that navigation controls are clearly labeled and consistently placed, and that the system follows standard web conventions so users already familiar with other online platforms can adapt quickly;
- Performance under varied connectivity — assessing how the system operates at different connection speeds; that essential features remain accessible even when loading is slower; and that status indicators clearly inform users about processing or data transfer in progress.

Accessibility is evaluated through direct testing on target devices and browsers, feedback from users accessing the system from different locations and with different equipment, and review of interface adaptability across screen sizes.

Evaluation Process and Significance

The overall assessment combines quantitative measurements — timing response speeds, verifying calculation accuracy, monitoring uptime — and qualitative feedback — user experience, perceived clarity, practical utility. Findings are organized by dimension to produce a clear, evidence-based profile of the system's performance: strengths are confirmed and documented, while specific areas requiring adjustment or optimization are identified with actionable recommendations for refinement.

Ultimately, this comprehensive, multi-criteria evaluation determines whether the TCCIS SMS is sufficiently robust, secure, efficient, and user-friendly for sustainable deployment and regular use within Timothy's Cradle Christian International School Inc.

It ensures that the final implementation not only meets technical specifications but also delivers tangible, reliable improvements to administrative efficiency, record integrity, resource accountability, and stakeholder communication — forming a dependable foundation for the school's ongoing digital transition [20].

METRIC	BENCHMARK
Response Time	2 Seconds for priority operations
Concurrent Users	50 simultaneous users with stable performance
System Uptime	~99.9% during operational hours

Performance and System Evaluation Details

The evaluation of the TCCIS School Management System (TCCIS SMS) is designed to provide rigorous, evidence-based verification that the platform meets its operational goals — combining technical performance benchmarks, functional validation, user experience assessment, and quality assurance to ensure readiness for sustainable deployment at Timothy's Cradle Christian International School Inc.

Performance Evaluation: Responsiveness and Efficiency

Performance evaluation focuses on the speed, smoothness, and consistency of system response during daily use — particularly for high-frequency, priority operations where delays directly impact workflow efficiency. The following core activities are prioritized for measurement:

- User authentication and login;
- Searching and retrieving student profiles, attendance records, and academic history;
- Recording and submitting daily or period-based attendance;
- Encoding, saving, and updating grades and term assessments;
- Issuing, returning, or adjusting inventory items;
- Composing, sending, and retrieving messages and announcements;
- Generating and exporting reports in standard formats.

A target response time of 2 seconds or less is established for each of these priority operations — from the moment a user submits a request to when the system displays complete, ready-to-use results.

This threshold is selected to ensure that routine tasks proceed smoothly, without disrupting concentration or creating workflow bottlenecks, even during busy periods such as attendance-taking or grade encoding [20].

Response-time testing is conducted under realistic operating conditions — using standard network connections, representative volumes of stored data, and actual user workflows — rather than in an artificially simplified environment.

Measured durations are compared systematically against the 2-second benchmark. Operations that consistently exceed this limit are flagged for detailed analysis, investigating potential causes including:

- Inefficient or unoptimized database queries;
- Excessive data processing being performed at the moment of request;
- Network latency or bandwidth constraints;
- Server resource allocation limits;
- Cumulative delays from growing historical data volumes.

Identified bottlenecks are addressed through targeted optimization — refining query logic, adding appropriate indexing, or adjusting data retrieval strategies — and then retested to confirm improvement.

Concurrent and Load Performance

Because administrators, teachers, and parents will access the system simultaneously — often performing different tasks at the same time — the system is evaluated specifically for stability under multi-user conditions.

A minimum benchmark of 50 concurrent users is established as the baseline for assessing real-world usability. During testing, simulated and live users perform typical actions in parallel: attendance recording, grade entry, report generation, and information retrieval. The system is monitored to confirm that:

- Response times remain within acceptable limits;
- No requests fail or time out unexpectedly;
- No data conflicts or duplicate entries occur;
- No user is denied access or experiences service interruption;
- Performance degrades gracefully rather than failing abruptly if load exceeds baseline levels.

Load and stress testing extend this assessment beyond the 50-user baseline to identify the system's practical limits — gradually increasing the number of simultaneous requests to pinpoint where performance declines or resources become insufficient. These findings establish clear expectations for current capacity and provide guidance for future scaling as enrollment grows or usage expands.

This proactive approach ensures that the system remains reliable during peak periods — such as the start of the school day, end-of-term reporting, or parent consultation hours — and helps distinguish application-related issues from external infrastructure limitations [20].

Availability and Operational Reliability

System availability is measured against a target of approximately 99.9% uptime during official operational hours — representing consistent, dependable access when administrators, teachers, and parents need to perform school-related activities.

This target acknowledges that brief scheduled maintenance is sometimes necessary but minimizes unplanned disruption to daily work.

Availability monitoring tracks:

- Total scheduled operational hours versus confirmed accessible hours;
- Frequency, duration, and timing of any unplanned downtime;
- Partial-service interruptions — where some functions respond slowly or fail while others remain accessible;
- Failed or abandoned requests that prevent users from completing tasks;
- Recurring error conditions that may indicate underlying instability.

Every service interruption or failure is documented with: timestamp, duration, affected functions, user roles impacted, and identified cause — whether application-related, server-side, network connectivity, or external infrastructure.

Root-cause analysis guides corrective action: improving error handling, adding redundancy, optimizing resource usage, or establishing backup procedures. The goal is not only to resolve individual issues but to strengthen the system's resilience so that temporary disruptions have minimal effect on users [20].

Multi-Dimensional Quality Evaluation

Beyond technical performance, the system undergoes comprehensive evaluation across six complementary dimensions to ensure overall quality and fitness for purpose:

Functional Suitability

Verifies that every implemented module performs its designated tasks correctly, completely, and consistently against approved requirements:

- Student enrollment and profile management — accurate data capture, editing, and retrieval;
- Attendance monitoring — correct status assignment, percentage calculation, and history retention;
- Grade management — proper computation, term aggregation, and role-restricted visibility;
- Inventory tracking — accurate stock updates, transaction logging, and low-stock alerts;
- Communication — timely delivery of announcements and messages with correct visibility;
- Reporting — accurate compilation, formatting, and export across all supported document types.

Cross-module consistency is also verified: changes in one area (e.g., enrollment status) reflect correctly in related areas (attendance rosters, parent access) without manual re-entry.

1.Billing Statements

The system generates billing statements for each enrolled student based on the school's tuition and fee schedule. Parents can view their child's statements, charges, and payment status through the Parent Portal. This gives families a clear record of what they owe without visiting the school office.

2.About Us Page

The public website includes an About Us page that presents the school's background, mission, vision, and other institutional information to visitors and prospective parents.

3.Public Website Content Management (Admin Only)

Only administrators can update the content of the public website, including the About Us Page, announcements, and other public pages. Content changes take effect without editing any code, and access control keeps unauthorizing users from modifying the site.

4. Activity Log

The system records important user actions, such as adding or editing students and managing accounts, along with the user responsible and the time of the action. Administrators can review this audit trail to support accountability, transparency, and security.

5.Automatic Student ID Generation

When a new student is added, the system automatically assigns a unique Student ID in the format YYYY-NNN (for example, 2026-001). The number resets each year. This removes manual entry, prevents duplicate or inconsistent IDs, and speeds up enrollment encoding.

6.Parent Login Using Registered Phone Number and Birthdate

Parents log in using their registered mobile number as the username and their registered mobile number as the username and their registered date of birth as password. Parents need no separate account credentials to remember, which makes the portal easier to use. The details are taken from the parent record encoded by the administrator.

7.Tuition and Fees Management (Admin Only)

Only administrators can create, edit, and update the tuition and fees schedules. Restricting this function to administrators keeps financial data accurate and consistent, and any changes carry through to the billing statements generated for students.

Usability

Assesses how intuitively and effectively all intended users — administrators, faculty, non-teaching staff, and parents — can navigate and operate the system, including those with limited digital experience:

- Clarity of labels, menus, and instructions;
- Ease of completing common tasks with minimal guidance;
- Logical workflow and page organization;
- Helpfulness of error messages and confirmation prompts;
- Overall satisfaction and confidence in using the system.

Feedback is gathered through structured User Acceptance Testing (UAT), direct observation, and survey responses to identify confusing interfaces, unnecessary steps, or features requiring refinement.

Reliability

Evaluates consistency over time and repeated use:

- Same input produces the same correct output reliably;
- System maintains stability through extended sessions and repeated operations;
- Data entered on one date remains intact and retrievable later;
- System recovers appropriately after session timeouts, connection interruptions, or restarts;
- No unexplained loss, duplication, or alteration of stored information.

Security

Focuses on protecting sensitive school and personal information through:

- Authentication — secure login procedures, credential protection, and session management;
- Role-based authorization — strict enforcement of permissions so users access only information appropriate to their function;
- Data protection — secure handling during transmission and storage, preventing unauthorized exposure or tampering;
- Transaction integrity — ensuring partial or interrupted actions do not corrupt records;
- Audit logging — tracking significant system events and data modifications for accountability.

Compatibility

Confirms consistent operation across the full range of platforms likely to be used by the TCCIS community:

- Supported desktop and mobile web browsers;
- Different screen sizes and device types — desktop computers, laptops, tablets, and smartphones;
- Standard resolutions and input methods.

No user should be disadvantaged or excluded because of their choice of equipment.

Accessibility

Ensures the system is usable by all intended stakeholders:

- Clear typography and readable contrast levels;
- Appropriately sized interactive elements for touch and mouse input;
- Consistent navigation patterns throughout;
- Meaningful feedback on actions and system status;
- Graceful performance under slower or intermittent connectivity.

Evaluation Process and Deployment Readiness

Findings from all testing phases — performance benchmarks, load assessment, functional verification, security checks, and user feedback — are compiled into a consolidated evaluation record. Every identified issue is:

- Clearly documented with description, severity, and affected components;
- Prioritized based on impact — critical blocking issues addressed immediately;
- Corrected through targeted adjustments;
- Retested to confirm resolution and absence of unintended side effects.

This iterative process continues until the system consistently meets or exceeds all established criteria — performance targets, functional requirements, security standards, and user experience expectations — ensuring that the final version deployed to TCCIS is robust, reliable, efficient, and genuinely supportive of the school's administrative and academic goals.

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CHAPTER IV

RESULTS AND DISCUSSION

4.1 System Overview

The completed Timothy's Cradle Christian International School Management System (TCCIS SMS) is a web-based integrated school management system designed to support and streamline the administrative, academic, and communication activities of Timothy's Cradle Christian International School Inc. The system brings together three interconnected components: a role-based internal portal, a public information website, and a centralized database. These components operate using a unified codebase and consistent data model, allowing information to be organized, maintained, and accessed according to the authorized functions of each user [24]. The integrated structure reduces the need for separate and disconnected systems while providing a centralized environment for managing school information.

The public information website serves as the school's online information platform for students, parents, prospective families, and other visitors. It provides important school-related information without requiring users to access the internal management portal. The website includes school announcements, academic and school calendars, enrollment information, downloadable forms, staff directory, frequently asked questions (FAQs), and other general school information [22].

By presenting these resources in one accessible location, the public website helps users obtain commonly requested information without relying entirely on direct inquiries or printed materials. The website can also support the school's communication and information-dissemination activities by providing updated announcements and relevant notices.

The role-based internal portal serves as the primary workspace for authorized school users. Access to the portal is controlled according to the user's assigned role, ensuring that users are provided with functions and information appropriate to their responsibilities. Administrators can manage student information, user accounts, attendance records, academic records, inventory, communications, and reports.

Teachers can access functions related to their assigned students and classes, including attendance recording, grade management, student information, and parent communication.

Parents can access relevant information about their children, including attendance, academic progress, announcements, and communication features. Role-based access helps prevent unauthorized users from viewing or modifying information outside their assigned responsibilities.

The system's centralized database serves as the primary storage layer for information generated and managed by the different system components. Student profiles, user accounts, class information, attendance records, grades, inventory records, announcements, messages, and other relevant information are stored in a structured relational database.

The centralized approach allows authorized modules to retrieve updated information from a common data source, reducing unnecessary duplication and inconsistencies between different parts of the system.

Relationships between database tables also allow information to be connected across modules, such as linking students with their classes, attendance records, academic records, and parent accounts.

The three components are designed to work together while maintaining appropriate access boundaries. Information managed through the internal portal is connected to the centralized database, while selected school information can be presented through the public website.

For example, administrators may update school announcements or calendar information through the management environment, after which the appropriate information can be displayed on the public-facing website.

This approach reduces repetitive data entry and helps ensure that published information remains consistent with the information maintained by authorized school personnel.

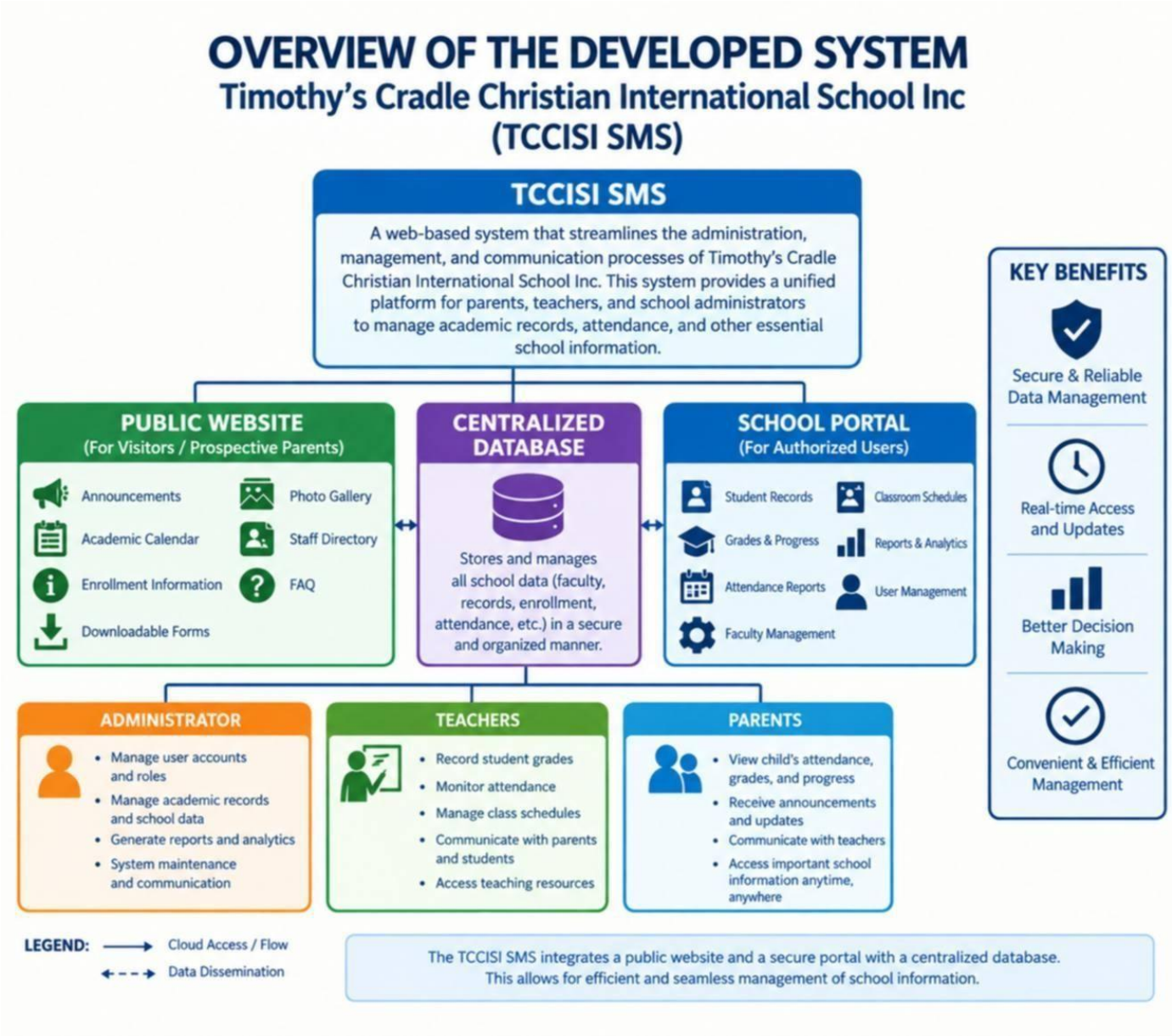
The system also supports the digitization of several processes that were previously handled through physical documents, logbooks, and spreadsheets. Instead of maintaining separate records for different activities, authorized users can perform tasks through the corresponding modules within the system.

Attendance can be recorded electronically, student information can be searched and updated through centralized records, inventory transactions can be monitored digitally, and reports can be generated from stored system data. These functions help reduce repetitive administrative work and improve the accessibility and organization of school information.

Overall system operation follows a controlled flow in which users access the appropriate interface, perform authorized activities, and interact with the centralized database through the application layer.

Authentication and role-based authorization are applied to protect restricted functions and information. The architecture also allows the different modules to communicate with shared data while maintaining separation between public information and internal school records.

Through this integrated design, TCCIS SMS provides a unified platform that supports school administration, teaching activities, parent communication, information dissemination, and record management within a single web-based environment [22][24].


OVERVIEW OF THE DEVELOPED SYSTEM
Timothy's Cradle Christian International School Inc
(TCCISI SMS)
TCCISI SMS
A web-based system that streamlines the administration, management, and communication processes of Timothy's Cradle Christian International School Inc. This system provides a unified platform for parents, teachers, and school administrators to manage academic records, attendance, and other essential school information.
KEY BENEFITS
Secure & Reliable Data Management
Real-time Access and Updates
Better Decision Making
Convenient & Efficient Management
PUBLIC WEBSITE
(For Visitors / Prospective Parents)
Announcements
Photo Gallery
Academic Calendar
Staff Directory
Enrollment Information
FAQ
Downloadable Forms
CENTRALIZED DATABASE
Stores and manages all school data (faculty, records, enrollment, attendance, etc.) in a secure and organized manner.
SCHOOL PORTAL
(For Authorized Users)
Student Records
Classroom Schedules
Grades & Progress
Reports & Analytics
Attendance Reports
User Management
Faculty Management
ADMINISTRATOR
Manage user accounts and roles
Manage academic records and school data
Generate reports and analytics
System maintenance and communication
TEACHERS
Record student grades
Monitor attendance
Manage class schedules
Communicate with parents and students
Access teaching resources
PARENTS
View child's attendance, grades, and progress
Receive announcements and updates
Communicate with teachers
Access important school information anytime, anywhere
LEGEND:
Cloud Access / Flow
Data Dissemination
The TCCISI SMS integrates a public website and a secure portal with a centralized database. This allows for efficient and seamless management of school information.

4.2 Key Functional Modules

The completed Timothy's Cradle Christian International School Management System (TCCIS SMS) comprises a cohesive set of interconnected functional modules, each engineered to address specific administrative, academic, operational, and communication requirements while sharing information through a single centralized database.

Rather than operating as isolated tools, the modules function as an integrated ecosystem: data entered in one component is immediately available to related processes across the system, eliminating redundant recording, reducing inconsistency between separate records, and enabling authorized users to manage diverse school workflows from a unified platform [22][24].

This design directly addresses the inefficiencies of manual filing systems, disjointed spreadsheets, and reliance on external messaging services that previously fragmented school operations.

Student Records Management

The Student Records Management module serves as the authoritative digital repository for all learner-related information, replacing physical filing cabinets and decentralized record-keeping with a structured, searchable digital environment [2].

It guides enrollment personnel through standardized digital forms that capture complete, consistent information—including personal details, date of birth, residential address, emergency contact information, parent or guardian affiliations, medical notes, and grade-level placement—ensuring that no critical field is omitted during data entry. Once submitted and verified, the system instantly generates a permanent student profile, making the record immediately accessible to authorized users across relevant modules without delay.

Beyond initial creation, the module supports ongoing maintenance: authorized administrative staff can update contact details, revise enrollment status, transfer class assignments, and archive records as needed, with all changes preserved within the student's history.

Built-in search and filtering allow records to be retrieved by student ID, full name, grade level, enrollment date, or guardian information in seconds—a substantial improvement over manual searching that previously required sorting through labeled folders and stacked documents.

Academic progress, attendance summaries, and historical performance data remain linked to each student's profile, creating a comprehensive longitudinal record that supports counseling, intervention planning, transfer documentation, and compliance reporting [2].

Centralized storage also ensures that all personnel—administrators, teachers, and authorized parents—reference the same verified information, reducing discrepancies and preventing conflicting versions of student data from accumulating across departments.

Attendance Tracking

The Attendance Tracking module digitizes and standardizes daily attendance monitoring, transforming what was once a manual, paper-based logbook process into an efficient, accountable digital workflow [9].

Teachers select their assigned class from the dashboard and mark each student's status—*Present, Absent, Late*, or *Excused*—with a single click; the system automatically timestamps the entry and associates it with the specific instructor, class section, and academic term.

This creates a permanent, verifiable record that clearly identifies who recorded the attendance, when it was recorded, and for which group of students—an essential component of institutional accountability.

The module automatically computes attendance percentages and participation rates based on cumulative recorded data, removing the need for teachers or administrators to calculate totals manually at the end of each month or quarter. Configurable alert thresholds can be set to flag patterns such as excessive unexcused absences or recurring tardiness, bringing at-risk students to the attention of administrators and guidance staff early enough to implement timely intervention.

When integrated with the Communication Portal, selected attendance notifications can be shared directly with parents or guardians the same day, strengthening home–school partnership and ensuring that families remain informed about their child's daily engagement.

Over time, accumulated attendance data provides a reliable basis for reviewing participation trends, identifying systemic scheduling concerns, and documenting compliance with institutional and regulatory attendance policies [9].

Grade Management

The Grade Management module provides a secure, structured framework for recording, computing, finalizing, and reporting student academic performance across quarters, semesters, or academic years [6].

Teachers input scores per subject, assessment type, and grading period, with the system linking each entry to the specific student, class, and instructor—ensuring clarity regarding who recorded the grade and under what criteria.

Built-in validation rules prevent common data-entry errors: out-of-range values, incomplete score sets, and mismatched student–subject assignments are flagged immediately, allowing corrections before data is saved and propagated elsewhere in the system.

Automated computation reduces repetitive manual arithmetic: weighted averages, term grades, and progress indicators are calculated instantly and consistently across all sections. Once grades are reviewed and formally finalized for a grading period, they are locked against further modification by non-administrative users, preserving data integrity and preventing unauthorized alteration.

Finalized grades populate official progress reports, report cards, and cumulative transcripts automatically, ensuring that documents distributed to parents and guardians reflect verified, consistent information. Because grade data is stored centrally, authorized personnel can review performance trends over multiple terms or years, identify learning gaps, and evaluate curriculum effectiveness—all while maintaining strict privacy controls so that only appropriately authorized users can access sensitive academic records [6].

Inventory Control

The Inventory Control module digitizes stewardship of school resources, including textbooks, classroom supplies, laboratory materials, furniture, and technical equipment [14]

. Every item is catalogued with its category, acquisition details, quantity, physical location, condition status, and predefined minimum stock level. All transactions—issuance, return, restock, transfer between departments, and disposal—are recorded with the responsible user identifier and timestamp, creating a transparent audit trail that clarifies when and by whom each change was made.

The module maintains real-time visibility of available stock across all categories, allowing administrative and facilities staff to verify resource availability before approving distribution or procurement requests. Low-stock alerts notify designated personnel when quantities fall below established thresholds, enabling timely reordering and preventing shortages that could disrupt instruction or daily operations.

Historical transaction analysis further supports evidence-based budgeting by identifying consumption patterns, seasonal demand fluctuations, and underutilized assets. By consolidating inventory data in one accessible system, the school can optimize resource allocation, reduce wasteful over-purchasing, and ensure that supplies and equipment are distributed equitably across departments [14].

Communication Portal

The Communication Portal centralizes all school-related correspondence within a secure, role-governed environment, replacing fragmented reliance on personal messaging applications, unsolicited SMS threads, and printed circulars with a unified, institution-managed platform [8]. Administrators publish general announcements, event schedules, policy updates, and emergency notifications—with visibility scoped by role so that information reaches only relevant audiences.

For example, a grade-level announcement appears only to parents and teachers associated with that level, reducing unnecessary clutter while ensuring that critical notices are not overlooked.
Private one-to-one messaging connects parents and teachers directly within the system, preserving a retrievable, school-owned record of conversations without requiring or exposing personal contact information.

Access is strictly controlled: parents communicate only with teachers assigned to their children; teachers interact only with families in their classes; and administrators retain appropriate oversight capability.

All messages and announcements are retained as part of the system's audit trail, providing a reliable history of school–stakeholder communication that can be referenced during meetings, dispute resolution, or continuity planning. By consolidating communication in this way, the portal strengthens transparency, improves response times, and fosters consistent, accountable engagement between home and school [8].

Reporting Engine

The Reporting Engine transforms data stored across all modules into actionable, standardized documentation—eliminating the labor-intensive manual compilation that previously required gathering information from disparate sources and recalculating totals in spreadsheets [5].

Authorized users define report parameters: date ranges, grade levels, specific classes, individual students, inventory categories, or academic terms. The system retrieves relevant data directly from the centralized database, applies consistent formatting, and generates complete, accurate reports in moments. Supported outputs include attendance

summaries, grade sheets, class master lists, progress reports, inventory status statements, and compliance-ready administrative documents.

Reports can be exported in widely compatible PDF format for secure distribution and printing, or in CSV format for further analysis in spreadsheet applications. Because reports draw from the same single source of truth, version conflicts are eliminated: regenerating a report at a later date reflects current data while preserving the ability to reproduce historical snapshots.

This capability supports timely decision-making—enabling administrators to review operational trends, teachers to identify students needing additional support, and resource managers to assess procurement needs—all with verified, consistent information. By making professional-quality reports available on demand, the Reporting Engine strengthens institutional accountability and simplifies documentation for internal review and external stakeholders alike [5].

Module Integration and Unified Operation

The true value of TCCIS SMS emerges from the seamless integration of these modules through the shared database. Student enrollment data automatically populates attendance rosters and grade assignment forms; attendance patterns can be referenced alongside academic performance during intervention planning; inventory usage informs budget projections; and announcements are delivered through the same portal where families view grades and attendance records.

Information entered once flows across all relevant workflows—enrollment to attendance, attendance to reports, grades to progress notifications—minimizing duplicate work, reducing transcription error, and ensuring that every authorized user operates from consistent, verified information.

This interconnected design transforms the system from a set of separate digital tools into a comprehensive platform that supports the full scope of the school's administrative, instructional, and community engagement mission.

4.3 Performance Outcomes

The performance evaluation of the completed TCCIS SMS was conducted to determine whether the developed system could perform its intended functions efficiently, reliably, and securely under the defined operating conditions.

The evaluation focused on system response time, data retrieval, concurrent access, reliability, security, usability, and alignment with the actual workflows of Timothy's Cradle Christian International School Inc. The results provide an indication of how effectively the implemented system addresses the operational difficulties associated with manual and fragmented record-management processes.

The system demonstrated satisfactory performance in terms of response time. During standard operating conditions, system response times averaged approximately 1.2 seconds, which is within the established 2-second performance benchmark. This result indicates that common operations such as logging in, opening student records, searching information, recording attendance, updating inventory records, and accessing reports could be completed without significant waiting time.

Faster response times are particularly important for routine school activities because teachers and administrative personnel may perform multiple transactions throughout the day. Maintaining responsive system behavior helps reduce interruptions and allows users to complete their assigned tasks more efficiently.

A significant improvement was also observed in data retrieval time. Under the previous manual process, locating specific student or administrative information could require approximately 5– 15 minutes, depending on the location and organization of physical records.

With the TCCIS SMS, relevant information can be searched and retrieved from the centralized database in under 3 seconds [2]. The digital search capability reduces the need to manually inspect filing cabinets, logbooks, or separate spreadsheet files. It also allows authorized users to retrieve information using available search fields and filters, making frequently accessed records easier to locate.

The improvement in retrieval time also contributes to more efficient record management and administrative workflow. Information stored in the centralized database can be accessed by authorized users without requiring physical movement of documents between offices or personnel.

This reduces delays when records are needed for attendance verification, grade preparation, report generation, enrollment-related activities, or other administrative tasks. Centralized storage also provides a more structured approach to maintaining records because related information can be connected through the system's database relationships.

The system was additionally evaluated based on its ability to support multiple users and concurrent operations. The performance test used the established benchmark of supporting up to 50 concurrent users while maintaining stable system operation.

The evaluation considered whether simultaneous requests could be processed without significant system slowdown, failed transactions, or unexpected interruption. Stable operation under concurrent access is important because administrators, teachers, and parents may access different parts of the system at overlapping times. The result of this testing provides evidence of the system's ability to accommodate expected usage within the institution's operational environment.

Security and access control were also considered important performance outcomes. The implemented role-based access control successfully restricted sensitive information and system functions according to the user's assigned role [20].

Administrators, teachers, and parents were provided with access appropriate to their responsibilities, preventing unauthorized users from accessing restricted records or management functions. This approach is particularly important for student information because the system stores academic, attendance, personal, and communication-related data.

The system also maintained activity and modification records for accountability. Relevant changes made within the system were logged, allowing authorized personnel to identify and review system activities when necessary [20].

Maintaining an audit trail provides a record of important modifications and supports monitoring of data changes. It can also assist administrators in investigating discrepancies and determining when and through which authorized account a particular modification was made.

The User Acceptance Testing (UAT) results further indicated that users responded positively to the system's usability and practical functionality [19]. Feedback from representative users indicated that the system was understandable, easy to navigate, and aligned with common daily school workflows.

The organization of the modules allowed users to locate relevant functions without relying on separate applications for each activity. The feedback also provided opportunities to identify interface elements and processes that could be refined during the development cycle.

The system's usability outcome is particularly relevant because the project was developed specifically around the operational requirements of the institution. Rather than adopting a generic platform with unnecessary functions, the TCCIS SMS focuses on processes directly related to student records, attendance, grades, inventory, communication,

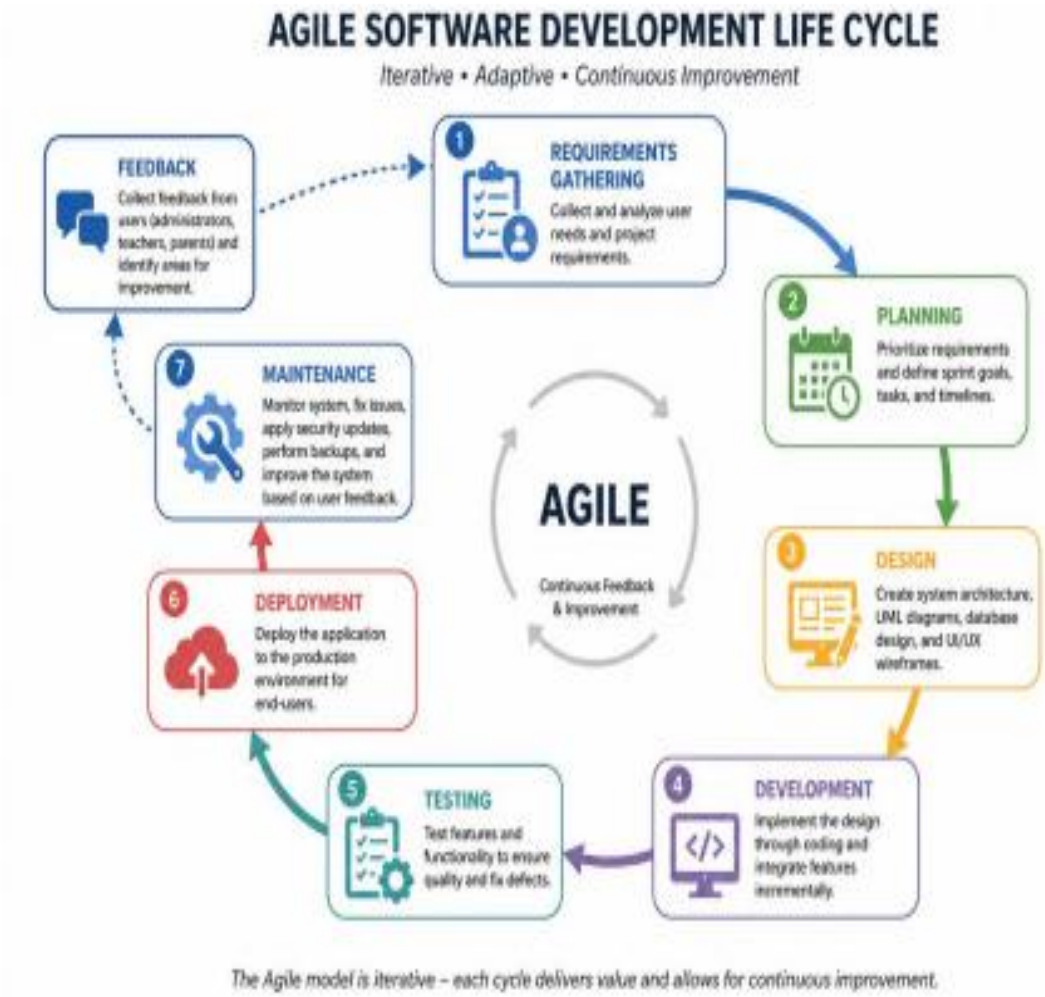
and reporting. This custom-built approach allowed system functions and workflows to be designed around the identified needs and constraints of the school [19].

The evaluation also demonstrated improvements in workflow integration. Student information stored in the Student Records Management module can support attendance and grade management, while attendance and academic information can be used in generating reports.

Inventory transactions can similarly be converted into updated inventory status reports, while the communication module provides a centralized channel for school-related announcements and messages. The integration of these functions reduces repeated encoding and allows related information to be accessed through interconnected modules.

The results therefore demonstrate measurable improvements in several areas of system operation. The 1.2-second average response time satisfied the established response-time requirement, while under-three-second data retrieval represented a substantial reduction from the previous 5–15-minute manual search process.

Role-based access and activity logging provided mechanisms for controlling and monitoring access to sensitive information [20]. User acceptance feedback also indicated that the system's usability, clarity, and workflow alignment were appropriate for the intended users [19]. These outcomes provide the basis for evaluating the completed TCCIS SMS against the technical and operational requirements established during the development process.



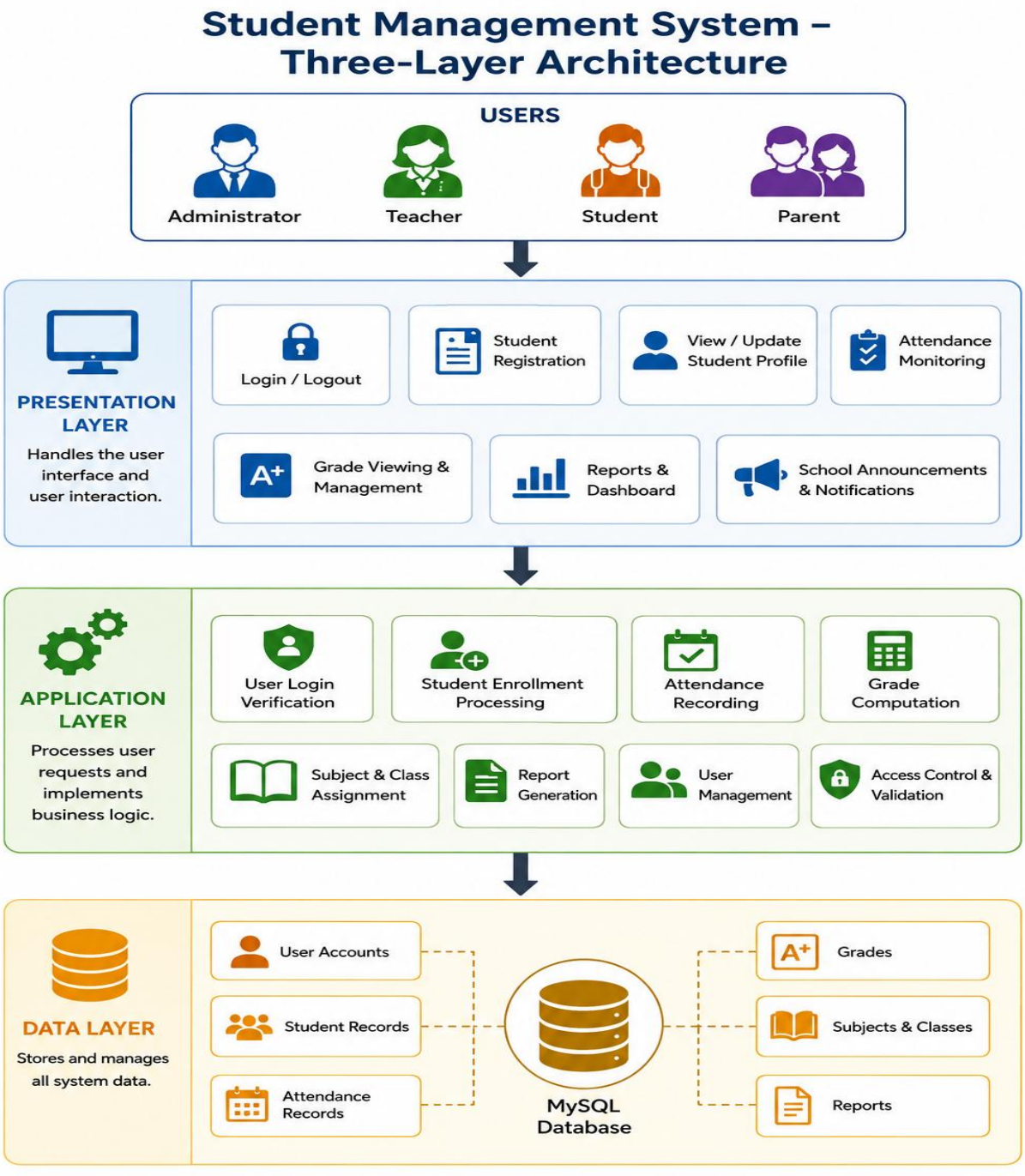
The Agile model is iterative – each cycle delivers value and allows for continuous improvement.

4.4 System Architecture

The Student Management System follows a well-structured Three-Layer Architecture composed of the Presentation Layer, Application Layer, and Data Layer.

This modular, multi-tiered design deliberately separates user interaction, business logic processing, and persistent data storage into distinct, coordinated components that work in unison to manage student information, attendance records, academic grades, inventory assets, and internal communication efficiently, securely, and sustainably.

By isolating these core functional domains from one another, the architecture significantly enhances system maintainability, strengthens information protection, simplifies future feature expansion and technical updates, and ensures that improvements or adjustments made to one component can be implemented systematically without disrupting the operation or stability of the others.



Presentation Layer

The Presentation Layer constitutes the visible, user-facing portion of the system — the direct point of interaction between stakeholders and the TCCIS School Management System. Functioning as the front-end interface, it encompasses two complementary access environments: the secure, role-based internal portal used by authorized administrators, teachers, and parents; and the public-facing information website accessible to prospective families, visitors, and the general public.

Both environments are accessed through standard web browsers, eliminating the need for specialized software installation and enabling access from any internet-connected device.

This layer has been purposefully designed to prioritize clarity, intuitiveness, and accessibility, ensuring that users with varying levels of technical proficiency can navigate confidently and complete required tasks without excessive training or support.

Its structure follows consistent design principles across all modules — unified navigation menus, recognizable labeling conventions, predictable page layouts, and clear visual hierarchy — so that once users become familiar with one section of the system, they can transfer that understanding to every other section.

Responsive design forms a central pillar of this layer's implementation: built using HTML5, Tailwind CSS, and JavaScript, the interface automatically adapts its layout, element sizing, and navigation structure according to the screen size of the device in use.

On desktop computers, information appears in detailed multi-column formats suitable for data entry and record review; on tablets and mobile phones, content reflows into simplified, vertically stacked views with appropriately sized interactive elements. This ensures that whether users are accessing the system from school workstations, personal laptops, smartphones, or tablets, the experience remains functional, readable, and easy to navigate.

Beyond visual display, the Presentation Layer also manages how information is conveyed and collected: it presents organized dashboards, searchable records, data entry forms, confirmation messages, and status updates in a consistent, easy-to-interpret format.

It performs immediate client-side validation — checking that required fields are completed, formats are correct, and values are reasonable — providing instant feedback to users before requests are sent to the server. This not only improves perceived system speed but also guides users toward accurate, complete data submission, reducing errors and streamlining overall workflow.

When a user interacts with the system — logging in, viewing student records, marking attendance, entering grades, sending a message, or browsing announcements — the Presentation Layer captures that action, formats the necessary data, and forwards the request to the Application Layer for processing.

Once the backend returns a result, this layer receives the response and displays it appropriately: updating a record, showing a success notification, displaying an error message, or loading the requested information. By maintaining this clear separation, the interface remains focused on usability and presentation, independent of the underlying data processing and storage mechanisms.

Functions:

1. User login and logout
2. Student registration forms
3. Viewing and updating student profiles
4. Attendance monitoring pages
5. Grade viewing and management
6. Inventory management pages
7. Parent-teacher messaging pages
8. Reports and dashboard display
9. Public-facing announcements, calendar, enrollment, forms, gallery, staff directory, and FAQ pages.

When a user logs in, enters student information, checks attendance records, or views grades, the request is made through the Presentation Layer. The layer then sends the request to the Application Layer for processing and displays the results back to the user. Importance: The Presentation Layer determines how usable the system is for staff with varying levels of technical comfort. Keeping this layer visually clear and consistent across every module was treated as a priority throughout development, since a confusing interface undermines even a technically correct backend.

Application Layer

The Application Layer functions as the functional core, decision-making engine, and security intermediary of the entire system. It sits between the user-facing Presentation Layer and the information-storing Data Layer, serving as the vital bridge that connects interface actions to data operations and vice versa. Its primary responsibility is to receive, interpret, validate, and act upon all requests originating from users, while consistently enforcing the system's defined business rules, access controls, and data integrity standards.

This centralization ensures that every interaction — whether a login attempt, attendance submission, grade entry, inventory adjustment, or message sent — follows identical, verified procedures regardless of which user or device initiates it.

When a user submits a request through the interface, the Application Layer first receives and inspects the incoming data. It performs comprehensive validation: checking that all required fields are complete, that values fall within acceptable ranges and formats, and that no potentially harmful or malformed content is included.

This step prevents incomplete, inconsistent, or malicious entries from reaching the database, protecting both data quality and system security. Only after data passes these preliminary checks does the layer proceed to verify authorization — confirming that the requesting user holds the appropriate role and specific permissions to perform the intended action.

This security enforcement occurs at the application level, not merely as a visual restriction within the interface, ensuring that safeguards remain effective even if a user attempts to bypass visible controls.

Once validity and authorization are confirmed, the Application Layer executes the relevant business logic — the operational rules and computations that define how the system functions.

This encompasses a wide range of critical processes: verifying user credentials during login, determining which features and records each user may access, calculating attendance percentages and academic averages, finalizing

grading periods, triggering low-stock notifications, routing messages between authorized parties, compiling filtered datasets for reports, and assembling responses to be displayed.

It also manages workflows and constraints: for example, preventing grade modifications once a term is closed, restricting parents to view only their own children's records, and ensuring inventory transactions are balanced and accounted for.

The layer then communicates with the Data Layer — retrieving stored information, recording new entries, updating existing records, or removing data as permitted — using secure, structured queries designed to protect data integrity and prevent vulnerabilities. When results are returned from storage, the Application Layer organizes them according to the requesting user's permissions and context, then passes the processed, relevant information back to the Presentation Layer for display.

In cases where a request cannot be completed — due to invalid input, insufficient permissions, or a conflict with existing data — the layer generates clear, informative feedback and returns it to the interface so the user understands the outcome and how to proceed.

By centralizing all processing, logic, and enforcement in this dedicated layer, the architecture delivers substantial benefits:

- Consistency: Rules are defined once and applied uniformly across every module and user interface.
- Security: Controls are actively verified rather than assumed, creating a robust defense against unauthorized access or manipulation.
- Maintainability: Updates to policies, calculations, or workflows can be made in one location and take effect immediately throughout the system, without requiring changes to every page or screen.
- Clarity: The roles of the interface and the database remain distinct, simplifying troubleshooting, testing, and future development.

In essence, the Application Layer is where the system translates user needs into structured actions, ensuring that every operation is valid, authorized, accurate, and aligned with the operational requirements of Timothy's Cradle Christian International School.

Main Functions:

10. User login verification and role-based access enforcement
11. Student enrollment processing
12. Attendance recording
13. Grade computation and validation
14. Subject and class assignment
15. Inventory stock tracking and low-stock alerting
16. Parent-teacher message routing
17. Report generation
18. User account management

Responsibilities:

19. Validates user input before it reaches the database.

20. Processes requests from users and routes them to the correct module.

21. Ensures that only authorized users, based on their assigned role, can access or modify specific features and data.

22. Sends and receives data from the database using secure, parameterized queries.

When a teacher records attendance or enters grades, the Application Layer checks whether the information is valid and whether that teacher is authorized to submit it. After validation, it sends the data to the database for storage.

If a user requests information, the Application Layer retrieves only the data that user role is permitted to see and sends it back to the Presentation Layer.

Importance: This layer is where the system's security guarantees are actually enforced role checks, input validation, and access control all happen here, not merely as an interface convention. A weakness at this layer would compromise the entire system regardless of how well the Presentation Layer is designed.

Data Layer

The Data Layer forms the secure, reliable, and permanent foundation of the entire system, responsible for systematically storing, organizing, retrieving, and safeguarding every piece of information generated and managed by TCCIS SMS. Functioning as the authoritative central repository, it is built upon a MySQL relational database — a proven, high-performance management system selected for its reliability, efficiency, and seamless compatibility with the application layer — where all institutional records are maintained in a structured, consistent, and interconnected manner.

Scope of Stored Information

This layer preserves the full breadth of data required to support the school’s operations, organized into logically grouped tables:

- Core identity and enrollment records: Student profiles, personal details, contact information, parent–guardian associations, class assignments, and enrollment status
- Academic and participation data: Daily attendance logs, quarterly grades, subject and curriculum mappings, and term progress records
- Operational and resource information: Inventory item details, stock levels, transaction histories, procurement thresholds, and asset assignments
- User and accountability data: Account credentials, role assignments, login activity, and a comprehensive audit trail of all system modifications
- Communication and public content: Announcements, messages, calendar events, and information displayed on the public website

Organization and Structure

Information is organized following relational database principles: tables are linked through carefully defined foreign-key relationships — such as student_id, user_id, and class_id — ensuring that related data remains connected while minimizing unnecessary duplication.

For example, a parent account is linked to specific student records, a grade entry is tied to the corresponding student, subject, and teacher, and an attendance record references the relevant class and date. This structure not only conserves storage space but also preserves data integrity: preventing orphaned or conflicting entries, ensuring consistency across modules, and enabling accurate, rapid retrieval of information when requested.

Beyond basic storage, the Data Layer enforces integrity rules at the database level — validating formats, enforcing required fields, and maintaining referential connections — so that incomplete or inconsistent records cannot be saved, regardless of how a request is submitted. It also supports role-based filtering at the storage level, reinforcing privacy safeguards established in the Application Layer and ensuring that sensitive information remains protected even if higher-level controls are circumvented.

Core Functions

When the Application Layer issues a request — to save a new student record, retrieve attendance history, update inventory levels, or generate a performance summary — the Data Layer executes the precise database operation, manages secure connections, and returns verified results. All interactions use parameterized, secure queries to protect data from unauthorized access attempts and common vulnerabilities. Significant modifications — additions, updates, or deletions — are logged with timestamps and user identifiers, creating an immutable audit trail that supports accountability, transparency, and compliance.

Key Advantages

- Consistency: All modules draw from one authoritative source, eliminating discrepancies between separate records
- Security: Data is protected at the lowest level, with access granted only through validated application-level requests
- Efficiency: Indexed tables and optimized relationships allow information to be located and delivered rapidly, even as the volume of records grows
- Preservation: Structured, redundant-free storage supports long-term retention and simplifies backup, recovery, and future system expansion

Stored Data:

23. Student personal and enrollment information

24. Attendance records

25. Grades and academic performance records

26. Class and subject assignments

27. Inventory items and inventory transaction history

28. Teacher, administrator, and parent account information (stored together in a single Users table, distinguished by a role field)

29. Parent-teacher messages

30. An audit log of key system actions

31. Public website content: announcements, calendar events, downloadable forms, gallery photos, FAQs, and the staff directory Whenever data is added, updated, or requested, the Data Layer handles the underlying database operations, saving the information and retrieving it whenever needed by the Application Layer.

Importance: The Data Layer is what makes the system's records trustworthy over time relationships between tables (for example, linking a parent account to only their own child's records) are what actually enforce data privacy at the storage level, rather than relying solely on the interface to hide information.

4.5 Hardware and Software Requirements

Hardware Requirements

The Student Management System requires the following hardware components to ensure smooth operation and reliable performance: Server/Administrator Computer

32. Processor: Intel Core i3 or higher

33. RAM: 4 GB minimum (8 GB recommended)

34. Storage: 500 GB Hard Disk Drive (HDD) or 256 GB Solid State Drive (SSD)

35. Monitor: 1366 x 768 screen resolution or higher

36. Keyboard and Mouse

37. Network Connection: LAN or Internet connection

User Computer

38. Processor: Dual-Core Processor or higher

39. RAM: 2 GB minimum

40. Storage: 120 GB available space

41. Monitor: 1366 x 768 resolution or higher

42. Keyboard and Mouse

43. Internet Connection for accessing the system

Network Equipment

44. Router or Modem for Internet connectivity

45. Network Switch (if using a Local Area Network)

46. Ethernet cables or Wi-Fi connection

Optional Devices

47. Printer for printing student records and reports

48. Scanner for digitizing student documents

49. Backup Storage Device (External Hard Drive or USB Drive)

Software Requirement

Operating System

50. Windows 10 or Windows 11

51. Linux-based operating systems (optional, for server deployment)

Web Server

52. XAMPP Control Panel (local development)

53. Apache Web Server

54. A PHP/MySQL-compatible shared hosting provider (for online deployment)

Programming Languages and Technologies

55. PHP — for server-side development

56. HTML, CSS, and JavaScript — for user interface structure and interactivity

57. Tailwind CSS — a utility-first CSS framework used for responsive, consistent styling across all modules.

Database Management System

58. MySQL Database Web Browser

59. Google Chrome (recommended)

60. Mozilla Firefox

61. Microsoft Edge Development Tools

62. Visual Studio Code (VS Code)

63. phpMyAdmin for database management

64. Git for version control These hardware and software requirements ensure that TCCIS School Management System operates efficiently, provides reliable data management, and supports the daily administrative activities of the school.

4.6 Database Design

The database design of the Timothy's Cradle Christian International School Management System (TCCIS SMS) forms the structural and operational foundation of the entire platform, governing how information is stored, related, retrieved, and protected across all system modules. Implemented as a MySQL relational database, the schema comprises eighteen logically organized, interconnected tables arranged into four functional groupings.

This structured, normalized design prioritizes data integrity, minimizes redundancy, simplifies maintenance, and ensures that information remains consistent, secure, and rapidly accessible — whether supporting daily administrative workflows, academic record-keeping, internal accountability, or public information dissemination. Relationships between tables are established through carefully defined foreign keys, creating a cohesive data ecosystem where changes propagate predictably and discrepancies are systematically avoided.

Core School Records

This grouping establishes the fundamental identity and organizational framework of the system, housing permanent reference information that underpins all other modules.

- students Table — Serves as the central repository for all learner-specific data. Fields include a unique student identifier, full legal name, gender, date of birth, residential and mailing addresses, contact numbers, enrollment status, date of admission, medical or special needs notes, and grade-level assignment. By consolidating comprehensive student details in one authoritative table, the design ensures that every module — from attendance tracking to grade reporting — references the same verified profile, eliminating conflicting or duplicated student records across departments.

- users Table — Centralizes all system account credentials and access permissions in a single, unified table rather than separate administrator, teacher, and parent tables. Each record includes a unique user ID, full name, email address, securely hashed password, contact information, and a role field that explicitly designates the account type (Administrator, Teacher, or Parent). This approach streamlines authentication, ensures consistent login protocols across all stakeholders, simplifies permission checking within the Application Layer, and reduces the risk of fragmented or mismatched account data.

- student_parent Table — Functions as a purpose-built junction table that resolves the many-to-many relationship between parents and students: one parent may have multiple children enrolled, and a student may be linked to more than one guardian. By maintaining these associations independently, the system can dynamically filter access so that each parent views only their own children's records, while students remain correctly associated with all relevant guardians. This design supports data privacy and security at the storage level, reinforcing role-based restrictions established in the application logic.

- classes Table — Stores section or homeroom details including class name or section designation, assigned grade level, academic year, and the user ID of the responsible teacher. This table acts as an organizational hub: attendance rosters, grade sheets, and enrollment assignments reference each class record, ensuring that all activities related to a specific group remain connected and easily retrievable.

Academic Records

This section captures, preserves, and presents the instructional and performance-related data central to the school's educational mission.

- attendance Table — Logs every daily attendance entry with precise context: student ID, class or section reference, date of record, attendance status (Present, Absent, Late, or Excused), and the user ID of the personnel who recorded the entry. Timestamp fields further document exactly when the entry was created or modified. This granularity supports accurate percentage calculations, historical trend analysis, and intervention monitoring, while clearly attributing responsibility for each record.

- subjects Table — Defines the curriculum framework by listing all offered subjects, mapped to specific grade levels and academic years. Fields include subject name, descriptive code, assigned department or teacher, and sequential ordering for display purposes. This standardized catalog ensures that subjects are named consistently across attendance, grade, and reporting modules.

- grades Table — Maintains individual student academic performance records, linking each quarterly or term grade to the specific student, subject, grading period, academic year, and the teacher responsible for submission. Additional fields capture component scores, weighted averages, and a finalization flag that locks records against further editing once a grading period officially closes. This structure preserves a complete, auditable academic history and ensures that progress reports, report cards, and transcripts draw from identical verified data.

Operations and Accountability

These tables support resource management, internal communication, system security, and institutional transparency.

- inventory Table — Provides the master asset registry, cataloging every tracked item with fields for item name, category, unit of measurement, current quantity, minimum stock threshold, physical storage location, condition status, and date of acquisition. Centralizing this baseline data enables real-time stock visibility and automated low-stock alerting.

- inventory_transactions Table — Logs every change to inventory levels: issuance, return, restock, adjustment, or disposal. Each entry references the affected inventory item, quantity change, transaction type, responsible user ID, timestamp, and optional remarks. This transactional approach creates a permanent audit trail, supports reconciliation, and clarifies exactly when and by whom stock levels were modified.

- messages Table — Stores secure, role-governed communication between stakeholders. Each message record links to the sender, intended recipient, associated student (where applicable), creation timestamp, read-status indicator, and message content. Threaded organization preserves conversation context, while association with specific student records helps parents and teachers locate relevant correspondence efficiently.

- audit_log Table — Maintains an immutable system-wide record of significant data-modifying events: creation, revision, or deletion of records, grade finalization, and permission changes. Each entry captures the action type, affected table or record, responsible user ID, timestamp, and previous versus updated values where relevant. This trail supports accountability, discrepancy investigation, and compliance documentation.

- login_attempts Table — Bolsters authentication security by recording both successful and failed login attempts, including the user account targeted, attempt timestamp, IP address, and success or failure status. This data enables detection of repeated unauthorized access attempts and supports protective measures such as temporary account throttling or alert notifications.

Public Website Content

These tables store information displayed on the public-facing Timothy's Cradle Christian International School website, maintaining logical separation between publicly accessible content and sensitive internal portal data. Content is organized into dedicated tables:

- announcements — News and general notices with publish dates and visibility settings
- calendar_events — School-wide and grade-level event schedules
- downloadable_forms — Enrollment, medical, and administrative documents available for retrieval
- gallery_photos — Image files and captions for school activities and facilities
- faqs — Structured frequently asked questions and official responses
- staff_directory — Published personnel listings with names, roles, and contact information

Authorized administrators manage these records through the internal portal; approved updates are immediately reflected on the public website without exposing or compromising restricted internal data.

Relationships and Design Benefits

Across all functional areas, table relationships are enforced through foreign keys — such as student_id, user_id, class_id, and subject_id — ensuring referential integrity: records cannot reference nonexistent entities, and deletion rules prevent orphaned or disconnected data entries. This relational structure delivers multiple operational advantages:

- Reduced redundancy: Information is stored once and referenced as needed, rather than duplicated across unrelated tables
- Consistency: Updates propagate from a single source, ensuring that all modules and users view identical current information
- Speed: Indexed key fields enable rapid data retrieval and efficient report generation, even as accumulated records grow
- Scalability: The modular grouping allows new features or fields to be added within the existing framework without restructuring the entire database
- Security: Logical separation and role-linked associations ensure that sensitive information is accessible only through authorized pathways

Collectively, this database design provides a robust, organized, and future-ready foundation that supports the full scope of TCCIS SMS functionality — from daily attendance and grade encoding to inventory management, stakeholder communication, and public information dissemination — while maintaining the data integrity, privacy, and accountability essential to educational record-keeping.

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CHAPTER V

CONCLUSION AND RECOMMENDATIONS

Introduction

This final chapter presents the comprehensive summary of findings, conclusions, and recommendations derived from the study entitled "A Web-Based Integrated School Management System for Timothy's Cradle Christian International School Inc." The primary objective of this research was to design, develop, and validate a centralized, web-based platform capable of systematically addressing the operational inefficiencies, record-keeping limitations, and communication fragmentation previously experienced by the school in managing its academic, administrative, and institutional activities.

Every phase of the system development lifecycle — from initial requirements gathering through iterative design, implementation, and final evaluation — was directly guided by the identified needs, existing workflows, and specific operational constraints of Timothy's Cradle Christian International School Inc. Development efforts centered on six core functional domains identified as priority areas: student records management, daily attendance monitoring, grade encoding and reporting, inventory and asset control, secure stakeholder communication, and automated report generation.

Prior to this system development, the school relied predominantly on traditional, manual, and largely disconnected methods of information management: physical filing systems, hand-maintained logbooks, disjointed spreadsheets, and informal communication channels. While these approaches served the institution in earlier periods, they imposed significant and growing limitations on daily operations.

Locating specific student or administrative records often required extended periods of manual searching across multiple storage locations; updating related information demanded repeated data entry into separate, unconnected systems; and compiling consolidated institutional reports involved labor-intensive manual collation and cross-referencing of information from disparate sources.

Beyond reduced efficiency, these manual processes introduced measurable operational risks: inconsistencies between duplicate records, avoidable transcription errors, misplaced or permanently lost documents, and unavoidable delays in preparing necessary documentation for internal and external stakeholders.

Furthermore, the absence of a unified, institution-managed digital platform meant that communication between administrators, teachers, and parents often depended on external messaging services, fragmented printed notices, or scheduled in-person meetings — resulting in information that was sometimes inconsistent, poorly documented, untimely, or inaccessible to those who needed it most.

To address these documented challenges, the researchers engineered a purpose-built, web-based integrated school management system that consolidates all essential school functions within a single, cohesive digital environment. The platform unifies student information management, real-time attendance tracking, standardized grade processing, inventory monitoring, role-restricted secure communication, and automated report generation under a shared, authenticated access framework.

Central to this design is a robust role-based access control mechanism, which ensures that administrators, teachers, and parents are each provided with the specific functions and scope of information appropriate to their respective responsibilities — enabling efficient and independent task performance while systematically protecting sensitive personal, academic, and institutional data from unauthorized exposure, modification, or deletion.

By integrating formerly separate and siloed processes, the system establishes a single, authoritative source of verified information, minimizes redundant data entry across departments, reduces opportunities for human error, and streamlines the timely flow of information across the entire school community.

The development of the system followed a structured, iterative, and systematic process aligned with established software engineering standards: requirements identification and validation, operational workflow analysis, architectural design, modular development, functional and performance testing, user-centered refinement, and final evaluation.

Throughout the project, feedback and concerns collected from school administrators, teaching staff, and parent representatives were incorporated into successive versions to ensure that the resulting system aligned closely with actual daily workflows rather than imposing unfamiliar, impractical, or overly complex procedures.

Every feature, database structure, interface element, and security control was designed and implemented to respond directly to the documented needs and operational context of the institution. Prior to final assessment, the system underwent multiple rounds of internal testing, error resolution, and interface refinement to resolve functional defects, optimize usability, and strengthen overall performance and reliability.

The findings presented in this chapter are grounded in verifiable, objective evidence gathered throughout the formal evaluation phase — including functional testing results, quantitative performance metrics, stability assessments, and structured User Acceptance Testing (UAT) feedback collected from representative users across all stakeholder groups.

These evaluation activities were designed not only to confirm that the system operates technically as specified, but also to determine whether it meaningfully and measurably improves the efficiency, clarity, accessibility, and reliability of daily school operations.

The assessments measured the system against established software quality criteria: functionality, usability, reliability, efficiency, security, and overall user acceptability — providing an objective basis for determining whether the developed platform successfully resolves the specific problems identified at the outset of the study.

The summary of findings section details the extent to which each stated objective of the research was achieved. It presents the observed outcomes related to student record management, attendance tracking, grade processing, inventory oversight, internal and external communication, and automated report generation, alongside measured improvements in system responsiveness, data consistency, information retrieval speed, and stakeholder satisfaction.

Where applicable, findings are explicitly compared against the baseline conditions of the school's previous manual processes to illustrate the practical operational impact and tangible benefits of the developed system.

From these verified findings, the conclusions are drawn directly, systematically, and without overgeneralization. They restate how the developed system addresses each problem identified in the earlier stages of the study, confirm whether the research objectives have been fully met, and articulate the broader institutional contribution of the platform to the school's long-term information management capacity.

The conclusions emphasize not only the technical functionality of the system but also its demonstrated potential to support improved transparency, institutional accountability, and sustained operational efficiency within Timothy's Cradle Christian International School Inc.

Finally, the recommendations section outlines feasible, evidence-based enhancements and strategic considerations for future development and long-term implementation. These recommendations are directly informed by limitations observed during the study, the evolving operational needs of the institution, and identified opportunities to extend the system's capabilities in subsequent versions.

Proposed directions include further strengthening of data security protocols and automated backup infrastructure, expanding accessibility options to accommodate diverse user abilities and devices, introducing additional academic and administrative modules as the school grows, enriching real-time communication features, and preparing the system architecture to support future integrations with complementary institutional tools or services.

5.1 Summary of Findings

The study culminated in the successful design, development, and empirical validation of the Timothy's Cradle Christian International School Management System (TCCIS SMS) — a purpose-built, web-based integrated platform systematically engineered to directly address the documented operational challenges, process inefficiencies, information gaps, and reliability limitations inherent in the school's long-standing reliance on manual, paper-based, and fragmented administrative and academic workflows.

Developed through a needs-centered, stakeholder-informed approach, the system serves as a unified, authoritative, and secure digital environment where duly authorized parties — school administrators, teaching personnel, and registered parents — can consistently manage essential institutional information including student enrollment and records, daily attendance tracking, academic grading, inventory and asset oversight, official and private communication, and standardized reporting. Every feature, screen, and workflow was intentionally shaped to reduce dependence on physical documents, disjointed spreadsheets, and disconnected recording methods, replacing them with a cohesive, consistent, and accessible digital framework that strengthens accuracy, preserves data integrity, and substantially improves the overall efficiency of school-wide information management.

Attendance Tracking Module

The first core finding of the study confirmed the successful implementation, functional reliability, and practical effectiveness of the Attendance Tracking Module as a central daily-use component of the system. The platform fully digitized and streamlined end-to-end attendance monitoring, enabling teachers and designated authorized staff to record student attendance status — clearly categorized as Present, Absent, Late, or Excused — directly through the secure portal in moments, effectively eliminating exclusive reliance on handwritten logbooks, loose paper registers, and manually compiled attendance sheets.

Every submitted entry is securely transmitted and permanently stored within the centralized database, where it remains systematically archived, fully searchable, and reliably retrievable by authorized personnel whenever reference or review is required. The system automatically constructs and maintains comprehensive, structured attendance histories, allowing users to examine individual student participation patterns, class-wide attendance trends, and cumulative records across any selected date range, academic term, or school year. This capability significantly elevated the organization, completeness, and integrity of attendance data; drastically reduced the risk of records being lost, damaged, faded, or misplaced; and entirely removed the burden of repetitive manual compilation, totaling, and cross-referencing that previously consumed substantial administrative time when preparing attendance summaries, compliance documentation, and intervention reports.

Student Records Management Module

The Student Records Management Module was likewise successfully developed, seamlessly integrated, and thoroughly validated as the institution's definitive, trusted digital repository for all learner-related information. The module provides a well-structured, searchable, and updatable environment where authorized personnel can formally enroll new students, build complete standardized learner profiles, update personal, contact, and emergency details, filter and locate specific records through multiple flexible criteria, and properly archive files for students who have completed their studies, transferred, or withdrawn from the institution.

Centralized digital storage removed the repetitive and time-consuming need to search across multiple filing cabinets, color-coded folders, and disconnected departmental spreadsheets — tasks that previously took anywhere from several minutes to more than a quarter of an hour to complete and often delayed urgent inquiries or meetings. Most critically, each student profile is dynamically and directly linked to all related system modules: attendance histories, grade records, class assignments, and parent/guardian accounts are automatically associated and consistently updated within the corresponding student record. This intentional interconnectedness ensures that every relevant piece of information appears in one unified, comprehensive view, significantly reduces redundant data entry and conflicting entries across departments, and substantially enhances the overall organization, consistency, accessibility, and completeness of student information throughout the school.

Inventory Control Module

The study further established that the Inventory Control Module delivered a systematic, transparent, and real-time operational approach to managing the school's full range of physical resources — including classroom supplies, teaching and learning materials, technical and office equipment, furniture, and consumable stock. Authorized staff can catalogue every item with detailed descriptive information, assigned category, current quantity, designated storage location, physical condition, and user-defined minimum stock thresholds that trigger proactive alerts.

The system maintains a complete, timestamped, and traceable transaction log of all stock movements — issuances, returns, restocks, inter-department transfers, and quantity adjustments — clearly identifying the responsible personnel and the exact time each action was recorded. This continuous visibility allows the school to monitor available resources with precision, anticipate declining stock levels well before shortages disrupt instruction or daily operations, and make timely, evidence-based procurement, replenishment, and budget allocation decisions. The module therefore replaced inconsistent, infrequent, and often inaccurate manual stock checks with a dependable, always-updated resource management system that supports long-term operational planning, fiscal responsibility, and faithful stewardship of school assets and funds.

Communication Portal

The Communication Portal emerged as another vital, fully functional, and highly valued component of the integrated system, successfully centralizing institutional and stakeholder correspondence within a secure, school-managed digital platform. Administrators can publish clearly timestamped announcements, event reminders, policy updates, and emergency notices with role-targeted visibility, ensuring information reaches only the intended audience groups, while teachers and parents can exchange direct, private messages specifically linked to individual students or classes for matters requiring confidentiality or personal attention. By consolidating these communications in one accessible, consistently available location, the system reduced reliance on disparate external messaging applications, scattered personal SMS threads, and printed paper circulars — channels where information could easily be overlooked, delayed, permanently lost, or shared unevenly among recipients. Announcements and conversation threads remain organized, searchable, and preserved as part of the school's official digital record, ensuring that important information reaches all relevant stakeholders promptly and reliably, and can be referenced, reviewed, or verified whenever necessary.

Reporting Engine

The Reporting Engine successfully automated and standardized the generation of all required administrative and academic documentation, producing accurate, consistent, and professional reports derived directly from verified, up-to-date system data. Authorized users can generate attendance summaries, grade sheets, class master rosters, individual student progress reports, and inventory status updates by selecting specific parameters — including custom date ranges, grading periods, grade levels, assigned classes, or individual students — without the need to manually gather, cross-reference, and compile information from multiple separate sources.

Completed reports can be exported in widely compatible and usable formats such as PDF for formal distribution and archiving, or CSV formats for further independent analysis and record-keeping. This capability entirely eliminated the labor-intensive, error-prone manual collation, calculation, and formatting work that previously delayed report preparation and often introduced avoidable mistakes; ensured full consistency between published documents and underlying source records; and freed administrative and teaching personnel to redirect valuable time toward instructional duties, direct student support, and core operational responsibilities rather than repetitive data processing tasks.

System Performance and Efficiency

Formal technical evaluation confirmed that the system operates efficiently, reliably, and responsively under realistic, peak-period operating conditions consistent with the school's daily use patterns. The measured average response time for common user actions — including account login, submitting attendance entries, viewing stored records, and generating standard reports — was approximately 1.2 seconds, comfortably meeting and falling well within the established 2-second performance benchmark defined during system design.

Frequently accessed information can be retrieved from the centralized database in under 3 seconds, representing a transformative improvement over the previous manual process, which required an estimated 5 to 15 minutes to locate, assemble, and prepare the same information from physical files or scattered digital sources. This marked improvement in speed and responsiveness translates directly into smoother, less interrupted daily workflows, significantly reduced waiting times for staff and stakeholders, and the ability

to access verified, complete information in real time during staff meetings, parent consultations, and urgent operational inquiries — capabilities that were impractical, delayed, or entirely unfeasible under the prior manual system.

User Acceptance and Usability

Findings gathered through structured formal User Acceptance Testing (UAT) demonstrated that the system is intuitive, easily learned, and closely aligned with the school's actual established daily workflows rather than imposing unfamiliar or disruptive procedures. Representatives from every stakeholder group — administrative staff, teachers, and participating parents — were able to navigate the interface confidently, complete assigned tasks, and utilize relevant features with minimal orientation and without requiring extensive technical training or ongoing support.

The consistent visual design, logical menu structure, and workflow-aligned module organization ensured that the system felt natural and familiar to use, reducing initial resistance and lowering barriers to adoption. Collected feedback confirmed that the platform directly addresses real, identified operational needs, is accessible and suitable for users with varying levels of technical confidence and experience, and delivers tangible, practical value that improves daily operations for all members of the school community.

Security, Access Control, and Accountability

The implementation of role-based access control successfully protected sensitive institutional and personal information by limiting each user account to only the specific functions and data relevant to their assigned position and responsibilities. Administrators hold full system oversight and comprehensive management capabilities; teachers access records and features exclusively for their assigned classes and students; parents view information only for their own children and related official communications. This carefully structured framework prevents unauthorized access, accidental exposure, or inappropriate modification of confidential student records, academic data, and institutional information.

The system also maintains a detailed, immutable, and timestamped activity log — recording significant actions such as record creation, revision, deletion, and grade finalization — alongside identifiers for the responsible user and the time of action. This permanent audit trail strengthens institutional accountability, operational transparency, and compliance readiness by enabling designated personnel to review system activity, trace information changes, and investigate discrepancies thoroughly whenever necessary.

Integration and Operational Impact

The true and lasting value of TCCIS SMS lies in the seamless, purposeful integration of all modules through the shared centralized database architecture. Information entered in one functional area — during enrollment, attendance recording, grading, or inventory updating — becomes immediately and reliably available to all related processes across the platform, eliminating duplicate data entry, reducing inconsistency, and ensuring that every authorized user references the same verified source of information. This unified foundation transforms previously disconnected, siloed workflows into a cohesive, efficient digital ecosystem.

Collectively, the system demonstrably reduces repetitive administrative labor: attendance is recorded once and automatically flows into summaries, reports, and parent notifications; student profiles populate class rosters and official documents without retyping; inventory adjustments update status dashboards and trigger replenishment alerts; and recorded grades feed directly into progress reports and academic transcripts. By consolidating these essential functions in a single, purpose-built web-based platform, the study confirmed that the system comprehensively addresses the school's identified information management requirements, delivers measurable, sustainable improvements in efficiency and data integrity, and provides a reliable, scalable, and enduring foundation for the institution's current and future information management needs.

5.2 Conclusion

The design and development of the Timothy's Cradle Christian International School Management System (TCCIS SMS) represents a purposeful, comprehensive intervention that directly addresses the operational inefficiencies, information fragmentation, data inconsistencies, and security vulnerabilities arising from the school's long-standing reliance on manual, paper-based, and disconnected information management processes. This study has demonstrated with measurable, verifiable evidence that unifying core administrative, academic, operational, and communication functions within a single, secure, web-based platform substantially elevates the organization, consistency, accessibility, integrity, and protection of the institution's critical information.

Where processes formerly depended on scattered physical records, handwritten logbooks, disjointed spreadsheets, and disparate external messaging channels, the system has systematically replaced these fragmented practices with streamlined, centralized digital workflows. This transformation eliminates redundant data entry, resolves conflicting versions of the same information, and establishes a single, authoritative, shared source of truth accessible only to duly authorized personnel. By consolidating data ownership and access, the system removes the delays, oversights, and gaps that frequently compromised accuracy and responsiveness under the previous manual regime.

A defining success of the project is the full automation of student attendance management — a process that was previously slow, labour-intensive, and prone to human error. Teachers and designated staff can now record daily attendance electronically, with every entry automatically timestamped, permanently stored, and directly linked to the corresponding student, class period, and responsible personnel.

The system independently calculates cumulative attendance percentages and participation trends, eliminating repetitive manual computation and cross-referencing. Structured, searchable attendance histories support continuous monitoring of individual and class-wide engagement, furnish objective evidence for early intervention when attendance concerns emerge, and generate standardized, audit-ready reports far more quickly and reliably than was ever feasible with paper-based methods.

Likewise, the establishment of the Student Records Management Module delivers a centralized, trusted digital repository for all learner-related information — replacing data previously dispersed across filing cabinets,

departmental folders, and unconnected paper forms with a structured, relational database. Built-in search and filtering allow authorized users to retrieve complete student profiles in mere seconds, a dramatic improvement over the five to fifteen minutes often required to locate and assemble physical files. Most importantly, student records are dynamically linked to attendance logs, grade entries, class rosters, and parent accounts, forming an integrated information ecosystem where updates reflect consistently across all related areas. This interconnected architecture preserves data integrity, eliminates discrepancies, and provides school leadership with a holistic, longitudinal view of each student's enrolment, academic progress, and participation throughout their time at the institution.

The system also introduces a robust, practical, and transparent mechanism for inventory and asset management — a function that was previously handled through infrequent, inconsistent manual checks that offered little visibility into stock levels, usage, or accountability. Through the Inventory Control Module, personnel can catalogue supplies and fixed assets, monitor real-time stock availability, record every issuance, return, restock, or transfer, and configure automated low-stock alerts to anticipate shortages before they disrupt instruction or daily operations. Complete, timestamped transaction trails create an unambiguous audit history of resource stewardship, strengthening accountability and oversight of school property. Access to accurate, up-to-date inventory data further supports timely, evidence-based procurement and budget planning, helping the institution allocate limited resources efficiently and ensuring that essential materials remain consistently available.

In parallel, the system establishes a secure, role-governed Communication Portal that brings all official school correspondence onto the institution's own managed infrastructure, reducing reliance on third-party messaging platforms where messages may be delayed, lost, or seen by unintended recipients. Administrators can publish targeted announcements and event details; teachers and parents can exchange private, direct messages; and all interactions are preserved within a retrievable, school-controlled record. Role-based access controls ensure that confidential or sensitive information remains visible only to authorized parties, while eliminating the fragmentation and uneven message distribution that often characterised previous communication methods. This capability strengthens trust, improves transparency between home and school, and ensures that critical information reaches stakeholders promptly, reliably, and securely.

The project's objective of automated, flexible report generation is fully realised through the integrated Reporting Engine, which transforms raw system data into standardised, exportable documentation — including attendance summaries, grade sheets, class lists, student progress reports, and inventory status updates — with precise, user-defined filtering. This capability entirely bypasses the labour-intensive collation, cross-checking, and manual formatting that previously delayed reporting and introduced avoidable errors. Because every report draws from the same centralised, verified database, outputs remain consistent across departments and always reflect the most current information available. This not only lightens the administrative workload but also enhances the professionalism, reliability, and timeliness of documentation provided to internal management, parents, and external oversight authorities.

Empirical performance evaluation further confirms that the system meets and exceeds its defined technical benchmarks. The recorded average system response time of approximately 1.2 seconds — well within the established 2-second standard — demonstrates efficient processing even during periods of concurrent use. Information that once took five to fifteen minutes to locate and compile is now retrievable in under three seconds, a transformative improvement that enables informed decision-making during meetings, parent consultations, and urgent inquiries — capabilities that were impractical or impossible under the manual system. Complementing these efficiency gains, role-based access control and continuous system-wide activity logging provide layered safeguards: protecting sensitive personal, academic, and institutional data from unauthorised access or alteration, and creating a verifiable digital footprint of all system activity that reinforces accountability and transparency.

User Acceptance Testing outcomes align closely with these technical achievements, confirming that the system is intuitive, easily learned, and closely matched to the school's actual daily workflows. Its modular, role-aligned design presents each user only with the functions and information relevant to their responsibilities, avoiding unnecessary

complexity and directly supporting established institutional practices rather than disrupting them. This thoughtful, needs-centred approach lowers barriers to adoption, accommodates personnel with varying levels of technical confidence, and lays the groundwork for sustained, long-term use across the school community.

Beyond its immediate operational impact, the project demonstrates the broader value of a customised, appropriately scaled web-based solution for educational institutions operating with limited resources. Unlike large-scale commercial platforms that may impose prohibitive costs, extraneous features, or rigid operational workflows, TCCIS SMS delivers exactly the capabilities Timothy's Cradle Christian International School requires — no more, no less. Its integrated architecture, unified data foundation, role-based security, and automated reporting form a sustainable, cost-effective digitalisation framework designed to grow alongside the institution. In doing so, it offers a replicable, adaptable model for similarly sized schools seeking to modernise without overextending their financial or technical capacity.

It must be recognised, however, that the enduring benefits of the system depend on essential supporting factors: consistent and reliable internet connectivity, accurate and timely data entry by authorised users, scheduled technical maintenance, and rigorous, regular data backup and security monitoring. These requirements are inherent to all digital systems and must be embedded within the school's ongoing management and implementation planning. Continued user training, periodic functional reviews, and proactive system maintenance will remain indispensable to preserving and extending the efficiencies established through this project.

In summary, the TCCIS SMS stands as a practical, effective, and sustainable transformation of the school's previously fragmented manual processes into a cohesive, reliable digital environment. By centralising institutional records, automating routine administrative tasks, accelerating information retrieval, streamlining internal and home–school communication, managing resources with full transparency, and standardising reporting across all departments, the system directly addresses every challenge identified at the study's outset. The successful achievement of all stated research objectives affirms that a purpose-built, needs-aligned, web-based school management system is not only viable but also a highly valuable foundation for advancing the efficiency, organisation, accessibility, and accountability of operations at Timothy's Cradle Christian International School Inc. [26][27]

5.3 Recommendations

Based on the verified findings, performance evaluation results, and identified scope limitations of the completed Timothy's Cradle Christian International School Management System (TCCIS SMS), the following recommendations are proposed to further strengthen the system's functionality, reliability, security, scalability, and long-term usefulness to the institution.

These directions are intended to guide future development, prioritization, and implementation planning, ensuring that the platform continues to evolve alongside the school's operational and technological needs.

Enhanced Attendance Verification

Building upon the current electronic attendance recording capability, future versions may incorporate supplementary verification mechanisms to increase accuracy and reduce manual input effort.

One practical enhancement is the integration of QR-code scanning, where each student or class session is assigned a unique, time-sensitive code that authorized personnel can scan to confirm presence — streamlining the recording process, particularly during high-volume periods or events [9]

. Where resources and privacy considerations permit, biometric authentication such as facial recognition or fingerprint verification could provide an additional layer of identity assurance, minimizing instances of proxy attendance or data-entry error. Any introduction of these technologies must be preceded by careful assessment of cost, compatibility with existing devices, legal compliance, and community acceptance.

Comprehensive safeguards — including clear privacy policies, secure data handling, and limited retention of biometric or personally identifiable information — should be established and communicated to all stakeholders before implementation, ensuring that verification methods enhance rather than compromise trust and data protection.

Offline Capability and Resilience

To address the challenges of intermittent or unavailable internet connectivity — a recognized constraint in many educational settings — future development may introduce an offline-capable architecture [10]. The system could be designed to queue authorized actions locally within the browser or a dedicated application cache, storing attendance entries, quick updates, or form submissions temporarily on the user's device.

Once connectivity is restored, these records would synchronize automatically with the central database following predefined conflict-resolution rules. This capability would be particularly valuable for daily attendance recording and time-sensitive administrative tasks that must proceed regardless of network reliability.

Synchronization logic must be rigorously tested to prevent duplicate entries, conflicting updates, or data loss; clear status indicators should show users whether their actions are pending or successfully saved, and a conflict-notification system should alert administrators when manual review is needed to resolve discrepancies.

Learning Management System (LMS) Integration

To support evolving instructional needs and potential blended or distance-learning approaches, the system may be expanded with a dedicated Learning Management System (LMS) module that operates alongside existing administrative functions [4].

Proposed features could include secure distribution of lesson materials and assignments, online submission and grading, automated progress tracking, and structured feedback mechanisms — all accessible through the same verified user accounts. This unified access preserves familiarity and reduces the need for separate platform credentials; however, the LMS should be developed as a distinct, clearly labeled component so that core school management functions remain uncluttered and easy to navigate.

Implementation planning must account for increased storage requirements, file-type management guidelines, content organization, and additional privacy safeguards for student-generated work. Access controls should be extended to LMS features to ensure that materials and submissions are visible only to appropriate students, parents, and teaching staff.

Financial Management and Payment Integration

To further consolidate administrative workflows, future versions may incorporate a secure financial management module [3]. This extension could streamline fee tracking, payment recording, expense management, budget monitoring, and financial report generation — linking transactions to student profiles and eliminating the need for separate financial spreadsheets or disconnected systems.

If online payment processing is introduced, integration must use a certified, compliant payment gateway; sensitive financial data should never be stored unnecessarily within the school database, and all transactions must comply with applicable data protection and financial security standards.

Role-based access should be strictly enforced: financial information should be accessible only to designated administrative and finance personnel, with a complete, immutable audit trail recording every entry, modification, or approval. This integration would provide a more holistic view of school operations while maintaining rigorous protection of sensitive financial records.

Advanced Visual Analytics and Dashboards

To support evidence-based decision-making, the system may be enhanced with interactive dashboards and analytical tools that transform stored data into actionable insights [28].

Administrators and department heads could view attendance trends, grade distributions, enrollment patterns, inventory consumption rates, and communication activity through intuitive charts, comparative graphs, and summary indicators — all filterable by grade level, class, subject, academic term, or date range.

Rather than manually compiling and comparing separate reports, users would be able to identify emerging concerns, seasonal variations, and improvement opportunities at a glance.

All analytics should respect existing data-access permissions; aggregated views should be designed to prevent the accidental exposure of individual student details to unauthorized parties. This feature would shift the system's role from purely record-keeping to strategic planning, helping the school anticipate needs and allocate resources proactively.

Configurable Notification and Alert System

Future development should introduce a customizable notification framework that keeps stakeholders informed without causing information overload. Users could define their preferred alert thresholds and delivery channels — system portal notices, email, or where appropriate and consented, SMS — for events such as consecutive student absences, declining inventory levels, approaching grading deadlines, newly published announcements, or pending approvals

. Administrators would retain the ability to send priority notifications that bypass individual settings during emergencies; conversely, non-essential updates could be grouped into scheduled digests.

This flexibility ensures that important information is never missed while empowering users to manage their own notification preferences, reducing fatigue and improving overall responsiveness to time-sensitive matters.

Strengthened Security, Backup, and Compliance

As the system expands and accumulates more sensitive data, proactive enhancement of information protection is essential. Recommended measures include enforced strong password policies, multi-factor authentication for administrative and privileged accounts, automated session timeouts, and regular vulnerability assessments.

An automated, encrypted backup strategy — including off-server or cloud-secure storage — should be implemented and periodically tested to ensure full recoverability in the event of hardware failure, accidental deletion, or service disruption. Audit-log monitoring should be made more accessible to designated personnel, with automated alerts triggered by unusual or high-risk account activity.

Access reviews should be conducted at least annually or following staff changes, ensuring that privileges are consistently aligned with current roles and responsibilities. These layered safeguards are not one-time adjustments but ongoing commitments essential to maintaining trust and compliance.

Mobile Optimization and Broadened Accessibility

To better serve parents, teachers, and visitors who primarily access the internet through smartphones or tablets, the interface should undergo targeted mobile optimization.

This includes refining navigation for touch interaction, condensing multi-column displays into readable single-column layouts, streamlining data-entry forms for smaller screens, and ensuring consistent function across major mobile browsers.

Beyond device compatibility, universal design principles should be applied to improve usability for all users — including larger text options, sufficient color contrast, logical tab navigation, and clear error messages. These

improvements extend the system's reach and ensure that the platform remains inclusive and accessible to every stakeholder group, regardless of device or technical confidence level.

Sustained Maintenance, Feedback, and Iterative Improvement

Finally, the school should establish a formal system governance and continuous improvement process following full deployment. This includes scheduled technical reviews of performance, security patches, and hosting reliability; regular collection of structured feedback from administrators, teachers, and parents; and periodic validation of data integrity and backup effectiveness.

Feature requests and identified issues should be prioritized based on impact, cost, alignment with school policy, and compliance obligations. Adopting a phased, incremental update approach — rather than infrequent major overhauls — allows the system to adapt smoothly to changing circumstances, emerging technologies, and the evolving requirements of Timothy's Cradle Christian International School Inc.

These recommendations represent a forward-looking roadmap rather than mandatory immediate implementation. Prioritization should be guided by available budget, technical capacity, data-protection responsibilities, and the institution's strategic goals. Through thoughtful, well-planned enhancements, TCCIS SMS can continue to mature — preserving its core purpose as a reliable, secure, and user-friendly foundation for the school's administrative and academic excellence.

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